
Data Science Core Skills

Statistics

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Data Science Core Skills

Build your background before you arrive

This booklet is the **Statistics** track of *Data Science Core Skills*, a self-study course for students arriving on an MSc in Data Science. The whole course — all three tracks, with runnable code and revealable solutions — is at uh-pam.github.io/bootcamp.

Twelve units on the part of statistics that is about judgement rather than arithmetic: what question you are asking, what the data can answer, and what a number entitles you to say.

Two units draw on the Mathematics track: **S5** uses integration from M7, and **S11** uses least squares from M11.

The track is built around one habit, stated in the first unit and tested in the last: write the question down before you touch the data.

Every worked example uses one small table describing a fictional previous cohort: their prior background, hours of practice and final scores. **The data are invented.** They describe no real student and no real cohort.

1. What is the question?

Statistics · Unit 1

S1 · Statistics Reading: 7 pages With practice: ~ 1.5 hours Assumes: nothing — you can start here

Why this unit is here

Statistics is usually taught as a menu of procedures, and it is easy to come away thinking the skill is picking the right one. It is not. The procedures are the easy part; a library computes them for you in one line.

The hard part is knowing *what you are asking, about whom, and what the answer entitles you to say*. Get that wrong and no amount of correct arithmetic saves you — you get a confident, reproducible, well-presented answer to a question nobody asked. This unit gives you the vocabulary for stating a statistical question properly, and one habit: **write the question down before you touch the data**.

1.1 Before you start

Nothing statistical. You should be able to read a table of data and say what one row of it means, work out an average, and read a percentage.

If you also want to *run* the code rather than read it, [P8](#) is as much pandas as you need. Nothing in the explanations depends on it.

Quick check. A table has one row per student, with columns for their prior background, hours of practice and final score. Five students scored 58, 62, 71, 64 and 65. What is their mean score, and what does one *column* of that table represent?

 Show the answer

The mean is $(58 + 62 + 71 + 64 + 65)/5 = 320/5 = 64$.

A column is one **variable** — one thing measured on every student. A row is one **unit** — everything measured on one student. Almost every confusion later in this track is really a confusion about which of those two you are talking about.

1.2 The idea

Here is a question of the kind people actually ask:

Does practice help?

It sounds answerable. It is not, and why not is the whole of this unit.

Help whom? The students in this table, or students in general? **Help how much, measured how?** Final score, pass rate, something else? **Help in what sense** — do people who practise more *tend to* score higher, or would a given person score higher *if they practised more*? Only the second of those is about cause.

Underneath all three is the gap statistics exists to manage: you have a table, and the claim you want to make is almost never about the table.

Throughout these notes we use one dataset: a synthetic record of a previous cohort who worked through this material.

```
import pandas as pd

cohort = pd.read_csv("data/cohort.csv")
print(f"{len(cohort)} rows, {cohort.shape[1]} columns")
cohort.head(3)
```

240 rows, 7 columns

	student_id	background	diagnostic_score	hours_practice	final_score	completed	started
0	S0001	Mathematics or Physics	89.1	12.9	83.6	True	2026-0
1	S0002	Life sciences	42.4	22.8	56.6	True	2026-0
2	S0003	Engineering	68.4	1.7	58.5	True	2026-0

Two hundred and forty students — and not one of them is the point. They have finished, and nothing you compute will change their marks. What you care about is students *like* them, including the ones arriving next year. Those rows are interesting only as evidence about a larger group you cannot see.

The gap has a standard vocabulary, worth learning properly.

- The **population** is the entire group your question is about. Here: something like *students starting an MSc in data science from a mixed set of prior backgrounds*. Note that you had to *decide* that. It was not in the file.
- The **sample** is the set of units you actually measured — the 240 rows.
- An **estimand** is the number you want, defined on the population: say μ , the mean final score across that whole population. It is one fixed number, and short of measuring every single unit you never see it.
- An **estimate** is the number you compute from the sample — here $\bar{x} = 65.3$, the mean of the 237 rows the worked example keeps. You do see it, and it is not μ .

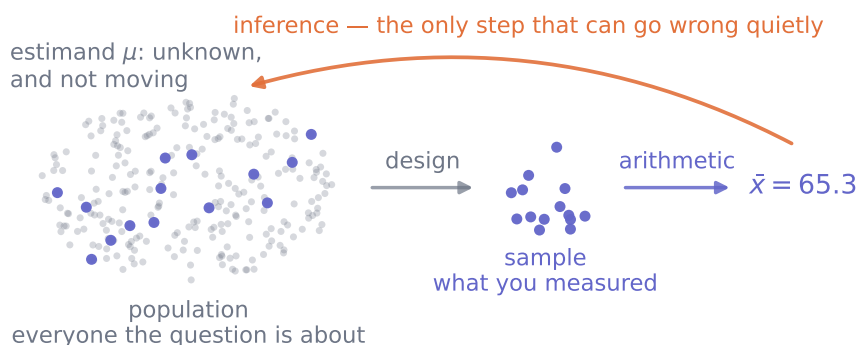


Figure 1: The shape of every statistical question. Arithmetic on a sample is safe and mechanical. The arrow that goes back — from the number you computed to a claim about everyone — is the one that can be wrong while everything on the page looks fine.

That picture splits the subject in two. **Describing** your sample is arithmetic: the mean of the 237 usable scores is 65.27, and no assumption is needed to say so — [S2](#) and [S3](#). **Inferring** something about

the population is a *claim*, it depends on how the sample was collected, and it is only ever true with a margin attached. Nearly everything from S4 onwards exists to make that margin honest.

1.3 The mechanics

Four things a usable question names

A question is ready to work on when you can name all four of these. Until then you cannot choose a method, and choosing one anyway wastes a week.

1. **The unit.** What does one row represent — a student, a submission, a week-of-a-student? Get this wrong and every count you report is wrong.
2. **The population.** Which units is the answer about? Not “the data”: the group you intend to speak about.
3. **The estimand.** A number defined on that population, stated precisely enough that someone else would compute the same thing. “How well students do” is not one; “the mean final score” is.
4. **The design.** How did units get from the population into your table? This is the part people skip, and it decides which conclusions are defensible.

One legitimate sharpening of our vague question:

Among students arriving from a *life sciences* background (population), taking a student as the unit, what is the **difference in mean final score** between those who practised *more than* the group median and those who practised *no more than* it (estimand)? The data are one past cohort’s recorded outcomes, with nobody assigned to practise (design).

Even the tie rule had to be stated: one life-sciences student practised for exactly the group median, and a definition saying “more” and “less” leaves them in neither group.

Longer, uglier, answerable — and it makes visible, in that last clause, something the original question hid.

Parameters and estimates

The convention is Greek letters for the population and Latin for the sample.

Table 1: The same quantity, in the two places it can live

Quantity	Population (parameter)	Sample (estimate)
mean	μ	$\bar{x}$
standard deviation	σ	s
proportion	p	$\hat{p}$

A convention rather than a law — you will meet $\hat{\mu}$ and $\hat{\theta}$ too, where the hat means “estimated from data” — but the distinction is real and load-bearing. **The parameter is fixed and, short of a census, unknown; the estimate is known and varies.** Draw a different 25 students and you will almost certainly compute a different mean; the population mean does not move, because it never depended on who you happened to ask.

That spread is not a defect in anyone’s work. It is what sampling *is*, and S7 puts a number on it.

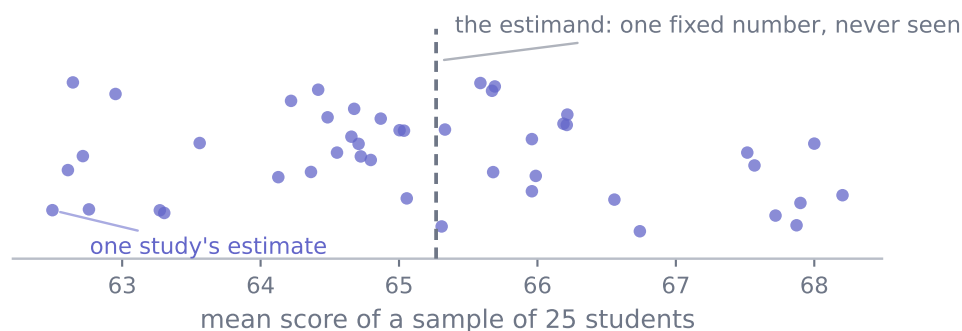


Figure 2: Forty-five imaginary studies, each measuring 25 students from the same population and reporting their mean score. The dashed line is the population mean. It does not move; the dots do. Every real study you will ever read is one dot, reported without the rest of the picture.

What kind of variable is it?

The type of a variable decides which arithmetic on it means anything.

Table 2: Five of the cohort's columns by type

Column	Type	What that permits
final_score	quantitative, continuous	mean, differences, spread
background	categorical, nominal	counts, proportions, mode
completed	categorical, binary	counts, proportions
started_on	date	ordering, gaps, counts per day
student_id	label	nothing — it is a name, not a number

Quantitative splits into continuous and discrete. A count — submissions, faults, siblings — is discrete: nothing lies between 3 and 4. Time, mass or a score on a fine scale is continuous, and the distinction drives the choice of model in [S6](#). It is about the quantity, not the file: `final_score` is recorded to one decimal, so the stored values are strictly discrete, but the thing measured varies continuously.

Categorical splits into nominal and ordinal. Nominal categories have no order — asking whether Computing is “more than” Engineering is meaningless. Ordinal ones are ordered but not evenly spaced: a satisfaction rating, a degree classification. A median is safe there; a *mean* assumes the gap from 1 to 2 matches the gap from 4 to 5, which nobody has checked and which is usually false. It is done constantly anyway; do it knowing that you are.

Where the sample came from

Three designs, in decreasing order of how much they let you say. A **census** measures every unit, so there is no inference left to do — rare, and usually impossible. A **simple random sample** gives every unit the same chance of selection, and is what the rest of this track assumes when it says “random”. A **convenience sample** is whoever was available: the people who replied, the users who opted in, the cohort whose records were kept.

“Random” alone is not enough. A design that samples one subgroup with probability 0.9 and another with probability 0.1 is perfectly random and systematically skewed; such designs are legitimate, and the analysis then has to weight each unit by the reciprocal of its selection probability. And note what even equal chances promise: not that any one sample resembles the population, but that the *procedure* has no systematic preference and that the variability is quantifiable.

Two consequences, both denied in practice constantly.

Sample size does not fix a bad design. If the mechanism that put units in your table is related to what you are measuring, more data makes the estimate more precisely wrong. A 1936 US election poll collected 2.4 million responses and called the wrong winner, beaten by a survey of a few thousand: two million of the wrong people is still the wrong people.

Observational is not experimental. In an experiment the investigator assigns the treatment — who practises more is decided by a coin, not by the student — so a difference in outcome can be laid at the treatment's door. Our cohort is observational, and students who chose more hours differ from those who chose fewer in every other way too.

1.4 Worked example

One vague question, three precise ones, two opposite answers.

Every analysis on this dataset starts by deciding which rows are usable. The file has three planted defects, described in P9.

```
usable = cohort.dropna(subset=["background", "final_score"])
usable = usable[usable["final_score"].between(0, 100)]

print(f"rows in the file {len(cohort)}    usable {len(usable)}    "
      f"excluded {len(cohort) - len(usable)}")
```

```
rows in the file 240    usable 237    excluded 3
```

One row has no recorded score, one has no background, and one scores 187 out of 100. Every unit in these notes that filters on score *and* group analyses these same 237 students.

Now three sharpenings of *does practice help?*

Question A — across the whole cohort. Split the 237 at the median number of hours — above it against at-or-below it — and compare mean final scores.

```
split = usable["hours_practice"].median()
more = usable[usable["hours_practice"] > split]["final_score"]
less = usable[usable["hours_practice"] <= split]["final_score"]
print(f"median hours of practice : {split}")
print(f"practised less  n={len(less):>3}    mean {less.mean():.2f}")
print(f"practised more  n={len(more):>3}    mean {more.mean():.2f}"
      f"    difference {more.mean() - less.mean():+.2f} marks")
```

```
median hours of practice : 15.2
practised less  n=121    mean 68.01
practised more  n=116    mean 62.40    difference -5.61 marks
```

The students who practised more scored **5.61 marks lower** on average. On this reading, practice looks harmful.

Question B — within each background. The same comparison, and the same tie rule, but each group split at its own median, so students are only ever compared with others who arrived in a similar position.

```
def practice_gap(frame):
    """Above-median students' mean score minus the rest's."""
    cut = frame["hours_practice"].median()
    above = frame["hours_practice"] > cut
    return frame[above]["final_score"].mean() -
        frame[~above]["final_score"].mean()

by_group = usable.groupby("background").apply(practice_gap, include_groups=False)
print(by_group.round(2).to_string())
```

background	
Computing	4.10
Engineering	5.14
Life sciences	7.63
Mathematics or Physics	5.24
Social sciences	7.61

In **every one of the five groups** the above-median students scored higher, by between 4.10 and 7.63 marks. On this reading, practice helps, and helps consistently.

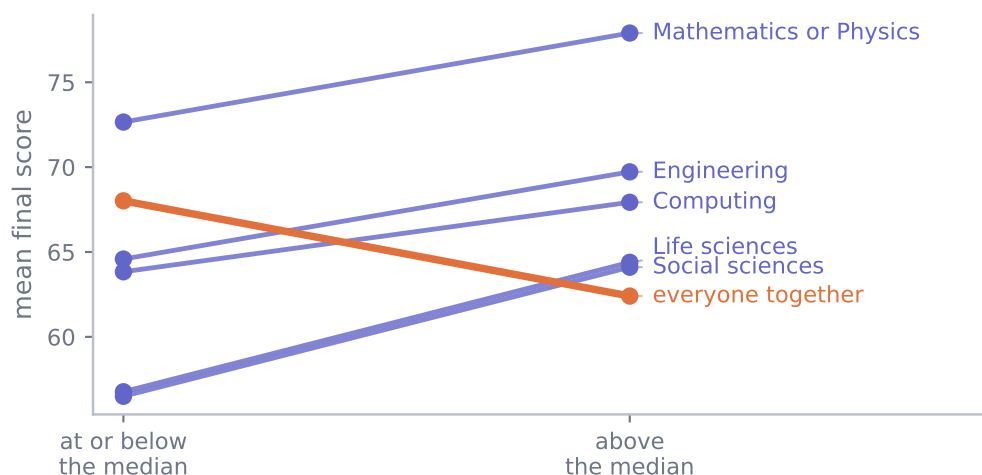


Figure 3: The same 237 students, summarised two ways. Each line joins the mean score of a set's at-or-below-median students to that of its above-median students. The five indigo lines split their own background group at that group's median; the orange line splits the whole cohort at the overall median (121 students against 116, the three exactly on 15.2 hours counting as at-or-below). Five lines up, one line down.

Both answers are correct arithmetic on the same 237 rows. They point in opposite directions because they answer different questions, and the reason is in the data.

```
columns = ["diagnostic_score", "hours_practice", "final_score"]
print(usable.groupby("background")[columns].mean()
      .sort_values("hours_practice").round(1).to_string())
```

	diagnostic_score	hours_practice	final_score
background			
Mathematics or Physics	89.3	8.0	75.3

Engineering	72.4	10.5	67.2
Computing	68.9	12.5	65.9
Life sciences	47.3	22.6	60.5
Social sciences	39.5	27.4	60.2

Students from Mathematics or Physics start furthest ahead and practise least; those from life and social sciences start furthest behind and practise most. So Question A compares a mostly-life-and-social-sciences half against a mostly-quantitative half: a comparison of *backgrounds* wearing the costume of a comparison of *practice*. Question B holds background fixed and asks the question you meant. The reversal has a name and a large literature, and S12 gives it both.

Question C — would practising more raise *your* score? This is what a student actually wants to know, and the data cannot answer it. Nobody assigned hours of practice, and whatever made a student choose more of them — anxiety, spare time, having further to go — is unmeasured. Question B is closer to the causal question than Question A, and it is still an association among students who chose their own hours.

Saying so is not a failure of the analysis. It is the analysis.

1.5 Try it yourself

Exercise 1 Classify each of these variables as quantitative continuous, quantitative discrete, categorical nominal, categorical ordinal, or a label:

- (a) a student's postcode;
- (b) the number of assignments a student submitted;
- (c) a satisfaction rating on a scale of 1 (poor) to 5 (excellent);
- (d) time taken to complete an exam, in minutes;
- (e) degree classification: third, 2:2, 2:1, first.

Show the solution

- (a) **Nominal categorical.** A postcode contains digits, which tempts people to treat it as a number, but the arithmetic is meaningless — there is no sense in which AL10 is twice AL5.
- (b) **Quantitative discrete.** A count. Nothing lies between 3 and 4 submissions.
- (c) **Ordinal categorical.** Ordered — 4 really is better than 3 — but the gaps are not known to be equal. The median is safe. The mean is a convention that assumes equal spacing; it is reported everywhere, and it is worth knowing that you are making an assumption when you report it.
- (d) **Quantitative continuous.** If it is recorded to the nearest minute the stored values are whole numbers, but the underlying quantity is continuous and you would model it that way.
- (e) **Ordinal categorical,** for the same reason as (c). Whether the gap between a 2:2 and a 2:1 is the same as the gap between a 2:1 and a first is not a question the scale answers.

Exercise 2 A colleague asks: “Do students who use the discussion forum do better?”

Rewrite it as a question you could work on. Name the unit, the population, the estimand and the design, and then name one conclusion the design will not support.

 Show the solution

There is no single right answer — that is the point of the exercise — but a usable one names all four. For example:

- **Unit:** one student.
- **Population:** students enrolled on this module in a given year. (If you want to speak about future cohorts, say so, and be ready to defend it.)
- **Estimand:** the difference in mean final score between students who posted at least once and students who never posted.
- **Design:** observational, from the module's own records; nobody was assigned to use the forum.

What it will not support: that using the forum *causes* better marks. The students who post are self-selected, and engagement, confidence and available time all plausibly drive both posting and marks. Also worth stating: “at least one post” is a decision you made, not a fact about the world, and the answer may depend on it. Write it down so the choice is visible rather than buried.

Exercise 3 A satisfaction survey is posted on a course forum. Of 900 enrolled students, 180 reply, and their mean rating is 4.2 out of 5.

- (a) What population is 4.2 an estimate for?
- (b) Name two mechanisms that could make it a poor estimate of the mean rating of all 900.
- (c) Would surveying twice as many students the same way fix the problem?

 Show the solution

- (a) Two different things are being run together, and separating them is the point. As a **description**, 4.2 is the mean rating of the 180 people who replied, and that needs no assumption at all. As an **estimate** it targets whichever population you nominate — nominate all 900 and 4.2 is an estimate of their mean rating, a biased one unless repliers resemble non-repliers *in respect of satisfaction*. Bias is a property of how good the estimate is, not of whether it is one. The population it describes for free and the population it is offered as evidence about are different, and that gap is the finding.
- (b) Any two of: **self-selection** — people with strong opinions, in either direction, are likelier to answer; **undercoverage** — the survey lives on a forum, so students who do not read the forum cannot reply, and forum readers are plausibly more engaged; **timing** — a survey posted in week 12 misses everyone who dropped out, who are unlikely to have been the satisfied ones.
- (c) **No**. Surveying more people the same way leaves the mechanism intact. The estimate would become more precise and no less biased: a tighter cluster around the wrong number. Only a change of *design* — chasing non-respondents, sampling from the enrolment register rather than the forum — helps. Note too that the ceiling here is 900: at some point you cannot buy precision with more people at all, and everything left is bias.

1.6 Check it in Python

Two claims in this unit are worth checking rather than believing: that the cleaning step removes three rows and no more, and that the reversal is real rather than a slip in the reading.

```

assert len(cohort) == 240, f"expected 240 rows, got {len(cohort)}"
assert len(usable) == 237, f"expected 237 usable, got {len(usable)}"

dropped = cohort.loc[~cohort.index.isin(usable.index)]
print(dropped[["student_id", "background", "hours_practice", "final_score"]])

```

	student_id	background	hours_practice	final_score
17	S0018	Computing	14.3	NaN
102	S0103	NaN	26.4	66.2
188	S0189	Life sciences	23.1	187.0

One missing score, one missing background, one impossible score. Exactly the three defects the dataset documents, and no accidental fourth.

Now the reversal. `practice_gap` is written once and applied to the whole cohort and to each group, so if the answers differ it is the data doing it rather than two slightly different calculations.

```

overall = practice_gap(usable)
print(f"whole cohort : {overall:+.2f}")
print(f"by group      : {by_group.min():+.2f} to {by_group.max():+.2f}")

```

```

whole cohort : -5.61
by group      : +4.10 to +7.63

```

A reasonable person, having seen the whole-cohort number, would assume the groups agree with it. Assert that, and watch it fail:

```

assert (by_group < 0).all(), (
    f"whole cohort gives {overall:+.2f}, but "
    f"{(by_group > 0).sum()} of {len(by_group)} groups disagree in sign"
)

```

```

AssertionError: whole cohort gives -5.61, but 5 of 5 groups disagree in sign

```

```

-----
AssertionError                                Traceback (most recent call last)
Cell In[11], line 1
----> 1 assert (by_group < 0).all(), (
      2     f"whole cohort gives {overall:+.2f}, but "
      3     f"{(by_group > 0).sum()} of {len(by_group)} groups disagree in sign"
      4 )

```

```

AssertionError: whole cohort gives -5.61, but 5 of 5 groups disagree in sign

```

That is the check doing its job. The assertion is not wrong about the arithmetic; it is wrong about the world, and it says so loudly instead of letting a false assumption travel unnoticed into a report.

The statement that survives is the one with both halves in it:

```

assert (by_group > 0).all(), "not every group shows a positive gap"
assert overall < 0, "the whole-cohort gap is not negative"
print(f"all {len(by_group)} groups positive (smallest {by_group.min():+.2f}), "
      f"cohort as a whole {overall:+.2f}")

```

all 5 groups positive (smallest +4.10), cohort as a whole -5.61

Both facts are true at once, and any sentence you write about this dataset has to be compatible with both.

1.7 Common wrong turns

Where this usually goes wrong

Choosing the method before stating the question “Which test should I use?” is not a first question, it is a fourth. Until the unit, population, estimand and design are written down, no test is the right one — and the answer is often that you do not need a test at all, you need an estimate with an interval on it.

Reporting a sample statistic as if it were the population value “The mean score was 65.3” describes 237 people. “We estimate the mean score of students on this course at about 65” is a claim about a population. The second needs a margin attached and the first does not; sliding from one to the other is how a description quietly becomes an overclaim.

Believing that a large sample makes a sample representative Size controls how much your estimate bounces around; the design controls what it is bouncing around. A million self-selected responses estimate the views of self-selecting people very precisely.

Reading an association as a lever In the cohort, practising more goes with scoring higher within a background. That does not establish that a given student adding five hours gains five marks. For that you would need the hours to have been assigned, or an argument about what else differs — and the argument is the hard part.

Averaging things that do not average The mean of a satisfaction rating assumes the rungs of the ladder are evenly spaced. The mean of a category code is nonsense dressed as a number. If a variable is nominal, count it; if it is ordinal, prefer the median and say what you did.

Letting the data pick the question Slicing until something interesting appears and then reporting the slice as though it had been the plan is the commonest way honest people produce false findings. Writing the question down first is a defence precisely because it is dated evidence of what you meant to ask.

1.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer — if you cannot say *why*, treat it as wrong.

1. Which of these is an estimand?
 - a. $\bar{x} = 65.3$, the mean score of the 237 usable rows
 - b. The mean final score of all students on this kind of course
 - c. The `final_score` column
 - d. The decision to exclude three defective rows
2. A dataset records, for each of 60 patients, their blood pressure at four visits. A colleague computes a confidence interval for mean blood pressure and reports it as based on “ $n = 240$ independent observations”. What has gone wrong, and what is the unit?
3. True or false: taking a random sample guarantees the sample resembles the population.
4. In the worked example, Question A and Question B gave answers with opposite signs. Which

one was calculated incorrectly?

5. You are given the mean of a variable coded 1 = Computing, 2 = Engineering, 3 = Life sciences, 4 = Mathematics or Physics, 5 = Social sciences. It is 2.87. What does that tell you?

 Show the answers

1. **(b).** An estimand is a number defined on the *population*. (a) is an estimate — computed, known, and liable to differ from one sample to the next. (c) is a variable, not a number. (d) is part of the design and analysis plan, which matters enormously but is not a quantity being estimated.
2. The unit has been confused with the measurement. There really are 240 rows and 240 measurements — that part is fine. What is wrong is *independent*: the **unit is the patient**, so there are only 60 independent units, and four readings from one person carry much less information than four readings from four people. Treating them as 240 independent observations overstates how much you know — S8 shows exactly what it costs.
3. **False**, and the distinction is worth being precise about. Random sampling makes the *procedure* well behaved: over many repetitions it neither favours nor disfavours any part of the population, and the variability is quantifiable. Any single random sample can still be unlucky, and small ones often are. What random sampling rules out is *systematic* mismatch, not bad luck.
4. **Neither.** Both are correct arithmetic and you can rerun them. They answer different questions — one compares students against the whole cohort, the other only against students from the same background — and there is no rule that two different questions must have answers of the same sign. The mistake would be to report either one as “the effect of practice”.
5. **Nothing.** The codes label a nominal variable, so their mean depends entirely on the arbitrary order in which the names were numbered. Renumber the backgrounds in reverse alphabetical order and the “mean” changes while the data do not. Count the categories or report proportions instead.

Where this shows up next

S2 takes one variable and describes it properly — centre, spread, shape — and shows why a mean on its own is rarely enough. S3 does the same for two variables at once, which is where the comparison in this unit’s worked example gets its proper treatment. S4 starts building the machinery that turns a description of a sample into a claim about a population. S12 returns to the reversal above, names it, and shows how to spot the pattern before it embarrasses you.

2. Describing one variable

Statistics · Unit 2

S2 · Statistics Reading: 6 pages With practice: 2 hours Assumes: S1

Why this unit is here

You have a column of numbers and one sentence in which to describe it. Whatever you put in that sentence decides what your reader notices — and what they never find out.

This is also the foundation floor. The confidence intervals, tests and regression models you will meet first are all built out of a mean and a standard deviation, so the places where those two numbers mislead are the places where everything built on top of them misleads too. Knowing what a summary hides is the difference between using these tools and being used by them.

2.1 Before you start

You should be able to:

- say what a variable is and what one row of a table represents — S1;
- tell a quantitative variable from a categorical one — S1;
- work out an average by hand.

Quick check. Six values: 4, 8, 8, 9, 12, 100. What is the mean, and what is the median?

 Show the answer

The mean is $141/6 = 23.5$. The median is the average of the two middle values once sorted, $(8 + 9)/2 = 8.5$.

They disagree by a factor of nearly three, and neither is wrong. Five of the six values are below the mean. That gap — and what it is telling you — is most of this unit.

2.2 The idea

Describing one variable comes down to three questions, in this order:

1. **Where is it?** A typical value.
2. **How spread out is it?** How far from typical a value usually falls.
3. **What shape is it?** Symmetric or lopsided, one peak or several, with or without stragglers.

Most people answer the first, occasionally the second, and skip the third. That is a mistake, and the following picture is why.

Every summary is a compression, and compression throws things away. The skill is knowing what each one throws away, so that you can choose a summary that keeps the part that matters and can say what you have lost.

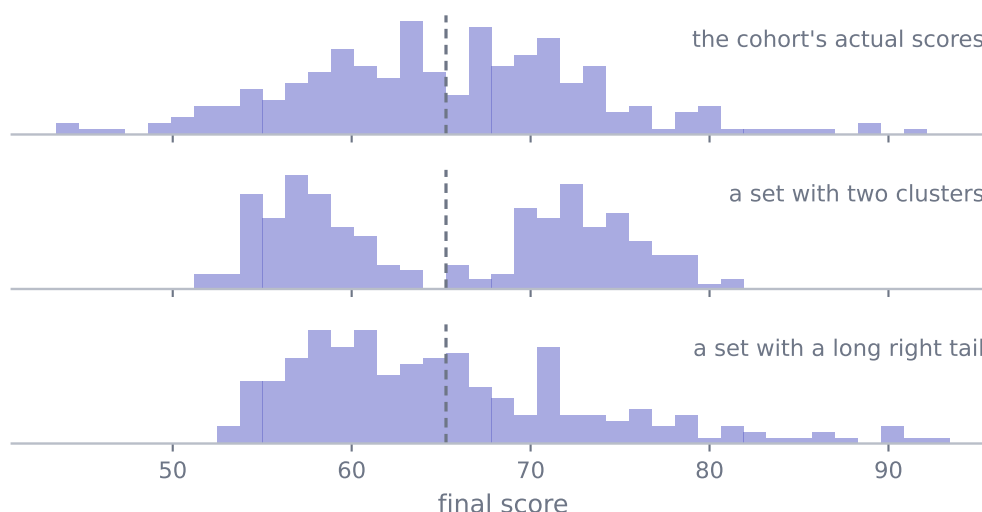


Figure 1: Three sets of 238 numbers with the same mean (65.3, dashed) and the same standard deviation (8.4). Only the top one is the cohort. A mean and a standard deviation are two numbers; the data is two hundred and thirty-eight. Two numbers cannot be the data.

2.3 The mechanics

Where is it?

Three answers, and they answer slightly different questions.

The **mean** is the balance point:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

(If $\sum$ is unfamiliar notation, [M1](#) unpacks it. It says “add these up”.) The mean uses every value, which is its strength and its weakness: every value gets a vote, including the ones that should not have one.

The **median** is the half-way point *by count*: sort the values, take the middle one, or the average of the two middle ones if n is even. Half the data is at or below it. Moving the largest observation from 92 to 920 does not move the median at all, because it is still the largest.

The **mode** is the most frequent value. It is the natural centre for a *nominal* variable, where neither the mean nor the median is available. For continuous data it is mostly an artefact of rounding — our scores are recorded to one decimal place, and the “most common score” is just whichever value happened to be recorded four times rather than three.

When the mean and the median disagree, the disagreement is information. It says the values on one side of the median sit further from it than the values on the other, so the mean has been pulled that way — and that is usually the first sign of a long tail or a handful of extreme observations.

Quantiles

The median generalises. The **p -quantile** is a value at or below which approximately a proportion p of the observations lie; the median is the 0.5-quantile. The 0.25- and 0.75-quantiles are the **quartiles**, Q_1 and Q_3 , and together with the minimum, median and maximum they form the **five-number summary**.

“Approximately” is doing real work in that definition. With n observations there is generally no value with exactly a proportion p below it, so software interpolates between the two neighbouring sorted values — and there is more than one defensible way to do it. On $\{2, 4, 7, 9\}$, asking for Q_1 gives 2.0, 3.0,

3.5 or 4.0 depending on which of NumPy's thirteen conventions you choose, and even on the cohort's 238 scores they do not all agree: the default returns $Q_1 = 59.4$ and two of the others return 59.3 and 59.325. The differences shrink as n grows and are usually invisible, but when a quartile is load-bearing, record which convention produced it.

How spread out is it?

The **range**, $\max - \min$, uses exactly two of your observations and throws away the rest. It also grows with sample size — measure more people and you will eventually meet a more extreme one — so it is not comparable between samples of different sizes.

The **interquartile range**, $IQR = Q_3 - Q_1$, is the width of the middle half. It ignores the tails entirely, which is exactly what you want when the tails are unreliable and exactly what you do not want when they are the point.

The **sample variance** measures spread by average squared distance from the mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2,$$

and the **sample standard deviation** s is its square root.

Two things about that formula deserve more than a nod.

Why square? Because the deviations $x_i - \bar{x}$ sum to zero by construction — that is what makes $\bar{x}$ the balance point — so their average is always 0 and carries no information. Squaring removes the signs. It also weights a deviation of 10 four times as heavily as one of 5, which is a real modelling choice and part of why s is sensitive to extreme values.

Why $n - 1$? Because the deviations are measured from $\bar{x}$ rather than from the population mean μ , and $\bar{x}$ sits in the middle of *this particular sample* by construction. That makes the squared deviations systematically a little too small. Dividing by $n - 1$ instead of n corrects it exactly, in the sense that under independent sampling with finite variance the long-run average of s^2 equals σ^2 . Note what that does *not* say: it does **not** make s an unbiased estimate of σ , because a square root is not a linear operation and averaging does not survive it.

Units matter. If scores are in marks then s^2 is in marks *squared*, which is not a quantity anyone has intuition about. s is back in marks and can be compared to the data. That is the only reason the standard deviation is reported more often than the variance.

What shape is it?

Skew is asymmetry: which side has the longer tail. A common heuristic is that $\bar{x} > \text{median}$ indicates a right (positive) skew.

It is a useful prompt and it is not a theorem. Take $\{0, 1, 1, 2, 2\}$: the mean is 1.2, the median is 1, so the rule says right-skewed — but the standard skewness coefficient works out at -0.34 , which says the opposite. Treat a gap between mean and median as a reason to plot the data, not as a conclusion about its shape.

Modality is how many peaks there are. Two peaks usually means two populations have been stacked on top of each other, and the right response is almost never a better summary of the whole — it is to split them and describe each. The cohort's scores are a mixture of five backgrounds, which is worth remembering every time you look at them.

Seeing it

A **histogram** bins the values and draws the counts. The bin width is a choice you make, and it changes the picture:

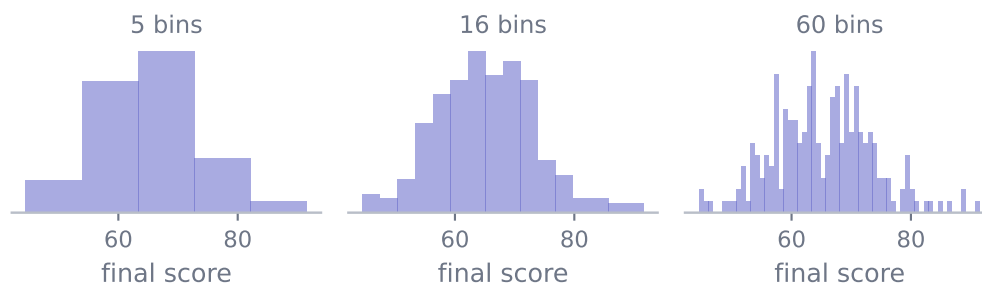


Figure 2: The same 238 scores at three bin widths. Too few bins and the shape is a rumour; too many and you are looking at which decimal each mark happened to land on. Nothing about the data changed between these three panels.

There is no correct answer, which is why you should look at more than one before believing any of them.

A **boxplot** draws the five-number summary. In the Tukey convention the box spans Q_1 to Q_3 with the median marked inside it, and the whiskers reach to the most extreme observation still within $1.5 \times \text{IQR}$ of the box. Anything further out is drawn as an individual point.

Two things people get wrong about that. The whiskers are **not** the minimum and maximum — they stop at the last point inside the fence. And a point drawn beyond a whisker is **not** thereby an error: it is a value that is far from the middle, which is a description, not a verdict. Other conventions put the whiskers at different places, so label which one you used.

2.4 Worked example

Describe the cohort's final scores, properly.

```
import numpy as np
import pandas as pd

cohort = pd.read_csv("data/cohort.csv")
scored = cohort[cohort["final_score"].between(0, 100)] # drops the missing and
↳ the impossible score
scores = scored["final_score"].to_numpy()

print(f"n      {scores.size}")
print(f"mean    {scores.mean():6.2f}      sd    {scores.std(ddof=1):6.2f}")
print(f"median   {np.median(scores):6.2f}      IQR   "
      f"{np.subtract(*np.quantile(scores, [0.75, 0.25])):6.2f}")
print("five-number summary " +
      " ".join(f"{q:.1f}" for q in np.quantile(scores, [0, .25, .5, .75, 1])))
```

```
n      238
mean    65.27      sd    8.38
median   64.80      IQR   11.28
five-number summary 44.5 59.4 64.8 70.7 91.6
```

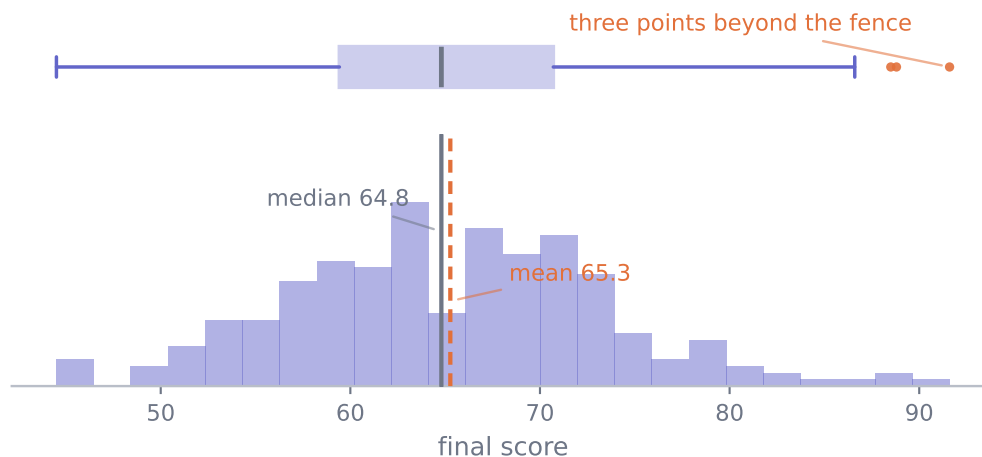


Figure 3: The 238 final scores as a histogram, with the same data as a Tukey boxplot above it on a shared scale. The box holds the middle half; the whiskers reach the furthest point within 1.5 IQR of it; the three orange points lie beyond that fence and are drawn individually.

A defensible sentence: *the 238 usable final scores have a mean of 65.3 marks and a standard deviation of 8.4; the middle half lies between 59.4 and 70.7, and the distribution is close to symmetric with a slight right lean.*

Now the point of insisting on the cleaning step in S1. Here is the same description with the impossible score of 187 left in.

```
with_defect = cohort.dropna(subset=["final_score"])["final_score"].to_numpy()
for name, values in [("cleaned ", scores), ("defect in", with_defect)]:
    print(f"{name} n={values.size} mean {values.mean():6.2f} "
          f"median {np.median(values):6.2f} sd {values.std(ddof=1):6.2f}")
```

```
cleaned      n=238   mean  65.27   median  64.80   sd    8.38
defect in    n=239   mean  65.78   median  64.90   sd   11.49
```

One bad row in 239 moves the median by a tenth of a mark and inflates the standard deviation by **37%**. That asymmetry is the whole argument for looking at both: the median barely noticed, and if you had computed only the mean and the standard deviation you would have had no signal that anything was wrong.

Finally, remember what this variable actually is. One student has no recorded background, so the breakdown below describes 237 of the 238 — the filter follows the question, and this question needs a group label.

```
grouped = scored.dropna(subset=["background"])
print(f"all {len(grouped)} together: sd {grouped['final_score'].std():.2f}\n")
print(grouped.groupby("background")["final_score"]
      .agg(["count", "mean", "median", "std"]).round(2).to_string())
```

```
all 237 together: sd 8.40
```

```
count  mean  median  std
background
Computing      69  65.85  66.70  6.60
```

Engineering	36	67.15	68.65	5.33
Life sciences	57	60.51	59.80	7.29
Mathematics or Physics	36	75.28	74.40	7.63
Social sciences	39	60.21	60.50	6.58

The standard deviation of those 237 students is 8.40, and no single group's is above 7.63. The extra spread is not variability *within* any group — it is the distance between the group means, from 60.2 to 75.3, showing up as spread once they are poured into one column. Describing the mixture as though it were one population is correct arithmetic and a poor description, and **S10** is where that gets taken seriously.

2.5 Try it yourself

Exercise 1 For the sample {2, 3, 4, 5, 6, 6, 7}, compute by hand: the mean, the median, the mode, the sample variance and the sample standard deviation.

Show the solution

The sum is 33 and $n = 7$, so $\bar{x} = 33/7 = 4.71$ (to 2 d.p.).

Sorted, the middle value is the fourth: the **median is 5**. The **mode is 6**, the only value appearing twice.

For the variance, square each deviation from 4.71:

$$(-2.71)^2 + (-1.71)^2 + (-0.71)^2 + (0.29)^2 + (1.29)^2 + (1.29)^2 + (2.29)^2 = 19.43$$

and divide by $n - 1 = 6$: $s^2 = 3.24$, so $s = 1.80$.

The commonest slip is dividing by 7. That gives 2.78, which is not wrong so much as a different quantity — it is the variance of these seven numbers *treated as the whole population*, and it is what `numpy.var` returns by default. (Its square root, 1.67, is what `numpy.std` returns by default; both take a `ddof` argument that switches the divisor to $n - 1$.)

Exercise 2 A department reports that the mean starting salary of its graduates is £42,000. The median is £31,000.

- What does the gap tell you about the distribution?
- Which number should the department publish, and which would a prospective student rather know?
- What would you need to see before trusting either?

Show the solution

- At least half the graduates earn £31,000 or less, and the mean sits £11,000 above that. So the salaries above the median must reach a very long way above it to drag the average up — a minority earning a great deal. For salaries that is nearly always a long right tail, and it is what you should expect before looking. Strictly, the gap tells you which way the mean has been pulled, not that the skewness coefficient is positive; the *What shape is it?* section has a five-value counterexample.
- A prospective student wants the **median**, which answers “what does a typical graduate earn?”. The mean answers “what would everyone earn if the total were shared equally?”, which is not the question. The department has an incentive to publish the larger number;

that incentive is exactly why you should ask which one you were given.

- (c) The five-number summary or a histogram, the sample size, and the response rate. A mean over 12 responses from 300 graduates is not a fact about the department, and — from **S1** — the people who reply are plausibly the ones with good news.

Exercise 3 A boxplot of exam marks has $Q_1 = 52$, median = 61, $Q_3 = 70$, whiskers at 38 and 86, and individual points drawn at 21, 98 and 99. Which of these follow?

- (a) The lowest mark was 38.
- (b) About a quarter of students scored above 70.
- (c) The students at 98 and 99 made a data-entry error.
- (d) The mean is 61.

Show the solution

Only **(b)**, and even that is “about” — quartiles are approximate for finite samples, and ties can make the counts uneven.

- (a) is false. The lower whisker reaches the most extreme observation *within the fence*, and the fence sits at $52 - 1.5 \times 18 = 25$, so 38 is merely the lowest mark above 25. The lowest mark is the point drawn at **21**. Had there been no point below the whisker, 38 would have been the minimum — but you would know that from the absence of a drawn point, not from the whisker.
- (b) does not follow at all. A point beyond the fence is a value far from the middle. That is a description, not a diagnosis, and treating it as one is how genuine top performers get deleted from datasets. (Check the arithmetic while you are here: the upper fence is $70 + 1.5 \times 18 = 97$, which is why 98 and 99 are drawn separately and 86 is not.)
- (c) does not follow. The median is 61; the mean could be anywhere, and the two high points suggest it sits somewhat above.

2.6 Check it in Python

Three claims from the mechanics, checked.

First, that the deviations from the mean really do cancel — this is what makes $\bar{x}$ the balance point, and why the variance has to square them.

```
deviations = scores - scores.mean()
print(f"sum of deviations from the mean : {deviations.sum():.3e}")
print(f"sum of deviations from the median: {(scores -
    - np.median(scores)).sum():.2f}")
```

```
sum of deviations from the mean : 9.095e-13
sum of deviations from the median: 112.30
```

Not exactly zero, but 10^{-13} on numbers of size 65 is floating-point noise, not a real residue — **P2** explains why. Never test a computed float with `==`. The deviations from the *median* do not cancel, and there is no reason they should.

Second, the variance formula, done by hand against the library:

```
by_hand = (deviations ** 2).sum() / (scores.size - 1)
library = scored["final_score"].var()
print(f"by hand {by_hand:.6f}    pandas {library:.6f}    "
      f"agree: {np.isclose(by_hand, library)}")
```

```
by hand 70.293592    pandas 70.293592    agree: True
```

Third — and this one is a genuine trap rather than a demonstration. NumPy and pandas disagree about the default divisor. Assert that they agree and watch it fail:

```
assert np.isclose(np.std(scores), scored["final_score"].std()), (
    f"numpy says {np.std(scores):.4f}, pandas says
    ↳ {scored['final_score'].std():.4f}"
)
```

```
AssertionError: numpy says 8.3665, pandas says 8.3841
```

```
-----
AssertionError                                Traceback (most recent call last)
Cell In[9], line 1
----> 1 assert np.isclose(np.std(scores), scored["final_score"].std()), (
      2     f"numpy says {np.std(scores):.4f}, pandas says
      ↳ {scored['final_score'].std():.4f}"
      3 )
AssertionError: numpy says 8.3665, pandas says 8.3841
```

Neither library is wrong. `numpy.std` divides by n by default and `pandas.std` divides by $n - 1$, and both document it. With $n = 238$ the gap is 0.018 marks and invisible; with $n = 5$ it is nearly 12%, and a figure or a test built on it would be quietly wrong. Say which you meant:

```
assert np.isclose(np.std(scores, ddof=1), scored["final_score"].std())
assert np.isclose(np.std(scores, ddof=0), scored["final_score"].std(ddof=0))
print(f"n=238: ddof=0 {np.std(scores, ddof=0):.4f}    ddof=1 {np.std(scores,
↳ ddof=1):.4f}")
small = scores[:5]
print(f"n=5: ddof=0 {np.std(small, ddof=0):.4f}    ddof=1 {np.std(small,
↳ ddof=1):.4f}")
```

```
n=238: ddof=0 8.3665    ddof=1 8.3841
n=5: ddof=0 9.9746    ddof=1 11.1519
```

Finally, robustness, which is easiest to believe once you have watched it:

```
damaged = scores.copy()
damaged[damaged.argmax()] = 1000
print(f"original mean {scores.mean():7.2f}    median {np.median(scores):.2f}")
print(f"damaged mean {damaged.mean():7.2f}    median {np.median(damaged):.2f}")
```

```
original mean 65.27    median 64.80
damaged mean 69.09    median 64.80
```

One value changed; the mean moves by nearly four marks and the median does not move at all.

2.7 Common wrong turns

Where this usually goes wrong

Reporting a mean with no measure of spread A reported mean of 65 is compatible with everyone scoring 65, and with half the cohort scoring 40 and half scoring 90. A centre without a spread is not a description. Report both, or report the five-number summary.

Reporting a mean for a strongly skewed variable Salaries, waiting times, house prices and counts of rare events are all right-skewed, and for all of them the mean answers a question nobody asked. Give the median, and give the mean too if you also give the shape.

Treating the range as a measure of spread It depends on exactly two observations and grows with sample size, so it is not comparable between samples. Use the IQR or the standard deviation.

Reading boxplot whiskers as the minimum and maximum They reach the most extreme point within 1.5 IQR of the box. If points are drawn beyond a whisker, the extreme is out there, not at the whisker's end.

Deleting points that fall beyond a fence The fence is a drawing convention, not a test for error. Investigate such a point — a score of 187 out of 100 is genuinely impossible and goes — but a score of 91.6 is a student who did well, and removing it is falsifying data.

Forgetting which divisor you used `numpy.std` uses n ; `pandas.std` and R use $n - 1$. On large samples nobody notices. On small ones the difference is large, and it propagates into every interval and test you compute afterwards.

Describing a mixture as though it were one population A histogram with two peaks is usually two groups. Splitting them and describing each is nearly always more informative than any single summary of the whole.

2.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

- Two variables both have mean 50 and standard deviation 10. What can you conclude about how similar their distributions are?
 - They are the same distribution
 - They have the same shape but may be shifted
 - Almost nothing — they can look completely different
 - One must be normal
- Every student's mark is increased by 5. What happens to the mean, the median and the standard deviation?
- In the cohort, the standard deviation of `final_score` across all 237 students with a recorded background is 8.40, while no single background's exceeds 7.63. How can the whole be more variable than every part?
- You compute a sample standard deviation of 12.0 using $n - 1$, then recompute using n and get 11.4. Which is "correct"?
- A variable has mean 40 and median 40. Is it symmetric?

 Show the answers

1. **(c).** This is the first figure in the unit. Same mean and same standard deviation is compatible with one peak, two peaks, or a tight cluster plus a long tail. (b) is the tempting answer and is wrong: matching two numbers constrains location and scale, not shape.
2. The mean and median both increase by 5. The standard deviation is **unchanged** — every deviation $x_i - \bar{x}$ is unchanged, because the mean moved by exactly as much as the data did. Spread is about differences, and adding a constant changes no difference.
3. Because the groups sit at different places. Spread within a group is small (each group's students resemble each other); the extra spread in the combined column is the distance *between* the group means, 60.2 to 75.3. This is the idea behind comparing groups in [S10](#), and it is why a mixture almost always looks more variable than its parts.
4. **Both, for different questions.** The $n - 1$ version estimates the spread of the population these observations came from; the n version describes the spread of exactly these observations, treated as everything there is. If you are doing inference you want $n - 1$. The real error is not knowing which one your code produced.
5. **Not necessarily.** Equal mean and median is what symmetry produces, but it does not imply it. Take $\{0, 3, 4, 4, 4, 6, 7\}$: the mean is 4 and the median is 4, and the set is plainly not symmetric about 4 — its skewness coefficient is -0.48 . Equal mean and median is evidence, not proof. Plot it.

Where this shows up next

[S3](#) does all of this for two variables at once and asks how they move together. [S6](#) puts named shapes to the histograms in this unit, and says which mechanisms produce which shape. [S7](#) takes the mean you have just computed and asks how much it would have differed had you measured 238 other students. [S8](#) turns that into the interval that goes next to every mean you report.

3. Describing two variables

Statistics · Unit 3

S3 · Statistics Reading: 5 pages With practice: 1.5 hours Assumes: S2

Why this unit is here

Almost no question worth asking is about one variable. *Does practice help? Do the students who arrive strongest finish strongest? Does this treatment work?* Every one of them is about how two things move together.

This unit gives you the standard summary of that — the correlation coefficient — and then spends most of its length on what the number cannot see. A correlation is one number standing in for a whole cloud of points, and the ways it misleads are specific, common, and easy to spot once you know to look.

3.1 Before you start

You should be able to:

- compute a mean and a standard deviation, and say what each is for — S2;
- read a histogram and a boxplot — S2;
- tell a quantitative variable from a categorical one — S1.

Quick check. The cohort's final scores have mean 65.3 and standard deviation 8.4. A student scores 82. How many standard deviations above the mean is that?

💡 Show the answer

$(82 - 65.3)/8.4 = 1.99$, so almost exactly two standard deviations above the mean.

That quantity has a name — the **standardised value**, or **z-score** — and this unit is built out of it. Subtracting the mean removes the location; dividing by the standard deviation removes the units and the scale. What is left is comparable across variables measured in completely different things.

3.2 The idea

Put the two variables on a scatterplot, then draw a vertical line at the mean of one and a horizontal line at the mean of the other. That divides the cloud into four quadrants, and the quadrants are the whole idea.

For each student, take how far their hours are from the mean hours, take how far their score is from the mean score, and multiply. A student who is above average on both gets a positive product; so does a student below average on both. A student above average on one and below on the other gets a negative one.

Add up all 237 products. If the cloud leans upwards the positives dominate and the total is positive; if it leans downwards the total is negative; if there is no lean the products cancel. That total, divided by

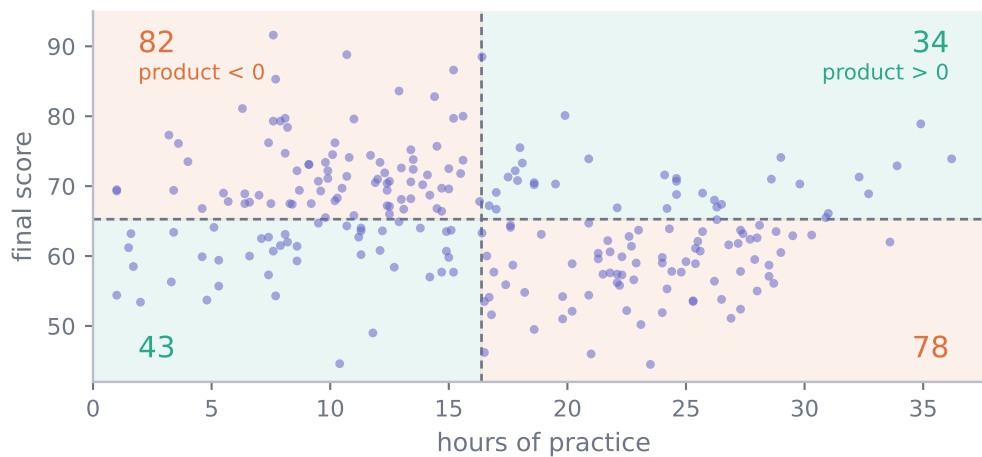


Figure 1: The 237 students, split at the mean hours of practice (16.4) and the mean final score (65.3). A point in the top-right or bottom-left agrees with both means in the same direction, and contributes a positive product; a point in the other two contributes a negative one. Here 160 points pull negative and 77 pull positive, so the total comes out below zero.

$n - 1$, is the **covariance**, and everything else in this unit is a repair to it.

3.3 The mechanics

Covariance, and why you cannot read it

$$\text{cov}(x, y) = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

The sign is informative. The size is not, because the covariance carries the units of both variables multiplied together — here marks $\times$ hours, a quantity nobody has intuition about. Worse, it depends on the units you happened to choose:

```
import numpy as np
import pandas as pd

cohort = pd.read_csv("data/cohort.csv")
usable = cohort.dropna(subset=["background", "final_score"])
usable = usable[usable["final_score"].between(0, 100)]
hours = usable["hours_practice"].to_numpy()
score = usable["final_score"].to_numpy()

print(f"cov in marks x hours    {np.cov(hours, score, ddof=1)[0, 1]:10.4f}")
print(f"cov in marks x minutes {np.cov(hours * 60, score, ddof=1)[0, 1]:10.4f}")
```

```
cov in marks x hours    -16.0477
cov in marks x minutes -962.8637
```

Nothing about the students changed. Recording practice in minutes rather than hours multiplied the covariance by 60. A summary that moves when you change your mind about units cannot be compared across studies.

Correlation is covariance made unitless

Divide by the standard deviation of each variable and the units cancel:

$$r = \frac{\text{cov}(x, y)}{s_x s_y} = \frac{1}{n-1} \sum_{i=1}^n z_{x,i} z_{y,i},$$

where $z_{x,i} = (x_i - \bar{x})/s_x$ is the standardised value from the quick check. The second form is worth carrying: **the correlation is the sum of the products of z-scores, divided by $n - 1$** . Standardise both variables, multiply them point by point, add up, divide by $n - 1$. (It is nearly the average of those products — the divisor is $n - 1$ rather than n , for the same reason the sample variance uses $n - 1$, and on the cohort the two differ by 0.001.)

This is **Pearson's correlation coefficient**, and it has four properties you should be able to state.

- It lies between -1 and $+1$, always.
- It is $+1$ exactly when the points lie on a straight line of positive slope, and -1 exactly when they lie on a straight line of negative slope.
- It is symmetric: $r(x, y) = r(y, x)$. It does not know which variable you think is the cause.
- It is unchanged by shifting or rescaling either variable — hours to minutes, Celsius to Fahrenheit, marks to percentages — except that multiplying by a negative number flips its sign.

What the number cannot see

r measures **linear** association and nothing else. That restriction is not a technicality; it is where most misreadings come from.

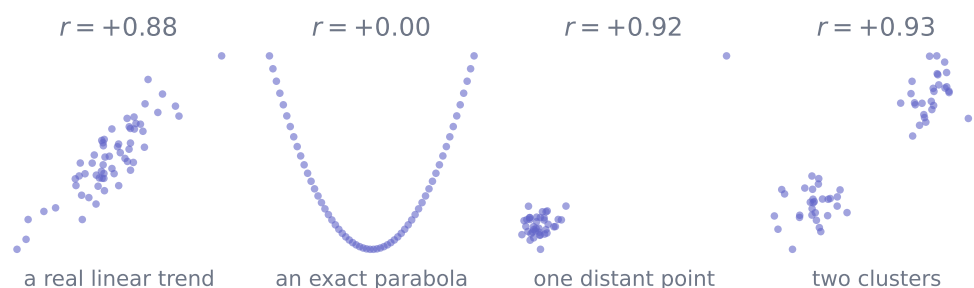


Figure 2: Four datasets. Three have correlations near $+0.9$ and nothing else in common; the fourth has a correlation of exactly zero and a perfect relationship — every y is determined by its x . One number cannot certify a shape.

Read those four again in order. The first is what people picture when they hear " $r = 0.9$ ". The second has a *perfect* relationship and a correlation of zero, because the parabola rises as fast on the right as it falls on the left and the products cancel exactly — $r = 0$ **does not mean the variables are unrelated**. The third is 39 points with no association and one point a long way off; delete that point and r collapses. The fourth is two clusters with no association inside either, and the correlation is measuring the gap between them, not anything about individual students.

The defence is the same in all four cases and takes five seconds: **plot it**.

When the relationship is monotone but not straight

Spearman's rank correlation is Pearson's r computed on the *ranks* rather than the values: replace the smallest x by 1, the next by 2, and so on, do the same for y , and correlate those. It measures whether

the relationship is consistently increasing or decreasing, without insisting it be a straight line, and one distant point can only ever be *the largest rank* rather than a value seventeen standard deviations out.

Use it when the relationship looks monotone but curved, when one or both variables are ordinal, or when extreme values are doing more work than you are comfortable with. Report which one you used — “correlation” unqualified means Pearson.

The other two combinations

Two variables need not both be quantitative.

Categorical with quantitative — the case that dominates real work — is a comparison of groups: report a summary of the quantitative variable within each category, and draw side-by-side boxplots on a shared scale. [S10](#) makes this into inference.

Categorical with categorical is a **contingency table** of counts, read as *conditional proportions*. The direction matters and is a decision about the question, not about the data: the proportion of engineers who completed, and the proportion of completers who were engineers, are different numbers answering different questions, and confusing them is the error [S4](#) is largely about.

3.4 Worked example

Practice against score, one number at a time.

```
from scipy import stats

r = np.corrcoef(hours, score)[0, 1]
rho = stats.spearmanr(hours, score).statistic
print(f"n = {hours.size}")
print(f"Pearson   r = {r:+.4f}")
print(f"Spearman   = {rho:+.4f}")
```

```
n = 237
Pearson   r = -0.2350
Spearman   = -0.2585
```

So a weak negative association: students who practise more tend, across the cohort as a whole, to score slightly lower. Spearman agrees in sign and is slightly stronger, which tells you the relationship is not being manufactured by a handful of extreme points.

i [M11](#) reports -0.2337 for the same pair of columns. It is not a different answer: that unit filters on `final_score` alone and keeps 238 rows, while this one also drops the student with no recorded background and keeps 237. One student out of 238 shifts the correlation by 0.0013 — enough to change the second decimal place, from -0.23 to -0.24 . If two numbers in these notes disagree, check the filter first.

Now split by background, which [S1](#) established is the variable doing the damage.

```
print(f"{'background':<24}{'n':>5}{'Pearson':>10}{'Spearman':>10}")
print(f"{'everyone':<24}{hours.size:>5}{r:>+10.2f}{rho:>+10.2f}")
for name, group in usable.groupby("background"):
    g_h, g_y = group["hours_practice"], group["final_score"]
    print(f"{name:<24}{len(group):>5}"
          f"{np.corrcoef(g_h, g_y)[0, 1]:>+10.2f}"
          f"{stats.spearmanr(g_h, g_y).statistic:>+10.2f}")
```

background	n	Pearson	Spearman
everyone	237	-0.24	-0.26
Computing	69	+0.43	+0.38
Engineering	36	+0.64	+0.58
Life sciences	57	+0.68	+0.62
Mathematics or Physics	36	+0.54	+0.51
Social sciences	39	+0.66	+0.66

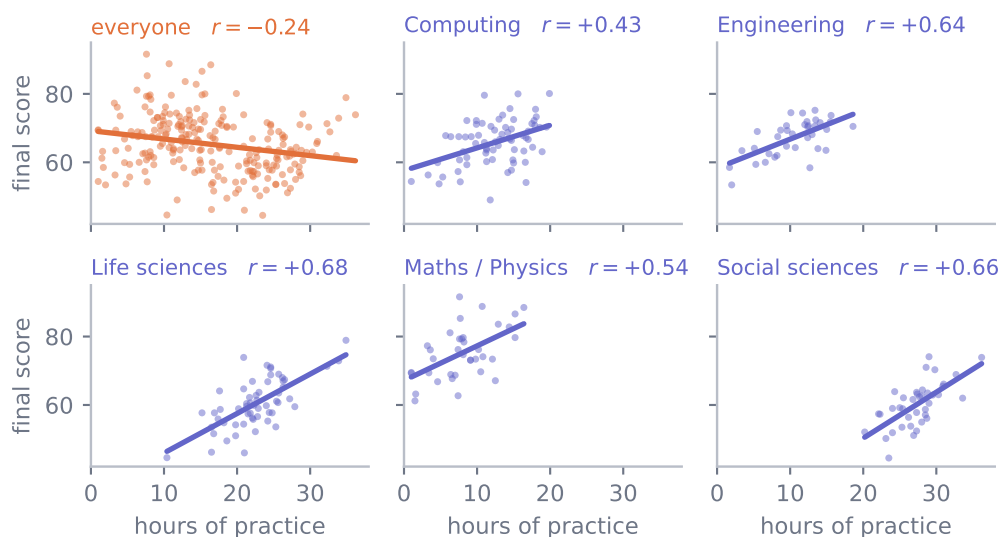


Figure 3: All 237 students in the first panel, then the same students split into their five backgrounds, on shared axes, with a straight line fitted to each panel. The first panel slopes down; the other five slope up. No observation was added, removed or altered — the five panels are a partition of the first.

One correlation of -0.24 ; five correlations between $+0.43$ and $+0.68$. The overall number is not wrong, and it is not a useful description of any student in the dataset, because no student is a member of “everyone” without also being a member of exactly one background.

That is the fourth panel of the four-dataset figure, in real data: a correlation computed across clusters is largely a statement about where the clusters sit. [S12](#) gives the pattern its name and its general treatment.

3.5 Try it yourself

Exercise 1 Two variables have $\text{cov}(x, y) = 24$, $s_x = 4$ and $s_y = 15$.

(a) What is r ?

(b) You now rescale x from metres to centimetres. What happens to the covariance, to s_x , and to r ?

 Show the solution

- (a) $r = 24/(4 \times 15) = 0.4$.
- (b) Multiplying x by 100 multiplies the covariance by 100 (to 2400) and s_x by 100 (to 400). The correlation is $2400/(400 \times 15) = 0.4$ — **unchanged**, because the same factor of 100 appears above and below the line. That invariance is the entire reason correlation is reported instead of covariance.

Exercise 2 A colleague finds $r = 0.02$ between two variables and concludes that they are unrelated. Give two distinct reasons the conclusion does not follow, and say what you would ask to see.

 Show the solution

- **The relationship may not be linear.** Any relation symmetric about the mean of x — a U-shape observed over a range centred on its vertex, for instance — produces $r \approx 0$ while y may be completely determined by x . The parabola panel in this unit has r exactly zero for that reason; observe the same parabola over an off-centre range and r climbs above 0.88.
- **The sample may be too small to see anything.** With $n = 8$, $r = 0.02$ is entirely consistent with a real and substantial association; r estimated from a small sample bounces around a great deal. S8 is how you find out.
- Or: **the population may be a mixture**, with opposite associations inside different groups that cancel when pooled — which is what happens in this unit's worked example, only in reverse.

What to ask for: **the scatterplot**, and the sample size. In that order.

Exercise 3 Ten students are ranked on two measures. Their x -ranks are 1 to 10 in order, and their y values are 1, 4, 9, 16, 25, 36, 49, 64, 81, 100 — that is, $y = x^2$ exactly.

Without computing anything, say what Spearman's rank correlation must be, and whether Pearson's r will be larger, smaller or equal.

 Show the solution

Spearman is **exactly** +1. Ranking $y = x^2$ for positive x gives back the x ranks in the same order, and Pearson's r on two identical rank sequences is 1. Spearman detects *any* strictly increasing relationship, however curved.

Pearson's r will be **smaller** than 1 — about 0.97 — because the points do not lie on a straight line, and $r = 1$ requires exactly that. It is high because $y = x^2$ is not far from straight over this range; over a wider range it would fall further.

The general lesson: Spearman equal to 1 with Pearson below it is the signature of a curved but consistently increasing relationship.

3.6 Check it in Python

First, the claim that correlation is the average product of z-scores. Standardise both columns by hand and compare.

```

z_hours = (hours - hours.mean()) / hours.std(ddof=1)
z_score = (score - score.mean()) / score.std(ddof=1)
by_hand = (z_hours * z_score).sum() / (hours.size - 1)
print(f"average product of z-scores {by_hand:+.6f}")
print(f"numpy corrcoef          {np.corrcoef(hours, score)[0, 1]:+.6f}")

```

```

average product of z-scores -0.235010
numpy corrcoef             -0.235010

```

Second, that changing units leaves r alone but not the covariance:

```

minutes = hours * 60
print(f"covariance hours {np.cov(hours, score, ddof=1)[0, 1]:>10.4f}"
      f"    minutes {np.cov(minutes, score, ddof=1)[0, 1]:>10.4f}")
print(f"correlation hours {np.corrcoef(hours, score)[0, 1]:>10.4f}"
      f"    minutes {np.corrcoef(minutes, score)[0, 1]:>10.4f}")

```

```

covariance hours    -16.0477    minutes    -962.8637
correlation hours    -0.2350    minutes     -0.2350

```

Third — the one worth running yourself. It is tempting to write a check that says “if two variables are related, the correlation will notice”. Watch it fail on a relationship that is not merely strong but *exact*:

```

x = np.linspace(-2, 2, 61)
y = x ** 2                                # y is completely determined by x
assert abs(np.corrcoef(x, y)[0, 1]) > 0.5, (
    f"y is an exact function of x, yet r = {np.corrcoef(x, y)[0, 1]:.6f}"
)

```

```

AssertionError: y is an exact function of x, yet r = -0.000000

```

```

-----
AssertionError                                Traceback (most recent call last)
Cell In[9], line 3
      1 x = np.linspace(-2, 2, 61)
      2 y = x ** 2                                # y is completely determined by x
----> 3 assert abs(np.corrcoef(x, y)[0, 1]) > 0.5, (
      4     f"y is an exact function of x, yet r = {np.corrcoef(x, y)[0, 1]:.6f}"
      5 )
AssertionError: y is an exact function of x, yet r = -0.000000

```

Every y here is fixed by its x with no noise at all, and the correlation is zero to machine precision. It is zero because the grid is symmetric about $x = 0$: for every point that contributes $+c$ to the sum of products there is a mirror point contributing $-c$. Shift the window off centre and the cancellation stops:

```

for lo, hi in [(-2, 2), (-1, 3), (0, 4)]:
    x = np.linspace(lo, hi, 61)
    print(f"x from {lo:>2} to {hi}:   r = {np.corrcoef(x, x ** 2)[0, 1]:+.4f}")

```

```
x from -2 to 2:  r = -0.0000
x from -1 to 3:  r = +0.8855
x from  0 to 4:  r = +0.9673
```

Same relationship, three different correlations, one of them zero. What r reports depends on the range you happened to observe, which is a good thing to know before quoting one.

3.7 Common wrong turns

Where this usually goes wrong

Reading $r = 0$ as “no relationship” It means no *linear* relationship. An exact parabola, observed symmetrically about its vertex, scores zero. Say “no linear association”, plot the data, and consider Spearman.

Quoting a covariance Its size depends on your choice of units, so it is not comparable with anyone else’s. Divide by the two standard deviations and quote r .

Treating r as a slope r is unitless and cannot tell you how many marks an hour of practice is worth. Two datasets with identical correlations can have wildly different slopes. The slope is S11’s business, and it carries units.

Computing r on a mixture If the data contain distinguishable groups, a pooled correlation partly measures the distance between the groups. Compute it within each group as well, always, and look at whether the answers agree.

Letting one point decide A single distant observation can create a correlation from nothing or destroy a real one. Plot the data; recompute without the point; if the answer changes, say so in the write-up rather than choosing the version you prefer.

Reading a correlation as a cause The coefficient is symmetric — it cannot even tell which variable came first. Everything S1 said about design applies here unchanged.

Reporting r without n A correlation from 8 observations and one from 8,000 are not the same kind of evidence, and the number alone does not distinguish them.

3.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

- Which of these *can* change the correlation between height and weight in a sample?
 - Converting height from centimetres to inches
 - Converting weight from kilograms to pounds
 - Adding 5 kg to everyone’s weight
 - Dropping the three tallest people from the sample
- In the cohort, $r = -0.24$ overall and between $+0.43$ and $+0.68$ within each background. Which of those numbers is the “real” correlation between practice and score?
- Spearman’s rank correlation for a pair of variables is $+1$ but Pearson’s is $+0.8$. What does that tell you about the relationship?
- A study reports $r = 0.62$ between two variables measured on 6 people and describes it as a strong relationship. Is that reading defensible?
- True or false: if $r = +0.9$, then a one-standard-deviation increase in x is associated with a

0.9-standard-deviation increase in y on average.

 Show the answers

1. **(d).** Correlation is invariant to shifting and positive rescaling, so (a), (b) and (c) leave it exactly unchanged — the units cancel in the definition. Dropping the three tallest people changes *which* points are in the sample, which is a different operation entirely; it usually reduces r by removing the widest part of the range. Note the “can”: it is not guaranteed to change anything. If every point lay exactly on a straight line, r would be 1 before and after. It is the only option that can move the number at all.
2. **Both, and neither is “the” correlation** — they are answers to two different questions, exactly as in S1. -0.24 describes the cohort pooled, and mostly reflects that quantitative-background students start ahead and practise less. The within-group figures describe students compared with others like them. If you had to report one number for “does practice go with better marks”, the within-group ones are closer to what was meant — but the honest answer reports both.
3. That the relationship is **monotone but curved**. Spearman being exactly 1 means the ranks agree perfectly: y increases every time x does. Pearson below 1 means the points do not lie on a straight line. Fit a straight line to this and the residuals will show a systematic bend.
4. **No.** With six observations, $r = 0.62$ is unremarkable: correlations of that size arise easily between unrelated variables in samples that small, and the estimate itself bounces around enormously — S8 shows how to put a width on it. Ask for the scatterplot, and ask how many pairs of variables were examined before this one was reported.
5. **True**, and it is the most useful reading of r there is. Written in standardised units, the fitted straight line has slope exactly r : on average a point one standard deviation above the mean in x sits r standard deviations above the mean in y . Note “on average” and “associated with” — this is a description of the cloud, not a prediction for any individual and not a claim about what would happen if you intervened.

Where this shows up next

S10 turns the group comparisons here into inference, with intervals on the differences. **S11** replaces the unitless r with a slope in marks per hour, and attaches an interval to it. **S12** names the reversal in this unit’s worked example and shows how to look for it before it finds you. **M12** takes the correlation matrix of several variables at once and asks which directions in the data carry the most variation.

4. Probability, conditioning and Bayes

Statistics · Unit 4

S4 · Statistics Reading: 6 pages With practice: ~ 2.5 hours Assumes: S1

Why this unit is here

From here on, every claim these notes make is a probability statement. A confidence level, a p -value, “the procedure captures the truth 95% of the time” — all of them are probabilities, and none of them mean what they appear to mean until you can handle conditioning.

There is one error in particular that this unit exists to prevent. It is swapping a conditional probability for its reverse: reading *the chance of a positive test given the disease* as *the chance of the disease given a positive test*, or *the chance of this data if the hypothesis were false* as *the chance the hypothesis is false*. It has sent people to prison. It is the same mistake every time, and once you can see it you cannot unsee it.

4.1 Before you start

You should be able to:

- turn a count into a proportion and back — S1;
- read a two-way table of counts — S1;
- work with fractions and percentages.

Quick check. Of the 237 students in the cohort, 69 scored 70 or more. What proportion is that, as a decimal and as a percentage?

💡 Show the answer

$69/237 = 0.291$, or 29.1%.

Keep hold of both the count and the total. Almost every confusion in this unit comes from losing track of what the denominator was.

4.2 The idea

A **probability** is a number between 0 and 1 attached to an event, measuring how often it happens. Take it, for now, as a long-run frequency: if you could repeat the situation many times, the probability is the fraction of times you would see the event. That is the reading frequentist statistics is built on, and it is why S8 can say things about *procedures* but not about individual results.

The whole subject then rests on one move: **conditioning**. Conditioning means restricting attention to a subset of the possibilities and asking the question again inside it. It changes the denominator, and changing the denominator changes the answer — often dramatically, and in a way people consistently fail to expect.

Here it is on the cohort. Take two facts about a student: they came from a Mathematics or Physics background, and they scored 70 or more. Twenty-four students have both.

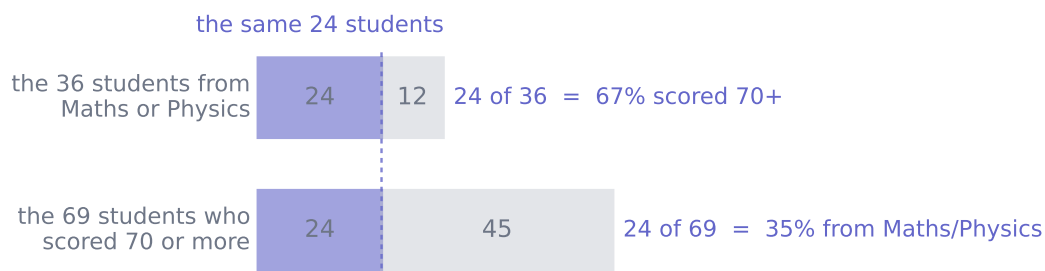


Figure 1: The same twenty-four students, counted against two different denominators. As a fraction of the 36 who came from Maths or Physics they are two thirds; as a fraction of the 69 who scored 70 or more they are about a third. Neither number is wrong, and they are not answers to the same question.

Two thirds and one third, from the same twenty-four people. The only thing that changed was which group we agreed to look at.

Write them properly and the difference is visible in the notation:

$$P(\text{scored } 70+ \mid \text{Maths/Physics}) = \frac{24}{36} = 0.67, \quad P(\text{Maths/Physics} \mid \text{scored } 70+) = \frac{24}{69} = 0.35.$$

The vertical bar is read “given”. **The thing after the bar is the denominator.** Swap the two sides and you have changed the question, not merely the phrasing.

4.3 The mechanics

The rules, all of them

An **event** is a set of outcomes; $P(A)$ is its probability. Everything follows from three facts: $0 \leq P(A) \leq 1$, the probability of “something happens” is 1, and the probabilities of outcomes that cannot both occur add up.

From those:

$$P(\text{not } A) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A \mid B) P(B)$$

The subtraction in the second is the whole content of it: add two events and you have counted their overlap twice, so take it off once. Read $\cup$ as “or” (inclusive) and $\cap$ as “and”.

Conditional probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{defined when } P(B) > 0.$$

In counts, this is exactly what the figure did: of the things in B , what fraction are also in A . **To condition is to change the denominator.**

Independent is not the same as mutually exclusive

A and B are **independent** when

$$P(A \cap B) = P(A) P(B),$$

equivalently (when $P(B) > 0$) when $P(A | B) = P(A)$: learning that B happened tells you nothing new about A .

They are **mutually exclusive** when they cannot both happen, $P(A \cap B) = 0$.

These are close to opposites, and confusing them is one of the two commonest errors in the whole unit.

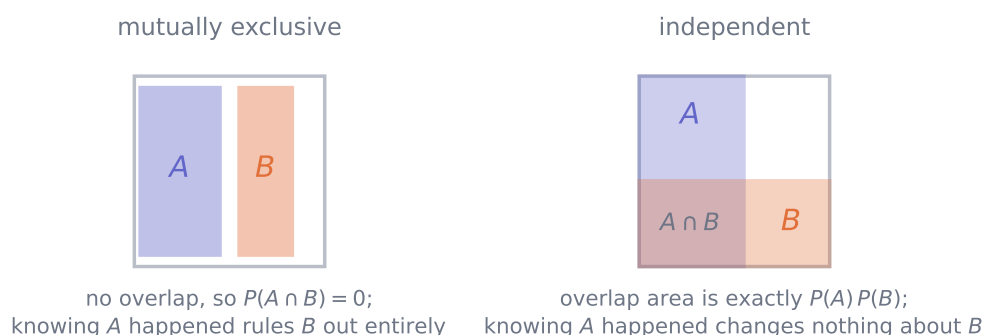


Figure 2: Two events drawn inside the space of all outcomes, with probability read as area. On the left they cannot both happen. On the right each is a strip across the whole space, so the overlap has area exactly $P(A)P(B)$ — which is what independence means.

If A and B are mutually exclusive and both have positive probability, they are as far from independent as two events can be: $P(A | B) = 0$, while $P(A) > 0$. Learning B has told you everything about A .

Splitting a probability up

If $B_1, \dots, B_k$ cover every possibility without overlapping — a **partition** — then

$$P(A) = \sum_{i=1}^k P(A | B_i) P(B_i).$$

This is the **law of total probability**, and in words it is unremarkable: an overall rate is the weighted average of the group rates, weighted by group size. It is the machinery behind S1's worked example, where a pooled answer disagreed with every group.

Bayes' theorem

Put the multiplication rule together both ways round — $P(A \cap B)$ equals both $P(A | B)P(B)$ and $P(B | A)P(A)$ — and rearrange:

$$P(B | A) = \frac{P(A | B) P(B)}{P(A)}.$$

That is **Bayes' theorem**. It is the machine for reversing a conditional, and the thing to notice is that you cannot reverse one without knowing $P(B)$ — the **base rate**. A conditional probability on its own does not determine its reverse.

That is not an abstraction, and the standard illustration is worth doing in counts rather than fractions, because in counts nobody gets it wrong. Take a disease that 1% of people have, and a test that is positive for 90% of people who have it and for 5% of people who do not. You test positive. What is the chance you have it?

Imagine 10,000 people.

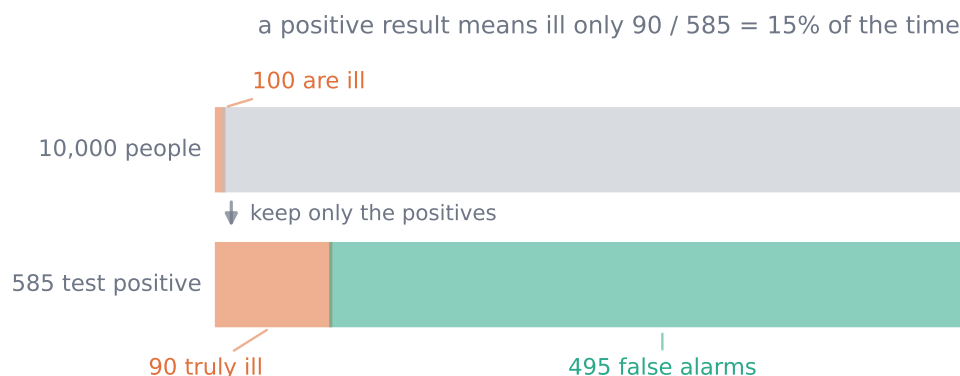


Figure 3: Ten thousand people, then only the 585 who test positive. The test is right nine times out of ten on people who are ill — but almost all the positives come from the enormous group who are not, so a positive result leaves you far more likely to be well than ill.

One hundred are ill and 90 of them test positive. Nine thousand nine hundred are well and 5% of them — 495 people — test positive anyway. So 585 people test positive and 90 of them are ill: $90/585 = 15\%$.

The test is good. The answer is 15% because the healthy group is a hundred times larger, so its small error rate still produces more positives than the ill group does in total.

Notice which dial matters. Push the sensitivity from 90% all the way to a perfect 100% and you gain ten true positives: $100/595 = 17\%$, barely a change. Cut the false-positive rate from 5% to 0.5% instead and the false alarms fall from 495 to 50, giving $90/140 = 64\%$. **When a condition is rare, the positive predictive value is governed by the false-positive rate and the base rate, and hardly at all by how well the test detects true cases.**

4.4 Worked example

The cohort's own screening test.

Every student sat a diagnostic on arrival. Treat “scored 70 or more on the diagnostic” as a test for “scored 70 or more at the end”, and ask how good a test it is.

```
import numpy as np
import pandas as pd

cohort = pd.read_csv("data/cohort.csv")
usable = cohort.dropna(subset=["background", "final_score"])
usable = usable[usable["final_score"].between(0, 100)]
usable = usable.assign(strong_start=usable["diagnostic_score"] >= 70,
```

```
strong_finish=usable["final_score"] >= 70)

print(pd.crosstab(usable["strong_start"], usable["strong_finish"], margins=True))
```

```
strong_finish  False  True  All
strong_start
False          121    22  143
True           47    47   94
All            168    69  237
```

Four counts and two margins, and every probability in this unit is a ratio of two of them.

```
n = len(usable)
both = ((usable["strong_start"]) & (usable["strong_finish"])).sum()
start = usable["strong_start"].sum()
finish = usable["strong_finish"].sum()

print(f"P(strong finish)                {finish / n:.4f}")
print(f"P(strong start)                 {start / n:.4f}")
print(f"P(strong start AND strong finish) {both / n:.4f}")
print(f"P(strong finish | strong start)   {both / start:.4f}    <-
  ~ {both}/{start}")
print(f"P(strong start | strong finish)   {both / finish:.4f}    <-
  ~ {both}/{finish}")
```

```
P(strong finish)                0.2911
P(strong start)                 0.3966
P(strong start AND strong finish) 0.1983
P(strong finish | strong start)   0.5000    <- 47/94
P(strong start | strong finish)   0.6812    <- 47/69
```

Read those last two lines slowly. The diagnostic flags 94 students, and exactly 47 of them go on to score 70 or more: **a positive flag is right half the time**. Yet the test is not useless, because 29% of students score 70+ overall, and among the flagged ones it is 50% — the flag has raised the odds considerably. It has simply not raised them to anything like certainty.

Now the same numbers as Bayes' theorem, to see the machine work:

$$P(\text{finish} | \text{start}) = \frac{P(\text{start} | \text{finish}) P(\text{finish})}{P(\text{start})} = \frac{0.6812 \times 0.2911}{0.3966} = 0.5000.$$

```
sensitivity = both / finish          # P(start | finish)
prior = finish / n                  # P(finish)
evidence = start / n                # P(start)
print(f"Bayes gives {sensitivity * prior / evidence:.4f}")
print(f"direct ratio {both / start:.4f}")
```

```
Bayes gives 0.5000
direct ratio 0.5000
```

Finally, are the two events independent? If they were, the probability of both would be the product of the two separate probabilities:

```
p_both = both / n
p_product = (start / n) * (finish / n)
print(f"P(start AND finish) {p_both:.4f}")
print(f"P(start) x P(finish) {p_product:.4f}")
print(f"ratio {p_both / p_product:.2f} – a strong start makes a strong finish "
      f"{p_both / p_product:.2f} times likelier than chance alone")
```

P(start AND finish) 0.1983

P(start) x P(finish) 0.1155

ratio 1.72 – a strong start makes a strong finish 1.72 times likelier than chance alone

Not independent, and nobody expected them to be. What the numbers add is a size: students who start strong finish strong 1.7 times as often as independence would predict, and still only half the time.

4.5 Try it yourself

Exercise 1 $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$.

- (a) Are A and B independent?
- (b) What is $P(A \cup B)$?
- (c) What is $P(A | B)$, and what is $P(B | A)$?

Show the solution

- (a) **Yes.** $P(A)P(B) = 0.5 \times 0.4 = 0.2$, which is exactly $P(A \cap B)$.
- (b) $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$. Forgetting to subtract the overlap gives 0.9, which is the commonest slip here.
- (c) $P(A | B) = 0.2/0.4 = 0.5$, and $P(B | A) = 0.2/0.5 = 0.4$. Both equal the unconditional probabilities, which is another way of saying the events are independent — and note that the two conditionals are *different numbers* even so. Independence does not make $P(A | B)$ and $P(B | A)$ equal; it makes each equal to its own unconditional probability.

Exercise 2 One in 200 people in a population has a certain condition. A screening test is positive for 99% of people who have it and for 2% of people who do not.

Work out, in counts, the probability that someone who tests positive has the condition. Then say what would have to change to push that probability above 50%.

Show the solution

Take 20,000 people. 100 have the condition and 99 of them test positive. 19,900 do not, and 2% of them — 398 people — test positive anyway. So 497 test positive and 99 have the condition:

$$P(\text{condition} | \text{positive}) = \frac{99}{497} = 0.199.$$

About one in five, from a test that is right 99% of the time on people who have the condition.

To push it past 50% you need the false positives to fall below 99. The sensitivity cannot help — it is already at 99% and even perfection would only add one true positive. What matters is the **false-positive rate**: dropping it from 2% to 0.4% leaves about 80 false positives and a probability of $99/179 = 0.55$. The other route is a higher **base rate**: screening a group already known to be at risk, rather than everyone. This is precisely why population-wide screening for rare conditions is a harder problem than it sounds.

Exercise 3 A barrister tells a jury: “The chance of this DNA match occurring if the defendant were innocent is one in a million. So there is only a one in a million chance that the defendant is innocent.”

Name the error, write both probabilities in symbols, and explain what else the jury would need to know.

 Show the solution

The error is swapping a conditional for its reverse — often called the **prosecutor’s fallacy**. The barrister has been given

$$P(\text{match} \mid \text{innocent}) = 10^{-6}$$

and has reported it as

$$P(\text{innocent} \mid \text{match}) = 10^{-6}.$$

Bayes’ theorem says those are equal only by coincidence, and to get from one to the other you need the **base rate**: how many people could plausibly have left that sample.

Suppose the pool of possible sources is 10 million people, one of whom is guilty. Among the 10 million innocent people, about 10 would match by chance. So there are roughly 11 matching people, one guilty and ten not, and the probability of innocence given a match is around $10/11$ — not one in a million. The number the jury was given is real; the conclusion drawn from it is off by a factor of about nine hundred thousand.

This is not a hypothetical error. It has produced wrongful convictions.

4.6 Check it in Python

Every probability in this unit is a ratio of counts, so all of it is checkable directly. First, that conditioning really is division by the new denominator:

```
flagged = usable[usable["strong_start"]]
print(f"P(finish | start) by filtering {flagged['strong_finish'].mean():.4f}")
print(f"P(finish | start) by division {both / start:.4f}")
```

```
P(finish | start) by filtering 0.5000
P(finish | start) by division 0.5000
```

Second — the mistake this unit exists to prevent. It is tempting to think that if a flag predicts an outcome well, the outcome predicts the flag equally well. Assert it and watch it fail:

```
p_finish_given_start = both / start
p_start_given_finish = both / finish
```

```
assert np.isclose(p_finish_given_start, p_start_given_finish), (
    f"P(finish|start) = {p_finish_given_start:.4f} but "
    f"P(start|finish) = {p_start_given_finish:.4f}"
)
```

AssertionError: P(finish|start) = 0.5000 but P(start|finish) = 0.6812

 AssertionError Traceback (most recent call last)
 Cell In[9], line 3

```
1 p_finish_given_start = both / start
2 p_start_given_finish = both / finish
----> 3 assert np.isclose(p_finish_given_start, p_start_given_finish), (
4     f"P(finish|start) = {p_finish_given_start:.4f} but "
5     f"P(start|finish) = {p_start_given_finish:.4f}"
6 )
```

AssertionError: P(finish|start) = 0.5000 but P(start|finish) = 0.6812

Both fractions have 47 on top. They differ because the denominators differ — 94 flagged students against 69 strong finishers. The two are equal only when the two events happen to be equally common, which is a coincidence rather than a rule:

```
print(f"P(start) {start / n:.4f} P(finish) {finish / n:.4f}")
print(f"ratio of the two conditionals {(both / start) / (both / finish):.4f}")
print(f"ratio of the two base rates {(finish / n) / (start / n):.4f}")
```

```
P(start) 0.3966 P(finish) 0.2911
ratio of the two conditionals 0.7340
ratio of the two base rates 0.7340
```

The two ratios agree, which is Bayes' theorem in one line: reversing a conditional multiplies it by the ratio of the base rates, and by nothing else.

Third, the addition rule, including the part everyone forgets:

```
either = ((usable["strong_start"]) | (usable["strong_finish"])).mean()
naive = start / n + finish / n
print(f"P(start OR finish) counted directly {either:.4f}")
print(f"P(start) + P(finish) {naive:.4f} too big by {naive - either:.4f}")
print(f"P(start AND finish) {both / n:.4f}")
```

```
P(start OR finish) counted directly 0.4895
P(start) + P(finish) 0.6878 too big by 0.1983
P(start AND finish) 0.1983
```

The amount by which the naive sum overshoots is exactly the overlap, which is what the $-P(A \cap B)$ in the addition rule is there to remove.

4.7 Common wrong turns

Where this usually goes wrong

Reading $P(A | B)$ as $P(B | A)$ The single most expensive error in applied statistics. Two thirds of the Maths/Physics students scored 70+, but only a third of the students who scored 70+ were from Maths/Physics. Say the denominator out loud every time: “of the 36, how many...”.

Ignoring the base rate A test that is right 90% of the time on people with a rare disease still produces mostly false alarms, because the healthy group is so much bigger. Before believing a positive result, ask how many of the thing there were to begin with.

Treating mutually exclusive as independent They are nearly opposite. Two events that cannot both happen are maximally informative about each other: if one occurred, the other definitely did not. Independence means the overlap is exactly $P(A)P(B)$, not zero.

Multiplying probabilities that are not independent $P(A \cap B) = P(A)P(B)$ only under independence. In general it is $P(A | B)P(B)$. Multiplying dependent probabilities — several test results from the same faulty instrument, several children in the same family — has produced some of the most notorious statistical miscarriages of justice on record.

Expecting independent trials to even out A fair coin that has come up heads six times has probability exactly $1/2$ of heads next. The coin has no memory, and “it is due a tail” is a statement about the gambler, not the coin.

Forgetting that $P(A \cup B)$ double-counts Adding two probabilities counts the overlap twice. Subtract it once. If the events are mutually exclusive there is no overlap and the addition is safe, which is why the shortcut works often enough to be dangerous.

4.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. $P(A) = 0.3$ and $P(B) = 0.6$. Which is impossible?
 - a. $P(A \cap B) = 0$
 - b. $P(A \cap B) = 0.18$
 - c. $P(A \cap B) = 0.3$
 - d. $P(A \cap B) = 0.4$
2. In the worked example, $P(\text{strong finish} | \text{strong start}) = 0.50$ while $P(\text{strong finish}) = 0.29$. Does the diagnostic score predict the final score?
3. A test for a condition affecting 1 in 10,000 people is positive for 99% of those who have it and 1% of those who do not. Roughly what fraction of positive results are correct?
4. Two events are mutually exclusive and both have probability 0.3. Are they independent?
5. A colleague says: “There is a 5% chance of getting a result this extreme if the drug does nothing, so there is a 95% chance the drug works.” What has gone wrong?

Show the answers

1. **(d).** An intersection cannot be larger than either event on its own, and $0.4 > 0.3 = P(A)$. Note that (a), (b) and (c) are all possible: 0 makes them mutually exclusive, 0.18 makes them independent, and 0.3 means A is entirely contained in B .
2. **Yes, usefully, and not decisively.** It nearly doubles the probability — from 29% to 50% — which is a real gain. But half of the flagged students do not go on to score 70+, so treating the flag as a verdict about an individual would be wrong half the time. Both statements

have to be made together.

3. **About 1%.** In a million people, 100 have the condition and 99 test positive; 999,900 do not and about 9,999 test positive anyway. That is roughly 10,098 positives of which 99 are correct — just under 1 in 100. If your first instinct was 99%, that is the base-rate error, and it is the normal instinct.
4. **No**, and this is the trap. $P(A \cap B) = 0$, while $P(A)P(B) = 0.09$. They are strongly *dependent*: knowing B happened tells you A certainly did not. Mutual exclusivity and independence can hold at once only if one of the events has probability zero.
5. The conditional has been reversed. The 5% is $P(\text{result this extreme} \mid \text{drug does nothing})$, and the colleague has reported it as $1 - P(\text{drug works} \mid \text{result})$. Getting from the first to the second needs a base rate — how plausible the drug was before the study — which the p -value does not contain. **S9** is entirely about this distinction.

Where this shows up next

S5 attaches probabilities to numbers rather than events, which is what makes them summable and averageable. **S6** names the standard families of those probabilities and says which mechanism produces which. **S9** is built on one conditional probability, and most of the trouble with p -values is people reading it backwards. **S12** returns to base rates, this time to explain why most published findings in a low-prior field are false.

5. Random variables

Statistics · Unit 5

S5 · Statistics Reading: 5 pages With practice: 2 hours Assumes: S4, M7

Why this unit is here

S4 attached probabilities to *events*: things that either happen or do not. That is enough for coins and diagnoses and not much else, because you cannot average an event or add two of them together.

A **random variable** attaches probabilities to *numbers*, and numbers can be added, averaged and rescaled. Doing so gives you two quantities — the expected value and the variance — that turn out to obey simple rules, and those rules are what make the rest of statistics possible. Every standard error you will ever compute is one line of algebra away from the variance rules in this unit.

5.1 Before you start

You should be able to:

- compute and interpret a conditional probability — S4;
- say why $P(A \cap B) = P(A)P(B)$ needs independence — S4;
- read an integral as the area under a curve — M7;
- compute a mean and a variance from data — S2.

Quick check. A fair six-sided die is rolled. What is the probability of rolling a 3, and what is the probability of rolling *at most* 3?

💡 Show the answer

$P(X = 3) = 1/6$, and $P(X \leq 3) = 3/6 = 1/2$.

You have just used both of the functions this unit is about. The first is the **probability mass function**, evaluated at one value; the second is the **cumulative distribution function**, which accumulates everything up to a value.

5.2 The idea

A **random variable** is a rule that assigns a number to each outcome of a random process. Roll a die, record the face: that is a random variable. Pick a student from the cohort at random, record their final score: so is that.

The convention is a capital letter for the variable and a lower-case letter for a value it might take, so $P(X = x)$ reads “the probability that the random variable X turns out to equal the particular number x ”. The distinction is worth respecting: X is the *procedure*, x is *one result*.

Random variables come in two kinds, and the difference is not a technicality. (Strictly there is a third possibility — a variable can be part lump, part smooth, like a rainfall total that is exactly zero on dry days and continuous otherwise. Such mixtures are real and are handled by combining the two cases; nothing else in these notes needs them.)

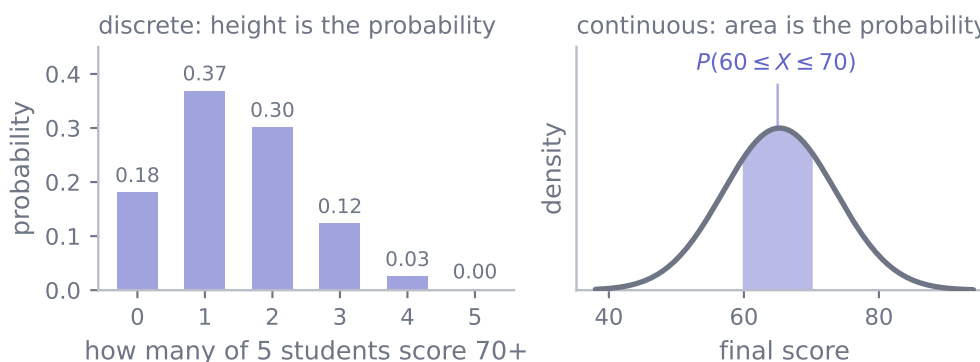


Figure 1: Two ways of carrying a probability. On the left each bar’s height is a probability and the six heights add to 1. On the right the curve’s height is not a probability at all — only the shaded area is, and the total area under the whole curve is 1.

A **discrete** random variable takes values you can list — 0, 1, 2, ... — and each one carries a probability of its own. The function $p(x) = P(X = x)$ is the **probability mass function**, and its values are genuine probabilities: between 0 and 1, adding to 1 across all the values.

A **continuous** random variable is one whose probabilities are given by a density, and for such a variable **the probability of any single exact value is zero**. That is a consequence of the definition rather than of having infinitely many values — a variable can have infinitely many possible values and still put a lump of probability on one of them, as the rainfall example does on zero. Under a density there are no lumps, so asking “what is the probability of a score of exactly 64.7?” has the answer 0, and that is not a defect in the question but a fact about the model.

What carries the probability instead is **area**. A continuous variable is described by a **probability density function** $f(x)$, and

$$P(a \leq X \leq b) = \int_a^b f(x) dx,$$

the area under the density between a and b . This is why M7 exists in these notes: a probability, for a continuous variable, is an integral, and it obeys the rules of integration because that is what it is.

Two consequences follow immediately. The total area under a density is 1. And $f(x)$ itself is *not* a probability — it can exceed 1 quite legitimately, as long as the width it applies over is small enough that the area does not.

5.3 The mechanics

The cumulative distribution function

One function works for both kinds:

$$F(x) = P(X \leq x).$$

It runs from 0 up to 1, never decreases, and for a continuous variable it is the accumulated area from the left — so F is an antiderivative of f , exactly as M7’s fundamental theorem describes. Every “probability of being below this value” that you will meet in S7 and S9 is a value of F , and every quantile is a value of F read backwards.

Expected value

$$E[X] = \sum_x x p(x) \quad (\text{discrete}) \quad E[X] = \int_{-\infty}^{\infty} x f(x) dx \quad (\text{continuous})$$

Read it as a weighted average: every value the variable can take, weighted by how likely it is. It is the long-run average of the results if you repeated the process for ever, and it is written μ when it belongs to a population — the μ of S1, which the sample mean $\bar{x}$ estimates.

each block's height is its probability; the pile balances at the mean

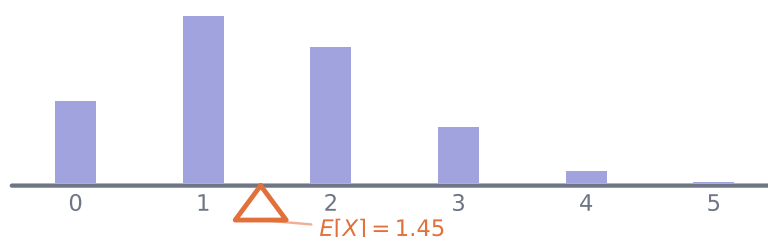


Figure 2: The left-hand distribution from the previous figure, treated as a pile of weights on a plank. The expected value is the point where it balances — which is why an expectation, unlike a median, is dragged about by values far from the centre.

Notice that $E[X] = 1.45$ is not a value the variable can take — you cannot have 1.45 students out of five. An expected value is a balance point, not a prediction.

Expectation is **linear**, and this is the most useful fact in the unit:

$$E[aX + b] = a E[X] + b, \quad E[X + Y] = E[X] + E[Y].$$

The second holds **whatever the relationship between X and Y** — no independence required. That is unusual and worth remembering, because the corresponding statement about variance is not free at all.

Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2,$$

with $\sigma = \sqrt{\text{Var}(X)}$ the standard deviation. Same idea as S2's s^2 , now attached to the distribution rather than to a sample. The second form is the one to compute with; the first is the one to understand.

The rules look similar to expectation's and are importantly different:

$$\text{Var}(aX + b) = a^2 \text{Var}(X), \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \text{ if } X \text{ and } Y \text{ are independent.}$$

Three things to take from that pair.

Adding a constant does nothing. Shifting every value by b moves the mean by b and leaves every deviation from the mean unchanged.

Scaling squares. Double a variable and its variance quadruples, so its standard deviation doubles. Variance lives in squared units; standard deviations are the ones that scale the way intuition expects.

The sum needs a condition, and it matters. The general statement is always $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$, with the covariance of S3 in it, and the two terms add cleanly exactly when that

covariance is zero. Independence guarantees a zero covariance, which is why it is the condition usually quoted, but it is stronger than necessary: two variables can be uncorrelated and still dependent. Assuming a zero covariance you do not have is how a standard error comes out too small, and a confidence interval too narrow.

The consequence everything else is built on

Put n independent copies of X , each with mean μ and variance σ^2 , and average them. Expectation is linear, so the mean of the average is μ . Variance scales by a^2 with $a = 1/n$, and adds across independent terms, so

$$\text{Var}(\bar{X}) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}, \quad \text{sd}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

That quantity $\sigma/\sqrt{n}$ is the **standard error** of the mean, and the $\sqrt{n}$ in its denominator is the reason precision is expensive: to halve the standard error you need four times the data. [S7](#) does nothing but explore this formula, and [S8](#) puts it to work. It fell out of two lines of algebra here.

5.4 Worked example

A student drawn at random.

Define X to be the final score of one student chosen uniformly at random from the cohort's 238 usable records. That is a genuine random variable, and because we can see the whole population it is one whose μ and σ we can compute exactly rather than estimate.

```
import numpy as np
import pandas as pd

cohort = pd.read_csv("data/cohort.csv")
scores = cohort.loc[cohort["final_score"].between(0, 100),
                  ~ "final_score"].to_numpy()

mu = scores.mean()
sigma = scores.std(ddof=0)      # ddof=0: these 238 ARE the population here
print(f"E[X]      = {mu:.4f}")
print(f"Var(X)    = {sigma**2:.4f}")
print(f"sd(X)     = {sigma:.4f}")
print(f"check: E[X^2] - mu^2 = {(scores**2).mean() - mu**2:.4f}")
```

```
E[X]      = 65.2718
Var(X)    = 69.9982
sd(X)     = 8.3665
check: E[X^2] - mu^2 = 69.9982
```

Note the `ddof=0`. [S2](#) used $n - 1$ because it was *estimating* a population from a sample; here the 238 records are the entire population we are drawing from, so there is nothing to correct for.

Now the rules. Rescale to a 0–1 fraction, $Y = X/100$:

```
fractions = scores / 100
print(f"E[Y]   predicted {mu / 100:.4f}      actual {fractions.mean():.4f}")
print(f"Var(Y) predicted {sigma**2 / 100**2:.6f}    actual
      {fractions.var(ddof=0):.6f}")
```

```
E[Y]   predicted 0.6527      actual 0.6527
Var(Y) predicted 0.007000    actual 0.007000
```

The variance fell by a factor of 100^2 , not 100. Squared units.

Now the part people get wrong. Draw *two* students independently and add their scores; compare that with drawing *one* student and doubling.

```
rng = np.random.default_rng(2026)
two_students = rng.choice(scores, size=(120_000, 2)).sum(axis=1)
one_doubled = 2 * rng.choice(scores, size=120_000)

print(f"X1 + X2 : mean {two_students.mean():7.2f}    sd {two_students.std():6.2f}"
      f"   predicted sd {np.sqrt(2) * sigma:.2f}")
print(f"2X      : mean {one_doubled.mean():7.2f}    sd {one_doubled.std():6.2f}"
      f"   predicted sd {2 * sigma:.2f}")
```

```
X1 + X2 : mean 130.52    sd 11.81    predicted sd 11.83
2X      : mean 130.54    sd 16.74    predicted sd 16.73
```

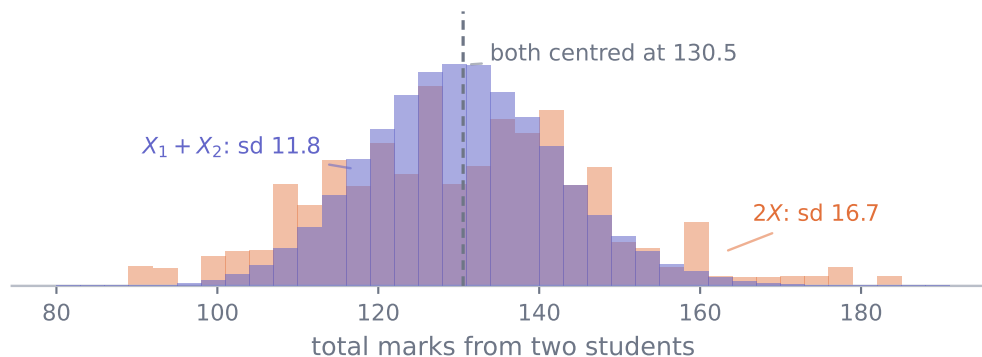


Figure 3: Two students' marks added together, against one student's mark doubled. Both have expected value 130.5, and the second is 40% more spread out. Independent errors partly cancel; a doubled error cannot cancel with anything.

Same expected total, and the doubled version is 40% wider. Both facts come straight from the rules: expectation is linear either way, but $\text{Var}(X_1 + X_2) = 2\sigma^2$ while $\text{Var}(2X) = 4\sigma^2$, and $\sqrt{4}/\sqrt{2} = \sqrt{2} = 1.41$.

The intuition behind the algebra: two independent students can be unlucky in opposite directions and partly cancel. One student, counted twice, cannot cancel with anything — whatever they did, it happens twice over.

5.5 Try it yourself

Exercise 1 A random variable X takes the value 0 with probability 0.7 and 1 with probability 0.3.

- (a) Find $E[X]$ and $\text{Var}(X)$.
 (b) Now let $S = X_1 + X_2 + \cdots + X_{10}$ be the sum of ten independent copies. Find $E[S]$ and $\text{Var}(S)$.

 Show the solution

- (a) $E[X] = 0(0.7) + 1(0.3) = 0.3$. For the variance, note that $X^2 = X$ here because $0^2 = 0$ and $1^2 = 1$, so $E[X^2] = 0.3$ and

$$\text{Var}(X) = 0.3 - 0.3^2 = 0.3 \times 0.7 = 0.21.$$

That shortcut generalises: a variable that is 1 with probability p and 0 otherwise has mean p and variance $p(1 - p)$.

- (b) $E[S] = 10 \times 0.3 = 3$, and by independence $\text{Var}(S) = 10 \times 0.21 = 2.1$, so $\text{sd}(S) = 1.45$.
 A common wrong answer is $\text{sd}(S) = 10 \times 0.458 = 4.58$ — multiplying the standard deviation by 10 rather than the variance. Standard deviations do not add; variances do.

Exercise 2 A continuous random variable has density $f(x) = 2x$ for $0 \leq x \leq 1$, and zero elsewhere.

- (a) Check that it really is a density.
 (b) Find $P(X \leq 0.5)$.
 (c) Find $E[X]$.

 Show the solution

- (a) A density must be non-negative — $2x \geq 0$ on $[0, 1]$ — and must enclose a total area of 1:

$$\int_0^1 2x \, dx = [x^2]_0^1 = 1. \checkmark$$

Note in passing that $f(1) = 2 > 1$. A density is allowed to exceed 1; only its integral is constrained.

- (b) $P(X \leq 0.5) = \int_0^{0.5} 2x \, dx = [x^2]_0^{0.5} = 0.25$. Not 0.5 — the density puts more weight on the right, so the left half of the range holds less than half the probability.
 (c) $E[X] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx = [\frac{2}{3}x^3]_0^1 = \frac{2}{3}$.
 Sanity check: $2/3$ is to the right of the middle of the range, which matches a density that rises towards 1.

Exercise 3 Two exam papers are each marked out of 50. A student's marks X and Y each have mean 30 and standard deviation 8.

- (a) What is the expected total, $E[X + Y]$?
 (b) What is $\text{sd}(X + Y)$ if the two marks are independent?
 (c) In reality a strong student tends to do well on both. Does the true standard deviation of the total go up or down relative to (b)?

 Show the solution

- (a) $E[X + Y] = 30 + 30 = 60$, and this needs **no** assumption about independence. Linearity of expectation is unconditional.
 (b) $\text{Var}(X + Y) = 64 + 64 = 128$, so $\text{sd} = \sqrt{128} = 11.3$. Note this is well below $8 + 8 = 16$.
 (c) **Up**. With positive correlation the covariance term is positive:

$$\text{Var}(X + Y) = 64 + 64 + 2 \text{Cov}(X, Y) > 128.$$

At the extreme of a perfect positive relationship, $Y = X$, the total is $2X$ with standard deviation 16 — the case in this unit's worked example. Independence is what lets the two marks partly cancel; correlation removes that cancellation. Assuming independence when it does not hold makes your stated uncertainty too small, which is the dangerous direction to be wrong in.

5.6 Check it in Python

The expected value is a long-run average, and that is a claim you can watch come true. Draw students one at a time and keep a running mean.

```
rng = np.random.default_rng(7)
draws = rng.choice(scores, size=2000)
running = np.cumsum(draws) / np.arange(1, draws.size + 1)

for k in [10, 100, 1000, 2000]:
    print(f"after {k:>4} draws: running mean {running[k-1]:.3f}    "
          f"gap from E[X] {running[k-1] - mu:+.3f}")
```

```
after   10 draws: running mean 61.170    gap from E[X] -4.102
after  100 draws: running mean 63.972    gap from E[X] -1.300
after 1000 draws: running mean 64.993    gap from E[X] -0.279
after 2000 draws: running mean 65.120    gap from E[X] -0.152
```

The gap shrinks, unsteadily — this is the **law of large numbers**, and note what it does and does not promise. The running mean converges to μ ; it does not promise the gap shrinks at every step, and it says nothing at all about any individual draw.

Next, the variance rules, all three at once:

```
pairs = rng.choice(scores, size=(200_000, 2))
X, Y = pairs[:, 0], pairs[:, 1]
print(f"Var(X)           {X.var():9.4f}    (population value {sigma**2:.4f})")
print(f"Var(X + 10)      {(X + 10).var():9.4f}    predicted {sigma**2:.4f}")
print(f"Var(3X)           {(3*X).var():9.4f}    predicted {9*sigma**2:.4f}")
print(f"Var(X + Y)        {(X + Y).var():9.4f}    predicted {2*sigma**2:.4f}")
```

```
Var(X)           69.8402    (population value 69.9982)
Var(X + 10)      69.8402    predicted 69.9982
Var(3X)          628.5615    predicted 629.9842
Var(X + Y)       139.5885    predicted 139.9965
```

And now the failure mode. The addition rule needs a zero covariance, and it is easy to write code that quietly violates it. Here Y is not a second draw at all — it is the same student again:

```

Y_same = X                                # NOT an independent second draw
total = X + Y_same
assert np.isclose(total.var(), 2 * sigma**2, rtol=0.02), (
    f"Var(X+Y) came out {total.var():.1f}, but the independence rule "
    f"predicts {2 * sigma**2:.1f}"
)

```

AssertionError: Var(X+Y) came out 279.4, but the independence rule predicts 140.0

```

-----
AssertionError                                Traceback (most recent call last)
Cell In[9], line 3
      1 Y_same = X                                # NOT an independent second draw
      2 total = X + Y_same
----> 3 assert np.isclose(total.var(), 2 * sigma**2, rtol=0.02), (
      4     f"Var(X+Y) came out {total.var():.1f}, but the independence rule "
      5     f"predicts {2 * sigma**2:.1f}"
      6 )

```

AssertionError: Var(X+Y) came out 279.4, but the independence rule predicts 140.0

Double, not doubled again: the variance is $4\sigma^2$ because $X + X = 2X$. In this cell the mistake is obvious. In a real analysis it looks like using the same patient twice, or treating repeated measurements as independent observations, and it inflates nothing visible — the mean comes out fine, and only the stated uncertainty is wrong.

```

print(f"Var(X + X)                                {(X + X).var():9.4f}")
print(f"Var(X) + Var(X) + 2Cov(X, X)              "
      f"{X.var() + X.var() + 2*np.cov(X, X, ddof=0)[0, 1]:9.4f}")

```

```

Var(X + X)                                279.3607
Var(X) + Var(X) + 2Cov(X, X)              279.3607

```

The general formula, with the covariance term restored, gets it right. The independence version is a special case in which that term happens to be zero.

5.7 Common wrong turns

Where this usually goes wrong

Reading a density as a probability $f(x)$ can be greater than 1. It is a probability *per unit of x* , and only an area under it — a density times a width — is a probability. For a continuous variable $P(X = x) = 0$ for every single x .

Adding standard deviations Variances add across independent variables; standard deviations do not. Two independent marks with sd 8 give a total with sd $\sqrt{128} = 11.3$, not 16. Convert to variance, add, convert back.

Assuming independence because the code drew twice Independence is a property of the *design*, not of the number of lines you wrote. Two measurements on the same person, two days from the same week, two students in the same tutorial group — none of these are independent, and each carries a covariance term that the simple addition rule leaves out.

Expecting the expected value $E[X] = 1.45$ students, $E[X] = 3.5$ on a die. An expectation is a balance point and often not a possible outcome. It is what the average converges to, not what you should predict for one trial.

Believing the law of large numbers applies to short runs It says the *average* settles down eventually. It does not say a run of bad luck will be compensated by a run of good luck; the earlier draws are simply diluted by the later ones.

Using $n - 1$ when you have the whole population The correction exists because $\bar{x}$ was estimated from the same data. If the values in front of you are the entire population, divide by n . Getting this wrong is harmless at $n = 238$ and material at $n = 5$.

5.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. X has mean 10 and standard deviation 3. What is the standard deviation of $4X - 7$?
a. 5 b. 12 c. 33 d. 144
2. A continuous random variable has $f(2) = 1.8$. Is that a problem?
3. X and Y each have variance 9. Is $\text{Var}(X + Y) = 18$?
4. You average n independent measurements. By what factor must n increase to halve the standard error?
5. True or false: $E[X + Y] = E[X] + E[Y]$ requires X and Y to be independent.

Show the answers

1. **(b), 12.** $\text{Var}(4X - 7) = 4^2 \times 9 = 144$, so the standard deviation is $\sqrt{144} = 12$. Equivalently, multiply the standard deviation by $|4|$. Option (d) is the variance, offered because reporting a variance where a standard deviation was asked for is the standard slip; the -7 affects neither.
2. **No.** A density is not a probability; it is a probability per unit of x . A value of 1.8 is fine provided the total area under f is 1 — for instance a variable spread uniformly over an interval of width 0.5 has density 2 throughout.
3. **Only if they are independent** (or, more weakly, uncorrelated). In general $\text{Var}(X + Y) = 9 + 9 + 2 \text{Cov}(X, Y)$, which can be anything from 0 (when $Y = -X$) to 36 (when $Y = X$).
4. **Four times.** The standard error is $\sigma / \sqrt{n}$, so halving it needs $\sqrt{n}$ to double, so n to quadruple. This is the single most useful piece of arithmetic in study design, and S8 returns to it.
5. **False**, and this is the asymmetry worth remembering. Linearity of expectation holds for *any* X and Y , however strongly related. It is the variance rule that needs independence.

Where this shows up next

S6 names the standard families of mass and density functions, and says which real mechanism produces which. S7 applies $\text{sd}(\bar{X}) = \sigma / \sqrt{n}$ to the sample mean and finds that its shape becomes normal whatever X looked like. S8 turns the standard error into an interval. S11 treats a fitted slope as a random variable and attaches a standard error to it by exactly the algebra in this unit.

6. Distributions chosen by mechanism

Statistics · Unit 6

S6 · Statistics Reading: 6 pages With practice: 2 hours Assumes: S5

Why this unit is here

A histogram tells you about the data you have. A **distribution** is a model, and a model lets you compute probabilities of things you have not seen: how often five failures in a row would happen by chance, how far from the mean an observation has to be before it is surprising, what a 95% interval should be wide.

The trap is choosing the distribution by looking at the histogram. Almost every family in this unit is the consequence of a **mechanism** — a story about how the numbers were generated — and the shape is what falls out of the story. Pick by the story and you can defend the choice and test it. Pick by the shape and you have drawn a curve that fits the data you happen to have.

6.1 Before you start

You should be able to:

- distinguish a probability mass function from a density — S5;
- compute an expected value and a variance — S5;
- read a histogram — S2.

Quick check. A random variable takes the value 1 with probability p and 0 otherwise. What are its mean and its variance?

💡 Show the answer

$E[X] = p$ and $\text{Var}(X) = p(1 - p)$.

This is the **Bernoulli** distribution, and it is the atom the first half of this unit is built from. Note that the variance is largest at $p = 0.5$ and vanishes at $p = 0$ or 1: a coin you cannot predict is the most variable one there is.

6.2 The idea

Six families cover most of what you will meet, and each of them answers a specific question about a specific kind of process.

Read the captions rather than the shapes. “Count of successes in n fixed trials” is a claim about your data-collection process that you can check. “Looks a bit like the second one” is not.

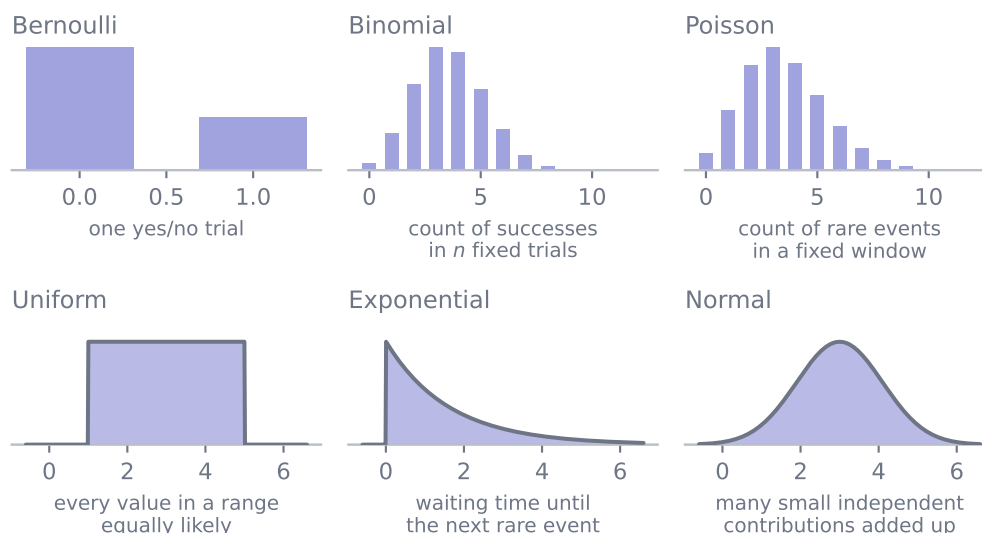


Figure 1: Six standard distributions, each with the mechanism that produces it. The top row is discrete — the bars are probabilities. The bottom row is continuous — the areas are probabilities. You choose between them by asking which story your data came from, not by matching the picture to a histogram.

6.3 The mechanics

Bernoulli and Binomial

A **Bernoulli** variable is one trial with two outcomes: $P(X = 1) = p$, $P(X = 0) = 1 - p$, mean p , variance $p(1 - p)$.

Add n **independent** Bernoulli trials that all share the same p and you count successes. That is the **Binomial**:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

with $E[X] = np$ and $\text{Var}(X) = np(1 - p)$ — which follow from S5's rules in one line each, since expectation and (under independence) variance both add.

The binomial makes two assumptions, and both fail in practice more often than people check. **The trials must be independent:** students in the same tutorial group, or repeat visits by the same customer, are not. **The probability must be the same every trial:** if p drifts — a machine wearing out, a policy changing midway — the count is no longer binomial.

It is worth being precise about which way each failure pushes the spread, because the obvious guess is wrong for one of them. Trials that stay independent but have *different* fixed probabilities $p_1, \dots, p_n$ give a count with variance $\sum p_i(1 - p_i)$, which is **smaller** than $n\bar{p}(1 - \bar{p})$ — spread is greatest when every trial is equally uncertain. What produces *overdispersion*, the far more common complaint, is failure of the first assumption: trials that are positively related, so that they tend to succeed or fail together.

Poisson

Events happen at a constant average rate λ per window, independently of each other, and you count how many fall in one window. That is the **Poisson**:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

with the striking property

$$E[X] = \text{Var}(X) = \lambda.$$

That equality is a **testable prediction**, and it is the first thing to check when someone proposes a Poisson model. Real count data — accidents, faults, support tickets — is usually *overdispersed*, with variance well above the mean, because the events cluster or the rate is not constant. If your counts have mean 4 and variance 15, the Poisson is already refuted, and no amount of curve-fitting repairs that.

Uniform

Every value in $[a, b]$ is equally likely: $f(x) = 1/(b - a)$ on that interval and zero outside it. Mean $(a + b)/2$, variance $(b - a)^2/12$. It is the distribution of “I know the range and nothing else”, and it is rarer in real data than beginners expect — genuine measurements almost always have a middle they prefer.

Exponential

The **waiting time** until the next event, when events arrive as a Poisson process at rate λ :

$$f(x) = \lambda e^{-\lambda x} \quad (x \geq 0), \quad E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Its defining property is **memorylessness**: having waited ten minutes already tells you nothing about how much longer you will wait. That is a very strong assumption. It is roughly true of radioactive decay and roughly false of almost everything involving wear, queues or human behaviour — a machine that has run for ten years is *not* as good as new.

Normal

The **normal** distribution arises when many small independent contributions add up, which is why it is everywhere: heights, measurement errors, exam marks built from many questions.

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

It is symmetric about μ , and the famous fractions follow:

Table 1: The 68–95–99.7 rule

within	of the mean	contains about
$\mu \pm 1\sigma$	one standard deviation	68%
$\mu \pm 2\sigma$	two standard deviations	95%
$\mu \pm 3\sigma$	three standard deviations	99.7%

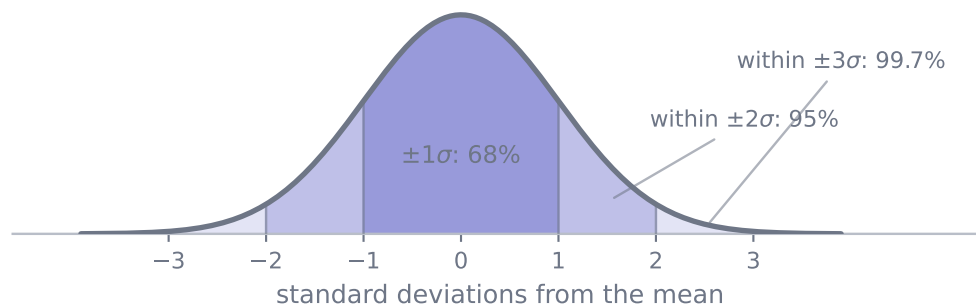


Figure 2: The 68–95–99.7 rule. Each band is an area under the density, so each is a probability — and the outermost band is thin enough that a three-sigma observation is genuinely rare, about 3 in 1000.

Two warnings that matter more than the formula. First, **the normal density has no elementary antiderivative** — the areas above cannot be worked out with the techniques of M7, which is why every statistics book once carried a table and every library now carries `norm.cdf`. Second, the normal has thin tails, so when the real distribution has heavier ones it badly underestimates how often extreme events happen. Financial returns are the standard cautionary tale. The error can run the other way — the cohort's diagnostic score, below, has *lighter* tails than a normal, so a normal model would over-predict its extremes — and the general point is that a fit judged in the middle tells you very little about either tail.

Choosing, then checking

The procedure is always the same. Ask what generated the data, pick the family whose mechanism matches, then **check the fit against a prediction the family makes** — the mean-equals-variance property for a Poisson, the 68–95–99.7 fractions for a normal, a quantile-quantile plot for anything continuous.

A **Q-Q plot** sorts your data and plots each value against the value the model says should sit at that position. If the model is right the points fall on a straight line; the ways they leave it tell you how it is wrong.

6.4 Worked example

Two questions about the cohort, two different families.

```
import numpy as np
import pandas as pd
from scipy import stats

cohort = pd.read_csv("data/cohort.csv")
scored = cohort[cohort["final_score"].between(0, 100)]
scores = scored["final_score"].to_numpy()
diagnostic = scored["diagnostic_score"].to_numpy()
```

Question one: of 20 students picked at random, how many score 70 or more?

The mechanism is right there: 20 independent draws, each either a success or not, with the same

probability each time. That is a binomial, and the only thing to supply is p .

```
p = (scores >= 70).mean()
n = 20
print(f"p = {(scores >= 70).sum()}/{scores.size} = {p:.4f}")
print(f"Binomial(20, {p:.3f}): mean {n*p:.2f} variance {n*p*(1-p):.2f}"
      f" sd {np.sqrt(n*p*(1-p)):.2f}")
print()
for k in [2, 4, 6, 8, 10]:
    print(f" P(exactly {k}>2) = {stats.binom.pmf(k, n, p):.4f}"
          f" P(at most {k}>2) = {stats.binom.cdf(k, n, p):.4f}")
```

```
p = 69/238 = 0.2899
Binomial(20, 0.290): mean 5.80 variance 4.12 sd 2.03
```

P(exactly 2) = 0.0336	P(at most 2) = 0.0434
P(exactly 4) = 0.1430	P(at most 4) = 0.2688
P(exactly 6) = 0.1907	P(at most 6) = 0.6463
P(exactly 8) = 0.1033	P(at most 8) = 0.9054
P(exactly 10) = 0.0253	P(at most 10) = 0.9869

So about six of twenty, and getting ten or more would be unusual but not extraordinary — $1 - F(9) = 3.8\%$ of the time. Notice that this is a claim about groups of twenty we never observed, produced entirely by the model.

The independence assumption deserves a moment. Twenty students drawn at random from a large population are effectively independent. Twenty students from the same tutorial group are not, and the binomial would understate the spread.

Question two: is the final score normal?

Here there is a mechanism to appeal to — a final score is built from many questions, each contributing a little — so the normal is a reasonable *prior* guess. Now check it, and check a variable where the same guess fails.

```
for name, values in [("final score", scores), ("diagnostic score", diagnostic)]:
    m, s = values.mean(), values.std(ddof=1)
    inside = [(values > m - k*s) & (values < m + k*s)].mean() for k in (1, 2,
~ 3)]
    print(f"{name:<18} within 1sd {inside[0]:.3f} 2sd {inside[1]:.3f} "
          f"3sd {inside[2]:.3f}")
print(f"'normal model says':<18} within 1sd 0.683 2sd 0.954 3sd 0.997")
```

final score	within 1sd 0.710	2sd 0.954	3sd 0.996
diagnostic score	within 1sd 0.601	2sd 0.979	3sd 1.000
normal model says	within 1sd 0.683	2sd 0.954	3sd 0.997

The final score is close on all three. The diagnostic score is not: only 60% of it lies within one standard deviation of the mean, against the 68% a normal predicts, and too much sits in the second band. That is the signature of a distribution that is *flatter* than a normal — which it should be, because S2 showed the cohort is five background groups with well-separated centres, and the diagnostic separates them most sharply of all.

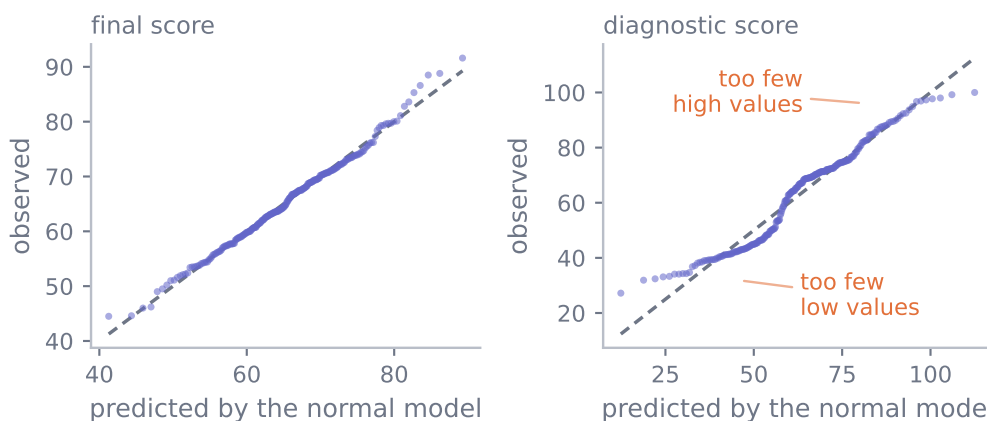


Figure 3: Quantile-quantile plots against a normal model. Each point puts an observed value against the value the model predicts for that position in the sorted order; a perfect fit lies on the dashed line. The final score follows it closely. The diagnostic score curves away at both ends: the low values are not as low as the model predicts and the high values not as high. That is the signature of a distribution with lighter tails than the normal, which is what a mixture of five well-separated groups produces.

The honest conclusion is the one to practise writing: *a normal model is adequate for the final score for the purposes of the intervals in S8, and not adequate for the diagnostic score, which is a mixture and should be handled by group*. Neither variable “is normal” — that is not a property data has. A model is adequate or not, for a stated purpose.

6.5 Try it yourself

Exercise 1 For each of these, name the distribution you would try first and say what mechanism justifies it.

- The number of heads in 50 tosses of a fair coin.
- The number of emails arriving at a server between 09:00 and 09:01.
- The time until a randomly chosen radioactive atom decays.
- The height of a randomly chosen adult.
- The number of students in a tutorial group of 12 who complete the course, where the tutor’s teaching affects everyone in the group.

💡 Show the solution

- Binomial(50, 0.5)** — a fixed number of independent trials with a constant success probability. Every assumption holds exactly, which is rare.
- Poisson** — a count of events in a fixed window at an approximately constant rate. Check the mean against the variance before trusting it: email arrives in bursts, and bursts produce overdispersion.
- Exponential** — the waiting time for a memoryless process. Radioactive decay is the one textbook example where memorylessness is genuinely true.
- Normal** — many small independent genetic and environmental contributions adding up. Note that adult height in a *mixed-sex* population is a mixture of two normals, and looks flatter than either, exactly like the diagnostic score in the worked example.
- Not binomial**, and this is the interesting one. The trials are not independent: a good or bad

tutor moves every student in the group the same way. The count still has a mean of $12p$, but its variance exceeds $12p(1 - p)$, so binomial intervals around it would be too narrow.

Exercise 2 A support desk records the number of tickets per hour over 200 hours. The mean is 4.1 and the variance is 14.8.

- (a) Is a Poisson model tenable?
- (b) Suggest two mechanisms that could produce this pattern.
- (c) What goes wrong if you use the Poisson anyway?

 Show the solution

- (a) **No.** A Poisson predicts mean = variance. Here the variance is more than three times the mean, which is the standard diagnosis of **overdispersion**.
- (b) Either the **rate is not constant** — busier at 10am than at 4pm, busier on Mondays, a spike when a release goes out — or the **events are not independent**, because one outage generates fifty tickets at once. Both produce the same symptom, and distinguishing them needs more than the two summary numbers.
- (c) Everything derived from the model understates the spread. A Poisson would put the standard deviation at $\sqrt{4.1} = 2.0$ when it is really $\sqrt{14.8} = 3.8$, so alert thresholds fire constantly, capacity plans are built on a quiet hour that does not exist, and any interval you quote is roughly half as wide as it should be.

Exercise 3 Exam marks have mean 65 and standard deviation 8, and a normal model is adequate.

- (a) Roughly what fraction of students score above 81?
- (b) A student scores 41. How unusual is that, and what would you check before saying so out loud?

 Show the solution

- (a) $81 = 65 + 2 \times 8$, so this is two standard deviations above the mean. About 95% lies within two standard deviations, leaving 5% outside, split evenly between the two tails: roughly **2.5%**.
- (b) $41 = 65 - 3 \times 8$, three standard deviations below. Under a normal model about 0.3% lies outside three standard deviations, so about **0.15%** below — roughly 1 student in 700. What to check first: whether the normal model is adequate *in the tail*. The 68–95–99.7 fractions are dominated by the middle of the distribution, and a model can match them well while getting the tails badly wrong — which is exactly where this question lives. With 240 students you have almost no information about a 1-in-700 event, so the honest answer is “unusual under this model, and this model is not well tested that far out”.

6.6 Check it in Python

The distributions in this unit make predictions, and every prediction here is one simulation away from being checked.

```
rng = np.random.default_rng(2026)
simulated = rng.binomial(n=20, p=p, size=200_000)
```

```
print(f"simulated mean {simulated.mean():.4f}    formula np      {n*p:.4f}")
print(f"simulated var  {simulated.var():.4f}    formula np(1-p) {n*p*(1-p):.4f}")
```

```
simulated mean 5.7879    formula np      5.7983
simulated var  4.1227    formula np(1-p) 4.1173
```

The Poisson's mean-equals-variance property, and then the same statistic on data that is not Poisson:

```
poisson_counts = rng.poisson(lam=4.0, size=200_000)
print(f"true Poisson:   mean {poisson_counts.mean():.3f}    "
      f"variance {poisson_counts.var():.3f}    ratio "
      f"↳ {poisson_counts.var()/poisson_counts.mean():.3f}")

# A rate that changes hour to hour: still counts, no longer Poisson.
rates = rng.uniform(1.0, 7.0, size=200_000)
mixed_counts = rng.poisson(lam=rates)
print(f"drifting rate: mean {mixed_counts.mean():.3f}    "
      f"variance {mixed_counts.var():.3f}    ratio "
      f"↳ {mixed_counts.var()/mixed_counts.mean():.3f}")
```

```
true Poisson:   mean 4.002    variance 4.006    ratio 1.001
drifting rate:  mean 4.005    variance 7.007    ratio 1.750
```

Both sets of counts have almost the same mean. Only the variance gives the second one away, which is why the ratio is the diagnostic worth computing every time.

Now the check that fails. It is tempting to treat “roughly bell-shaped” as “normal enough”, so assert the 68% rule on the diagnostic score and watch:

```
m, s = diagnostic.mean(), diagnostic.std(ddof=1)
within_one = ((diagnostic > m - s) & (diagnostic < m + s)).mean()
assert abs(within_one - 0.683) < 0.02, (
    f"a normal model predicts 68.3% within one sd; the diagnostic score has "
    f"{100*within_one:.1f}%"
)
```

```
AssertionError: a normal model predicts 68.3% within one sd; the diagnostic score
↳ has 60.1%
```

```
-----
AssertionError                                Traceback (most recent call last)
Cell In[9], line 3
      1 m, s = diagnostic.mean(), diagnostic.std(ddof=1)
      2 within_one = ((diagnostic > m - s) & (diagnostic < m + s)).mean()
----> 3 assert abs(within_one - 0.683) < 0.02, (
      4     f"a normal model predicts 68.3% within one sd; the diagnostic score
↳ has "
      5     f"{100*within_one:.1f}%"
      6 )
AssertionError: a normal model predicts 68.3% within one sd; the diagnostic score
↳ has 60.1%
```

Eight percentage points is a large miss, and it is in the direction that says “flatter than normal”. Run the same check on the final score and it passes:

```
m, s = scores.mean(), scores.std(ddof=1)
within_one = ((scores > m - s) & (scores < m + s)).mean()
print(f"final score: {100*within_one:.1f}% within one sd - "
      f"normal predicts 68.3%")
assert abs(within_one - 0.683) < 0.05
```

final score: 71.0% within one sd – normal predicts 68.3%

Note the tolerance had to be loosened to 5 percentage points for even the good case to pass. This is a coarse check, not a test, and S9 gives you the machinery to say how coarse.

6.7 Common wrong turns

Where this usually goes wrong

Choosing a distribution because the histogram looks like it With 30 observations almost any smooth family can be made to look plausible. Start from the mechanism — fixed trials, arrivals in a window, waiting times, many small additive effects — and use the histogram to *check*, not to choose.

Using a binomial when the trials are related Students in a class, measurements from one instrument, days in one week. The mean is still np , but the variance is not $np(1-p)$: positive dependence, the usual kind, inflates it, so everything built on the formula is too confident. (Negative dependence deflates it instead — drawing without replacement from a small pool is the standard example — which is safer but equally wrong.)

Using a Poisson without checking mean against variance It is one line of code and it refutes the model outright when it fails. Real counts are overdispersed far more often than not.

Assuming an exponential waiting time for something that wears out Memorylessness says a machine that has run ten years is as good as new. For radioactive decay that is true; for machinery, queues and people it is usually false and always worth stating explicitly.

Saying “the data are normally distributed” Data are not distributed; they are data. A *model* is adequate for a purpose. Say “a normal model fits well enough for this interval”, and name the check that persuaded you.

Trusting a normal model in the tails The 68–95–99.7 fractions are dominated by the middle. A model can match them and still be wrong by orders of magnitude about a 4-sigma event, and 4-sigma events are usually the ones that matter.

Expecting the central limit theorem to normalise your data It does no such thing. It says something about the distribution of the *sample mean* — S7 — and leaves the individual observations exactly as skewed as they were.

6.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. A count variable has mean 6 and variance 6.1. Which family is worth trying?
 - a. Binomial b. Poisson c. Normal d. Exponential

2. $X \sim \text{Binomial}(100, 0.2)$. What are $E[X]$ and $\text{sd}(X)$?
3. A variable is a mixture of two well-separated normal distributions **in roughly equal proportions**. Why does it fail the 68% check in the direction it does, and why does the “roughly equal” matter?
4. A colleague fits a normal model, finds it matches the 68–95–99.7 fractions well, and concludes the chance of an observation 5 standard deviations out is about 1 in 3.5 million. What is wrong with this?
5. True or false: if the individual observations are skewed, no normal-based method can be used.

 Show the answers

1. **(b), Poisson.** Mean and variance being nearly equal is the Poisson’s signature. (a) is possible but a binomial has variance $np(1 - p)$, which is below its mean np for every p strictly between 0 and 1; (c) and (d) are for continuous variables, and an exponential would have variance 36, not 6.
2. $E[X] = np = 20$ and $\text{Var}(X) = np(1 - p) = 100 \times 0.2 \times 0.8 = 16$, so $\text{sd}(X) = 4$. The commonest error is stopping at the variance and reporting 16.
3. Because a balanced mixture is **flatter** than a single normal with the same standard deviation. The overall standard deviation is inflated by the distance between the two centres, so the band $\mu \pm 1\sigma$ is wide, but the data are piled at the two centres rather than in the middle — leaving *fewer* than 68% inside. The cohort’s diagnostic score shows exactly this at 60%. The proportions matter because a *lopsided* mixture behaves in the opposite way. Put 99% of the weight at 0 and 1% at 100: the standard deviation is about 10, and roughly 99% of the values sit within one standard deviation of the mean. That is not a flat distribution at all — it is a tight one with a very heavy tail, and it fails the check in the other direction.
4. The fit was checked where the data are and the conclusion is drawn where they are not. Matching the central fractions says almost nothing about the tail beyond three standard deviations, and the number quoted — 1 in 3.5 million — is a property of the normal density, not of the data. Any dataset small enough to fit on a screen contains no evidence at all about 5-sigma events.
5. **False**, and this is the most useful thing in the next unit. Skewed observations can still give a sample *mean* whose distribution is close to normal, provided n is large enough — which is what [S7](#) is about. What you must not do is apply a normal model to the individual values.

Where this shows up next

[S7](#) explains why the normal turns up so often: sums and averages of almost anything tend towards it. [S8](#) uses the tail areas of a normal — and of its close relative the t — to build intervals. [S9](#) computes the probability of data at least as extreme as yours, which is a tail area of one of these families. [S10](#) uses the binomial and the chi-square when the outcome is a count rather than a measurement.

7. Sampling distributions and the CLT

Statistics · Unit 7

S7 · Statistics Reading: 5 pages With practice: 2 hours Assumes: S6

Why this unit is here

This is the hinge of the whole track. Everything before it describes data; everything after it makes claims about a population. The bridge between them is one idea: **a statistic computed from a sample is itself a random variable**, with a distribution of its own, and knowing that distribution is what lets you say how far your number might be from the truth.

The unit's two results are the standard error, $\sigma / \sqrt{n}$, and the central limit theorem, which says the sample mean becomes normally distributed almost whatever the population looks like. Together they are why a single study can say anything at all — and the limits of the second one are why some studies cannot.

7.1 Before you start

You should be able to:

- name a distribution from the mechanism that produced it — S6;
- state what a normal distribution's 68–95–99.7 fractions mean — S6;
- use $\text{Var}(aX) = a^2 \text{Var}(X)$ and the fact that variances add — S5;
- tell a population parameter from a sample estimate — S1.

Quick check. A population has standard deviation 12. You take a sample of 9 and compute its mean. What is the standard deviation of that mean, over repeated samples?

💡 Show the answer

$$\sigma / \sqrt{n} = 12 / \sqrt{9} = 12 / 3 = 4.$$

If you answered 12, you have given the spread of the *observations*, not of the *mean*. Distinguishing those two is what this unit is for, and everything from S8 onwards depends on it.

7.2 The idea

Take the cohort's 238 usable final scores and treat them, for the length of this unit, as an entire population whose mean we happen to know. Now imagine drawing 25 of them at random and computing the mean. Then do it again. And again.

Each of those means is a number you could have reported. Collect them and you have a **sampling distribution**: the distribution of a statistic across all the samples you might have taken.

Two things about that picture are the content of the unit.

The sampling distribution is much narrower. Individual students are spread over 8.4 marks; the means of 25 students are spread over 1.67. Averaging cancels some of the noise, and the cancellation

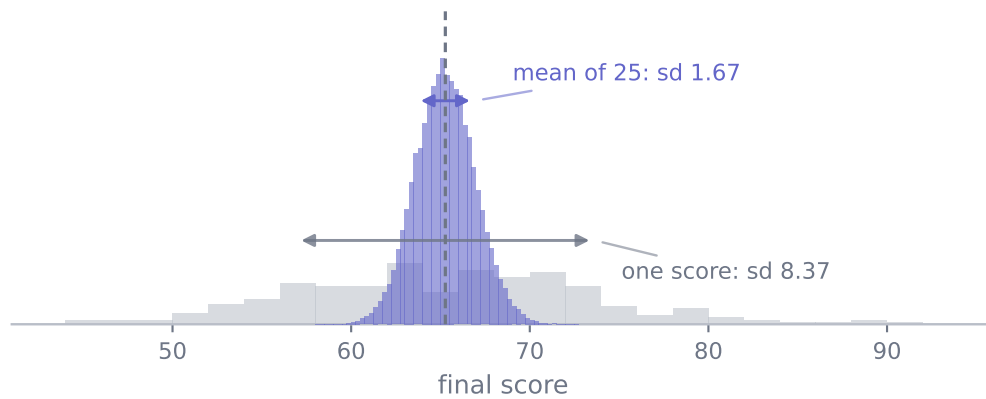


Figure 1: The same quantity, measured two ways. Behind, in grey, the 238 individual scores, spread over 8.4 marks. In front, the means of forty thousand samples of 25, spread over 1.67 marks — five times narrower, because five is the square root of 25. Both are centred in the same place.

is exactly the variance arithmetic of S5.

It is centred in the same place. The sample mean is *unbiased*: over many samples it neither systematically overshoots nor undershoots. That is a property of the procedure, not a guarantee about any single sample.

7.3 The mechanics

The standard error

For a mean of n independent draws from a population with mean μ and standard deviation σ , S5 gave

$$E[\bar{X}] = \mu, \quad \text{sd}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

That second quantity has its own name — the **standard error of the mean** — and keeping it separate from σ is the single most important piece of vocabulary in the subject.

Table 1: Two numbers that are not the same

	describes	changes with more data
standard deviation σ	how spread out the <i>observations</i> are	no
standard error $\sigma / \sqrt{n}$	how much the <i>mean</i> moves between samples	shrinks

Collect more data and the observations stay exactly as spread out as they were — people do not become more alike because you measured more of them. What improves is your knowledge of where the centre is.

In practice σ is unknown, so the standard error is *estimated* by $s / \sqrt{n}$ with the sample standard deviation of S2 in place of σ . That substitution is not free, and paying for it is what the t distribution in S8 is for.

The central limit theorem

The standard error tells you the *width* of the sampling distribution. The central limit theorem tells you its *shape*.

If $X_1, \dots, X_n$ are independent draws from any distribution with a finite mean μ and a finite variance σ^2 , then as n grows the distribution of $\bar{X}$ approaches a normal distribution with mean μ and standard deviation $\sigma / \sqrt{n}$.

Read what it does and does not say. It is a statement about the distribution of the **mean**, not about the data. Your observations remain exactly as skewed as they always were. And it requires **finite variance** — distributions with tails heavy enough to break that condition exist, and for them the theorem simply does not apply: $\sigma / \sqrt{n}$ has nothing to describe, and no sample size makes the sample mean normal. (The mean itself may still settle down — a finite mean is enough for that — but its distribution never becomes the one the theorem promises.)

Here it is happening. The population below is exponential, which is about as lopsided as a common distribution gets.

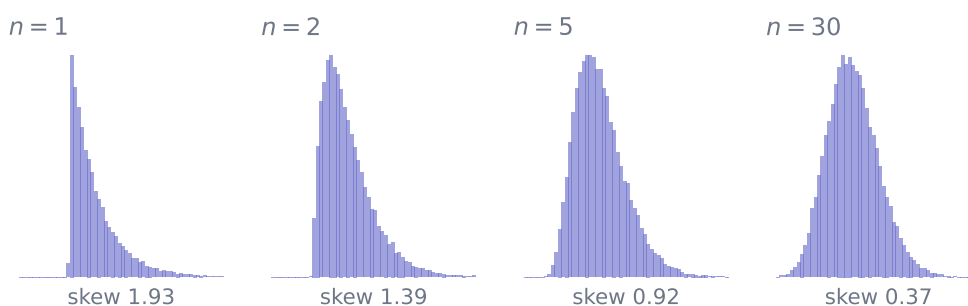


Figure 2: Sampling distributions of the mean of n draws from an exponential population, which is strongly right-skewed. Each panel is rescaled to the same width — the real widths shrink as $\sigma / \sqrt{n}$ — so that only the change in *shape* is visible. The lopsidedness falls steadily and by $n = 30$ the distribution is close to symmetric. The population itself, in the first panel, never changed.

“ n is big enough” is not a number

Textbooks often say $n = 30$ is enough. That is a rule of thumb calibrated on mildly skewed populations, and it is badly wrong for strongly skewed ones. How large n has to be depends on how far the population is from normal — and the worse the population, the larger the n , with no upper bound.

The practical rule that replaces the number: **look at your data’s skewness, and when in doubt use the bootstrap of S8**, which does not assume a shape for the sampling distribution at all.

Two more assumptions worth naming

Independence. Everything here assumes the n draws are independent. If they are not, the correct variance of the mean carries covariance terms:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \left(\sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \right).$$

The dependence you meet in practice — repeated measures, clustered data, a time series — is nearly always *positive*, so those terms add and the true standard error is larger than $\sigma / \sqrt{n}$, often much larger,

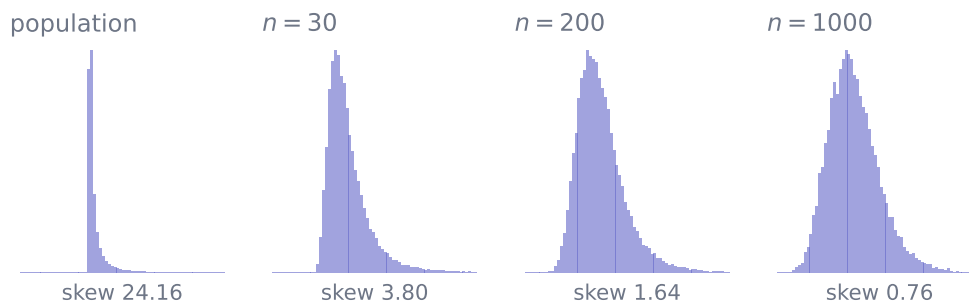


Figure 3: The same treatment applied to a much worse population: a lognormal whose skewness is about 24. Panels are again rescaled to a common width, and drawn over the same range as the figure above so the two can be compared directly. The convergence the theorem promises is real and it is slow — at $n = 1000$ the sample mean is still twice as lopsided as the exponential's was at $n = 30$.

and every interval built on it is too narrow. Negative dependence works the other way and makes the mean *more* precise than the formula claims; the next paragraph is an example. This is the assumption most often violated in practice and least often mentioned.

Sampling with replacement, or a small fraction. Drawing without replacement from a finite population makes the draws slightly dependent. If your sample is under about 5% of the population the effect is negligible; if you have sampled half the population, the standard error is genuinely smaller than the formula says, and there is a finite-population correction for it.

7.4 Worked example

Watching the formula come true on the cohort.

Treat the 238 usable scores as the population, and take forty thousand samples at each of three sizes.

```
import numpy as np
import pandas as pd
from scipy import stats

cohort = pd.read_csv("data/cohort.csv")
population = cohort.loc[cohort["final_score"].between(0, 100),
    ~ "final_score"].to_numpy()
mu, sigma = population.mean(), population.std(ddof=0)
rng = np.random.default_rng(2026)

print(f"population: n = {population.size}    mu = {mu:.4f}    sigma = {sigma:.4f}")
print()
print(f"{'sample size':>12}{'mean of means':>16}{'sd of means':>14}"
    f"{'sigma/sqrt(n)':>16}")
for n in [5, 25, 100]:
    draws = rng.choice(population, size=(40_000, n))
    sample_means = draws.mean(axis=1)
    print(f"{'n':>12}{sample_means.mean():>16.4f}{sample_means.std():>14.4f}"
        f"{'sigma / np.sqrt(n)':>16.4f}")
```

```
population: n = 238    mu = 65.2718    sigma = 8.3665
```

sample size	mean of means	sd of means	sigma/sqrt(n)
5	65.2603	3.7478	3.7416
25	65.2695	1.6733	1.6733
100	65.2731	0.8370	0.8366

Three things to take from that table.

The formula is exact, not approximate. $\sigma/\sqrt{n}$ is the true standard deviation of the sample mean for any population with finite variance, at any n . It is the *shape* that the central limit theorem makes approximate, never the width.

The means are unbiased at every sample size. Even at $n = 5$, where a single sample is nearly useless, the average of many such means lands on μ . Unbiasedness and precision are separate properties, and a procedure can have the first without the second.

Quadrupling n halves the standard error. From 25 to 100 the spread falls from 1.67 to 0.84. That is the $\sqrt{}$, and it is the arithmetic behind every complaint about how expensive precision is.

Now the shape. The cohort's scores are only mildly skewed, so the mean should become normal quickly:

```
for n in [5, 25, 100]:
    sample_means = rng.choice(population, size=(40_000, n)).mean(axis=1)
    centre, spread = sample_means.mean(), sample_means.std()
    inside = ((sample_means > centre - spread) & (sample_means < centre +
    ← spread)).mean()
    print(f"n = {n:>3}    skewness {stats.skew(sample_means):+.3f}    "
          f"within one sd {100*inside:.1f}%    (normal: 68.3%)")
```

```
n =   5    skewness +0.108    within one sd 68.3%    (normal: 68.3%)
n =  25    skewness +0.050    within one sd 68.7%    (normal: 68.3%)
n = 100    skewness +0.038    within one sd 68.4%    (normal: 68.3%)
```

Even at $n = 5$ the sampling distribution is within a percentage point of the normal fractions, because the population it came from was already close to symmetric. The lognormal in the previous figure needed more than a thousand. The theorem is the same in both cases; what differs is how long it takes to arrive.

7.5 Try it yourself

Exercise 1 A population has $\mu = 50$ and $\sigma = 20$.

- What is the standard error of the mean of a sample of 25?
- How large a sample would you need for a standard error of 1?
- A colleague reports that with $n = 400$ their sample standard deviation was
 - Is that plausible?

 Show the solution

- (a) $20/\sqrt{25} = 20/5 = 4$.
- (b) Set $20/\sqrt{n} = 1$, so $\sqrt{n} = 20$ and $n = 400$.
- (c) **Almost certainly they have computed the standard error and called it the sample standard deviation.** The sample standard deviation estimates $\sigma = 20$ and does not shrink with n ; the standard error at $n = 400$ is exactly $20/20 = 1$, and 2 is much closer to that than to 20.
Strictly, a sample standard deviation of 2 is not impossible. A population whose σ comes almost entirely from very rare enormous values can have $\sigma = 20$ while a typical sample of 400 — containing none of them — looks far tighter. Nothing in the question suggests such a population, and the mix-up is the far likelier explanation, but it is worth knowing that the sample standard deviation is itself an estimate with a distribution.

Exercise 2 Someone measures 40 patients, taking 5 blood-pressure readings from each, and computes a standard error as $s/\sqrt{200}$.

- (a) What has gone wrong?
- (b) Is the reported standard error too large or too small?
- (c) What should they do instead?

 Show the solution

- (a) The 200 readings are **not 200 independent draws**. Five readings from one patient are related to each other; the independent unit is the patient, and there are 40 of them.
- (b) **Too small.** The formula credits them with 200 independent pieces of information when they have closer to 40. How much closer depends on how strongly readings within a patient agree: if they agreed perfectly, 200 readings would carry exactly as much information as 40.
- (c) Reduce each patient to a single number — their mean of five readings — and compute the standard error from those 40 values. That is the simplest correct analysis. (More sophisticated methods use all 200 readings and model the within-patient correlation explicitly, which recovers a little more precision; they never recover the factor of $\sqrt{5}$ the naive calculation claimed.)

Exercise 3 Household income is strongly right-skewed. A survey of 50 households reports a mean income and a symmetric interval around it.

- (a) Which assumption is under strain?
- (b) Which direction would you expect the error to go?
- (c) Name two things you could do about it.

 Show the solution

- (a) Not the standard error — $\sigma/\sqrt{n}$ holds regardless of shape. The strained assumption is the **central limit theorem's approximation at $n = 50$** : with a strongly skewed population the sampling distribution of the mean is still lopsided at that size, so a *symmetric* interval does not match it.
- (b) **Coverage falls below the stated 95%, and the misses are nearly all of one kind.** Simulat-

ing a strongly right-skewed population at $n = 50$, the usual t interval contains the true mean about 90% of the time, and in almost every failure the whole interval sits *below* it. The mechanism is worth following. Most samples from a right-skewed population miss its rare large values, so their mean comes out too low — and because those same large values are what inflate the standard deviation, such a sample also has a small s and therefore a *narrow* interval. Too low and too narrow is exactly the combination that misses. The occasional sample that does catch a large value gets a high mean and a wide interval, and usually still covers.

- (c) Use the **bootstrap** (S8), which builds the sampling distribution empirically and produces an asymmetric interval if that is what the data imply. Or change the estimand: report the **median** income, which is what people usually meant anyway and is far less affected by the tail. Transforming to logs is a third option, but note it changes what you are estimating — the mean of the logs is not the log of the mean.

7.6 Check it in Python

The standard error formula is exact, so it should hold for a population of any shape at all. Check it on three:

```
populations = {
    "cohort scores": population,
    "exponential": rng.exponential(3.0, 200_000),
    "coin flips": rng.integers(0, 2, 200_000).astype(float),
}
n = 40
for name, pop in populations.items():
    sample_means = rng.choice(pop, size=(30_000, n)).mean(axis=1)
    print(f"{name:<15} sigma {pop.std():7.4f}   observed sd of mean "
          f"{sample_means.std():7.4f}   sigma/sqrt(n) "
          f"→ {pop.std()/np.sqrt(n):7.4f}")
```

cohort scores	sigma	8.3665	observed sd of mean	1.3174	sigma/sqrt(n)	1.3229
exponential	sigma	2.9853	observed sd of mean	0.4704	sigma/sqrt(n)	0.4720
coin flips	sigma	0.5000	observed sd of mean	0.0788	sigma/sqrt(n)	0.0791

Three completely different populations — one roughly bell-shaped, one strongly skewed, one taking only the values 0 and 1 — and the width formula is right in all three.

The *shape* is a different matter, and this is where the rule of thumb fails. Assert that a sample of 30 makes the mean approximately normal, and watch it fail on a population skewed enough to matter:

```
heavy = rng.lognormal(0.0, 1.6, 300_000)
means_of_30 = rng.choice(heavy, size=(40_000, 30)).mean(axis=1)
skewness = stats.skew(means_of_30)
assert abs(skewness) < 0.5, (
    f"n = 30 was supposed to be enough, but the sample mean still has "
    f"skewness {skewness:.2f} (a normal distribution has 0)"
)
```

AssertionError: n = 30 was supposed to be enough, but the sample mean still has
 ↳ skewness 5.52 (a normal distribution has 0)

```
-----
AssertionError                                Traceback (most recent call last)
Cell In[7], line 4
      2 means_of_30 = rng.choice(heavy, size=(40_000, 30)).mean(axis=1)
      3 skewness = stats.skew(means_of_30)
----> 4 assert abs(skewness) < 0.5, (
      5     f"n = 30 was supposed to be enough, but the sample mean still has "
      6     f"skewness {skewness:.2f} (a normal distribution has 0)"
      7 )
```

AssertionError: n = 30 was supposed to be enough, but the sample mean still has
 ↳ skewness 5.52 (a normal distribution has 0)

The same check on the cohort's scores passes comfortably at $n = 5$:

```
mild = rng.choice(population, size=(40_000, 5)).mean(axis=1)
print(f"cohort scores, n = 5: skewness of the sample mean
↳ {stats.skew(mild):+.3f}")
assert abs(stats.skew(mild)) < 0.5
```

cohort scores, n = 5: skewness of the sample mean +0.077

The difference is not the sample size. It is the population, and the only way to know which case you are in is to look at your data before choosing a method.

Finally, independence. Here is the same n with the draws deliberately made dependent — each sample is one student repeated:

```
n = 25
independent = rng.choice(population, size=(20_000, n)).mean(axis=1)
repeated = np.repeat(rng.choice(population, size=(20_000, 1)), n,
↳ axis=1).mean(axis=1)
print(f"25 independent students : sd of mean {independent.std():.4f}"
      f" formula {sigma/np.sqrt(n):.4f}")
print(f"one student, 25 times : sd of mean {repeated.std():.4f}"
      f" formula still claims {sigma/np.sqrt(n):.4f}")
```

25 independent students	: sd of mean	1.6817	formula	1.6733
one student, 25 times	: sd of mean	8.3464	formula still claims	1.6733

The second case is an extreme, but the direction is the one you should expect: positively dependent observations make the true standard error larger than the formula, and the formula gives no warning at all.

7.7 Common wrong turns

Where this usually goes wrong

Confusing the standard deviation with the standard error s describes the observations and does not shrink as you collect more. $s/\sqrt{n}$ describes the mean and does. Reporting one where the other was meant is the commonest numerical error in student work, and it is a factor of $\sqrt{n}$.

Thinking the CLT makes your data normal It makes the *sample mean* approximately normal. The observations keep whatever shape they had, and a normal model applied to them is as wrong after the theorem as before it.

Trusting $n = 30$ It is a rule of thumb for mildly skewed populations. For income, insurance claims, or anything with a long tail, thirty is nowhere near enough, and there is no sample size that works for every population.

Using $\sigma/\sqrt{n}$ on dependent observations Repeated measures, clustered samples and time series all break the assumption, and because their dependence is positive they all make the true error larger. Reduce to one number per independent unit, or use a method that models the dependence. (Negative dependence — sampling a large fraction of a finite population without replacement — makes the formula conservative instead.)

Reading unbiasedness as accuracy The sample mean is unbiased at $n = 5$, which sounds reassuring and means only that it is right *on average across many studies*. Your one study can still be a long way out. Unbiased and precise are separate virtues.

Forgetting the theorem needs a finite variance Some distributions have tails so heavy that the variance does not exist. The sample mean may still converge to the population mean, but $\sigma/\sqrt{n}$ does not exist to describe how fast, and the normal approximation never arrives. They are rare in teaching examples and not rare in finance.

7.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. A population has $\sigma = 30$. You increase your sample from 100 to 900. What happens to s and to the standard error?
 - a. Both fall by a factor of 3
 - b. s stays about 30; the standard error falls from 3 to 1
 - c. s falls from 3 to 1; the standard error stays about 30
 - d. Both stay the same
2. The cohort's scores are mildly skewed and the sample mean is nearly normal by $n = 5$. Household income is strongly skewed and the sample mean is still skewed at $n = 1000$. Do these two facts contradict the central limit theorem?
3. Is the formula $\text{sd}(\bar{X}) = \sigma/\sqrt{n}$ an approximation?
4. You have measured 60% of a small population. Is your standard error $s/\sqrt{n}$?
5. A study of 2,000 people reports a mean with a very narrow standard error. The 2,000 were recruited through one social-media post. What does the standard error tell you about how close the estimate is to the population mean?

Show the answers

1. **(b).** The sample standard deviation estimates σ , which is a property of the population and does not change: it stays near 30. The standard error is $30/\sqrt{n}$, falling from $30/10 = 3$ to $30/30 = 1$. Option (c) swaps the two, which is exactly the confusion the question is testing.

2. **No.** The theorem is a statement about a limit: the sampling distribution approaches normality as $n \rightarrow \infty$. It says nothing about *how fast*, and the rate depends on the population. Both observations are the theorem behaving exactly as stated.
3. **No.** It is exact for any population with a finite variance and any n , given independent draws. The approximation in this unit is the *shape* — the claim that the sampling distribution is normal — not the width.
4. **No.** Drawing 60% of a finite population without replacement makes the draws dependent, and in this direction it *helps*: the true standard error is smaller than $s/\sqrt{n}$, by a factor of roughly $\sqrt{1 - 0.6} = 0.63$. In the limit where you sample everybody there is no sampling error at all, because there is no sampling left.
5. **Almost nothing.** The standard error measures how much the estimate would bounce around between repeated samples *drawn the same way* — and every one of those samples would be recruited through the same social-media post, with the same people systematically missing. A narrow standard error around a biased estimate is a precise answer to the wrong question, which is **S1**'s lesson arriving with a number attached.

Where this shows up next

S8 turns the standard error into a confidence interval, and introduces the t distribution that pays for estimating σ . **S9** asks how far into the tail of this sampling distribution an observed result falls. **S10** builds the sampling distribution of a *difference* between two means. **S11** does the same for a fitted slope.

8. Confidence intervals and the bootstrap

Statistics · Unit 8

S8 · Statistics Reading: 5 pages With practice: ~ 2 hours Assumes: S7

Why this unit is here

Any number you calculate from a sample is one draw from a distribution of numbers you did not see. S7 established that. This unit is about saying it out loud.

The difference between reporting “the mean score was 60.5” and “the mean score was 60.5, and given how few people we measured, anything from about 58.6 to 62.5 would be consistent with what we observed” is the difference between a number and a finding. Almost every result you will produce or read for the rest of your degree comes with an interval attached, and knowing what that interval does and does not claim is most of the skill.

8.1 Before you start

You should be able to:

- explain what a sampling distribution is, and why the standard error is not the standard deviation — S7;
- calculate a sample mean and sample standard deviation — S2;
- say roughly what fraction of a normal distribution lies within two standard deviations of its centre — S6.

Quick check. A sample of 100 measurements has a standard deviation of 20. What is the standard error of the mean?

💡 Show the answer

$$SE = \frac{s}{\sqrt{n}} = \frac{20}{\sqrt{100}} = 2.$$

If you answered 20, reread S7 first — the distinction it draws is the one this entire unit is built on.

8.2 The idea

Take the cohort, and look only at the students who came from a life-sciences background.

```
import pandas as pd

cohort = pd.read_csv("data/cohort.csv")
usable = cohort.dropna(subset=["background", "final_score"])
usable = usable[usable["final_score"].between(0, 100)]

life = usable[usable["background"] == "Life sciences"]["final_score"]
```

```
print(f"n = {life.size}    mean = {life.mean():.2f}    sd =
      {life.std(ddof=1):.2f}")
```

```
n = 57    mean = 60.51    sd = 7.29
```

Their mean score is 60.51. But had 57 *different* life-sciences students walked through the door, the mean would have come out somewhere else — a bit higher, a bit lower. The number 60.51 is not wrong, it is simply one of many that could reasonably have happened.

A **confidence interval** is a range built from the sample, designed so that the *procedure* which builds it captures the true population value a stated fraction of the time.

Read that sentence again, because the emphasis is doing real work. The 95% in “95% confidence interval” is a property of the **method**, not of the particular interval sitting in front of you. Once you have computed (58.58, 62.45), the true mean either is in that range or it is not; nothing about it is 95% any more. What is true is that if you repeated the whole study many times, about 95% of the intervals produced this way would contain the answer.

That distinction sounds like pedantry and is not. It is the single most common misreading in applied statistics, and S9 depends on you having it straight.

8.3 The mechanics

For a mean, when the population standard deviation is unknown — which is essentially always — the interval is

$$\bar{x} \pm t_{1-\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$$

This is exact when the observations are independent draws from a normal population. For other populations it is an approximation, and how good an approximation depends on the sample size and on how skewed or outlier-prone the data are. It is not distribution-free.

Three parts, each doing a job:

- $\bar{x}$ is where the interval is centred: your best single guess.
- $s/\sqrt{n}$ is the standard error — how much $\bar{x}$ would bounce around between repeated samples.
- $t_{1-\alpha/2, n-1}$ is how many standard errors wide to make it. For 95% confidence it is about 2, and it is *slightly more* than 2 for small samples.

That last point is the only new idea here. If you knew the population standard deviation you would use 1.96, from the normal distribution. You do not know it; you estimated it with s , and that estimate is itself uncertain. The t distribution pays for that extra uncertainty by making the interval a little wider, and the smaller the sample the more it charges.

What the interval does not cover

A confidence level is a statement about sampling variation **and nothing else**. It assumes the model, the design and the arithmetic are right. It does not protect you against:

- a sample that was never representative of the population you care about;

- systematic measurement error, which shifts every observation the same way;
- observations that are not independent — repeated measures, clustered data;
- a bug in the code that produced it.

A narrow interval computed from a badly collected sample is a precise answer to the wrong question. Precision and correctness are different properties, and only one of them is visible in the output.

When there is no formula: the bootstrap

The formula above exists because a great deal of theory has been done for means. For a median, a correlation, a ratio of two quantities, or the difference between two percentiles, there may be no convenient formula at all.

The **bootstrap** sidesteps the theory. The reasoning is that your sample is the best picture you have of the population, so:

1. draw n rows from your sample **with replacement** — some rows appear twice, some not at all;
2. recompute the statistic on that resample;
3. repeat several thousand times;
4. take the middle 95% of the resulting values as your interval.

Sampling *with* replacement is essential. Without it you would draw the same n rows every time and see no variation at all.

The bootstrap is remarkably general, and it is not magic. It works badly for very small samples, for statistics near a boundary such as a maximum, and for strongly skewed distributions. It cannot repair bias — if your sample is unrepresentative, every resample of it is unrepresentative in the same way. And for paired or clustered data you must resample **whole units**: for a paired design, resample pairs, never individual measurements.

8.4 Worked example

An interval for the life-sciences mean, two ways.

By hand first. We have $n = 57$, $\bar{x} = 60.51$ and $s = 7.29$, so the standard error is $7.29/\sqrt{57} = 0.97$. For 95% confidence with 56 degrees of freedom the multiplier is 2.003, giving a half-width of $2.003 \times 0.97 = 1.93$ and an interval of 60.51 ± 1.93 .

```
import numpy as np
from scipy import stats

n = life.size
mean = life.mean()
se = life.std(ddof=1) / np.sqrt(n)
t_multiplier = stats.t.ppf(0.975, df=n - 1)

lower, upper = mean - t_multiplier * se, mean + t_multiplier * se
print(f"multiplier t(0.975, {n-1}) = {t_multiplier:.3f}")
print(f"standard error          = {se:.3f}")
print(f"95% interval for the mean = ({lower:.2f}, {upper:.2f})")
```

```
multiplier t(0.975, 56) = 2.003
```

standard error = 0.965
 95% interval for the mean = (58.58, 62.45)

Now the bootstrap, which needs no formula and no t distribution:

```
rng = np.random.default_rng(2026)
values = life.to_numpy()

resamples = rng.choice(values, size=(10_000, n), replace=True)
boot_means = resamples.mean(axis=1)

print(f"bootstrap interval = ({np.quantile(boot_means, 0.025):.2f}, "
      f"{np.quantile(boot_means, 0.975):.2f})")
```

bootstrap interval = (58.67, 62.46)

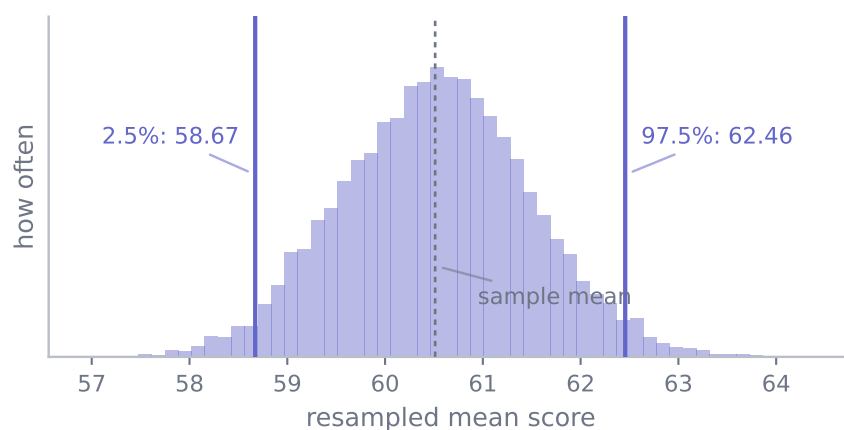


Figure 1: Ten thousand resampled means. The bootstrap interval is simply the middle 95% of this pile — the 2.5th and 97.5th percentiles, marked.

The two agree to within a few hundredths, which is what you should expect for a mean from a reasonably sized, reasonably symmetric sample. The bootstrap earns its keep on statistics where the formula runs out:

```
boot_medians = np.median(resamples, axis=1)
print(f"sample median = {np.median(values):.2f}")
print(f"95% interval for the median = ({np.quantile(boot_medians, 0.025):.2f}, "
      f"{np.quantile(boot_medians, 0.975):.2f})")
```

sample median = 59.80
 95% interval for the median = (57.70, 61.80)

Notice the median's interval is wider than the mean's. That is not a defect — the median throws away information about how far the extreme values lie, and it pays for that robustness in precision.

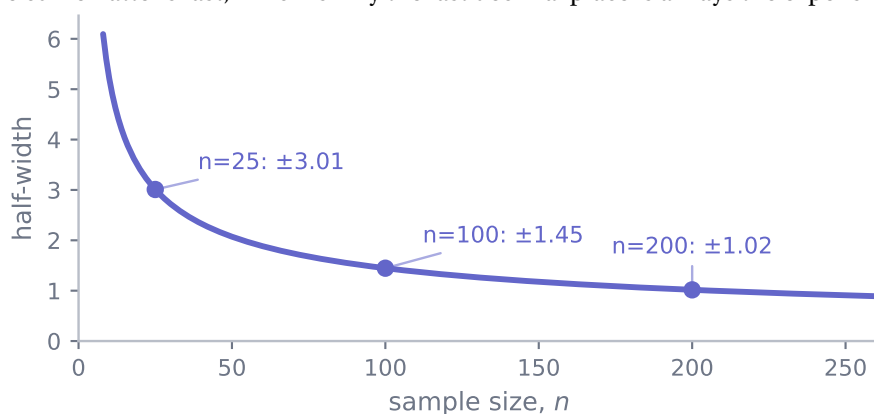
One last thing worth seeing. Compare two groups:

```
for group in ["Life sciences", "Mathematics or Physics"]:
    scores = usable[usable["background"] == group]["final_score"]
    k = scores.size
    err = scores.std(ddof=1) / np.sqrt(k)
    width = stats.t.ppf(0.975, df=k - 1) * err
    print(f"{group:<24} n={k:>3} mean={scores.mean():5.2f} ±{width:.2f}")
```

```
Life sciences          n= 57  mean=60.51  ±1.93
Mathematics or Physics n= 36  mean=75.28  ±2.58
```

```
[Text(0.5, 0, 'sample size, $n$'),
 Text(0, 0.5, 'half-width'),
 (0.0, 262.0),
 (0.0, 6.459629627947625)]
```

(a) Half-width of the interval against sample size, holding the spread of the data fixed. Precision improves only as $1/\sqrt{n}$: the curve flattens fast, which is why the last decimal place is always the expensive one.



(b)

Figure 2

The second group has a *wider* interval despite being a perfectly good sample. It has fewer students in it, and the width is driven by n at least as much as by the spread of the data.

8.5 Try it yourself

Exercise 1 A sample of 36 measurements has mean 50 and standard deviation 12. The multiplier $t_{0.975, 35}$ is 2.030. Compute the 95% interval, then say what would happen to its width if the sample were four times as large.

Show the solution

SE = $12/\sqrt{36} = 2$, so the half-width is $2.030 \times 2 = 4.06$ and the interval is (45.94, 54.06).

Quadrupling n to 144 divides the standard error by $\sqrt{4} = 2$, giving SE = 1. The multiplier also shrinks slightly, to about 1.977.

Be careful what follows from this. A *different* sample of 144 would have its own mean and its own standard deviation, so you cannot predict the actual interval. What you can say is that the

width would be a little under half. If the mean and standard deviation happened to come out at 50 and 12 again, the interval would be roughly (48.02, 51.98).

The lesson to carry: to halve an interval you need **four times** the data. This is why precision gets expensive quickly.

Exercise 2 A colleague reports a 95% confidence interval of (12.1, 12.3) for a mean and says the estimate must be reliable because the interval is so narrow. Give two distinct reasons that conclusion does not follow.

 Show the solution

Any two of:

- The interval only accounts for sampling variation. If the sample was drawn from the wrong population, or people who chose not to respond differ from those who did, the interval is narrowly centred on a biased value.
- Systematic measurement error shifts every observation the same way and does not widen the interval at all — a miscalibrated instrument produces a beautifully tight interval around the wrong number.
- If the observations are not independent — several readings per person, say — the true standard error is larger than $s/\sqrt{n}$, so the interval is narrower than it should be and its real coverage is below 95%.
- Narrowness is partly just a large n . It says nothing about whether the measured quantity is the one the question was about.

Exercise 3 You have 40 patients, each measured before and after a treatment, and you want a bootstrap interval for the mean change. What exactly should you resample, and what goes wrong if you resample the 80 individual measurements instead?

 Show the solution

Resample the **40 patients**, each carrying its own before-and-after pair, and recompute the mean difference on each resample.

Resampling the 80 measurements independently destroys the pairing. A patient's "before" could be drawn without their "after", or paired with someone else's. That throws away exactly the structure a paired design exists to exploit — that the two readings on one person are related — and gives a miscalibrated interval.

Which way it is wrong depends on the data. When before and after are positively correlated, as they usually are, breaking the pairing re-introduces all the between-patient variation the design was built to remove, and the interval comes out too wide. If they were negatively correlated it could come out too narrow. The safe statement is that the number no longer means what it claims.

The general rule: **resample whatever unit was independently sampled.**

8.6 Check it in Python

The claim that 95% of intervals contain the truth is checkable, and checking it is the fastest way to make the definition stop feeling like a word game.

We treat the whole usable cohort as a population whose mean we happen to know. Then we take a great many small samples, build an interval from each, and count.

```

population = usable["final_score"].to_numpy()
true_mean = population.mean()

rng = np.random.default_rng(2026)
sample_size, trials = 25, 4_000

samples = rng.choice(population, size=(trials, sample_size), replace=True)
means = samples.mean(axis=1)
half_widths = (
    stats.t.ppf(0.975, df=sample_size - 1)
    * samples.std(axis=1, ddof=1) / np.sqrt(sample_size)
)
contains = (means - half_widths <= true_mean) & (true_mean <= means +
    - half_widths)

print(f"true mean of the population      : {true_mean:.2f}")
print(f"intervals built                    : {trials}")
print(f"intervals containing the truth : {contains.sum()}  ({100 *
    - contains.mean():.1f}%)")

```

```

true mean of the population      : 65.27
intervals built                  : 4000
intervals containing the truth : 3801  (95.0%)

```

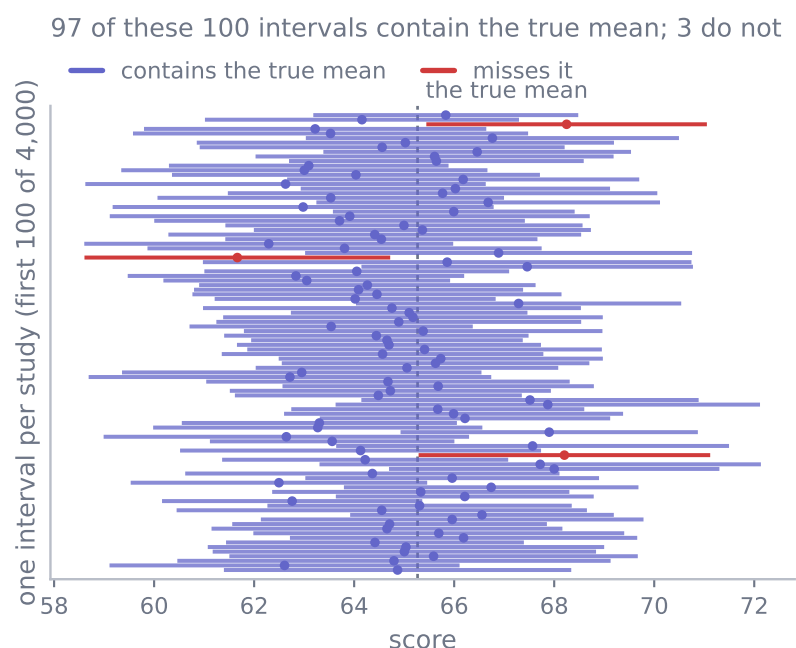


Figure 3: The first hundred of those intervals. Each bar is one study; the dotted line is the population mean, which in real life you would never know. The ones that miss are marked — and nothing about them looks different from the inside.

95.0% — close to 95.0% — close to the 95% the procedure promises, which is what you should expect when its assumptions roughly hold, as they do for this fairly symmetric population.

One simulation on one population does not prove the method always works. The t interval is not distribution-free, and on a badly skewed population with a small sample you would see coverage

noticeably below 95%. What this shows is that the promise is a real, checkable property rather than a form of words — and that you can check it yourself for any population you are worried about.

The important part is what the failures look like. Around 200 of those intervals missed entirely, and nothing about those 200 distinguishes them from the inside. Each was computed correctly from a legitimate sample; each simply landed somewhere unlucky. **You never know which kind you have**, and that is precisely why the 95% belongs to the procedure rather than to your interval.

Two variations worth running yourself. Change `0.975` to `0.995` and watch coverage rise towards 99% as the intervals widen. Change `sample_size` to 10 and watch the intervals get much wider while the coverage stays near 95% — the method keeps its promise, it just has less to work with.

8.7 Common wrong turns

Where this usually goes wrong

“There is a 95% probability the true mean lies in this interval” The commonest error in the subject. Before you collect data, the procedure has a 95% chance of producing an interval that covers the truth. After you have the numbers, the interval either covers it or it does not. The 95% describes the method across many hypothetical studies, not this one result.

Confusing the standard deviation with the standard error s describes how spread out the *observations* are. $s/\sqrt{n}$ describes how much the *mean* would move between samples. Adding data leaves the first roughly unchanged and shrinks the second.

Treating a narrow interval as a good analysis Width measures sampling precision only. Bias, bad design, dependence and bugs all leave the width untouched while making the centre wrong.

Expecting the bootstrap to fix a bad sample Every resample is drawn from your data. If the data are unrepresentative, so is all ten thousand of them. The bootstrap quantifies sampling variability; it cannot manufacture information that was never collected.

Resampling the wrong unit For paired, clustered or repeated-measures data, resample the independent unit — the patient, the school, the participant — never the individual rows.

Reporting the interval instead of the estimate An interval without its centre is hard to read, and a centre without its interval is overconfident. Report both, in the units of the thing measured.

8.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. A 95% interval for a mean is (4.2, 6.8). Which of these is a correct reading?
 - a. 95% of the observations lie between 4.2 and 6.8
 - b. There is a 95% probability the population mean is between 4.2 and 6.8
 - c. The interval came from a procedure that captures the population mean in 95% of repeated studies
 - d. 95% of future samples will have means between 4.2 and 6.8
2. You quadruple your sample size. Roughly what happens to the width of the interval, and why is it not halved by doubling instead?
3. Give one statistic for which you would reach for the bootstrap rather than a formula, and say why.
4. In the coverage simulation, about 200 intervals missed the true mean. Was anything wrong with

how those 200 were calculated?

5. A study measures the same 30 people on five occasions and treats all 150 readings as independent when computing a confidence interval. Is the interval likely to be too wide or too narrow?

 Show the answers

1. **(c).** (a) confuses the spread of the data with the precision of the mean — that would be roughly $\bar{x} \pm 2s$, not $\bar{x} \pm 2s/\sqrt{n}$. (b) is the standard misreading: after observation there is no probability left. (d) describes something else again, and is not what the procedure guarantees.
2. The width scales with $1/\sqrt{n}$, so quadrupling n roughly halves it. Doubling n divides the width by $\sqrt{2} \approx 1.41$, giving about a 29% reduction rather than 50%. Precision is bought at four times the cost you might expect.
3. Any statistic without a convenient closed-form standard error: a median, a correlation, an interquartile range, a ratio of two means, the difference between two 90th percentiles. The bootstrap needs only that you can compute the statistic on a resample.
4. **No.** Every one of them was computed correctly from a legitimate sample. They landed badly by chance, and that is the designed behaviour of a 95% procedure — a method that never missed would be far wider than necessary.
5. **Too narrow**, assuming — as is almost always the case for repeated measures — that readings on the same person are *positively* correlated. Five readings from one person then carry much less information than five readings from five people, so the effective sample size is nearer 30 than 150. Treating them as independent understates the standard error and the real coverage falls below 95%. Strictly the direction depends on the sign of that correlation; negative within-person correlation, which is rare, would push the other way.

Where this shows up next

S9 shows that a two-sided test and a confidence interval are the same calculation viewed from two sides: the test rejects exactly when the interval excludes the null value. **S10** builds intervals for the *difference* between two groups, which is almost always the quantity you actually care about. **S11** attaches an interval to a fitted slope, turning a line through a scatterplot into a claim with uncertainty attached.

9. Hypothesis tests

Statistics · Unit 9

S9 · Statistics Reading: 6 pages With practice: 2 hours Assumes: S8

Why this unit is here

A hypothesis test answers one narrow question: *if nothing were going on, how often would I see data at least as extreme as this?* That is a useful thing to know and it is much less than people take it for.

You need this unit for two reasons. Every paper you read for the rest of your degree reports p -values, and you cannot evaluate a claim you cannot decode. And the standard misreadings — that p is the probability the hypothesis is wrong, that a small p means a large effect, that a large p means no effect — are so widespread that being able to spot them is a genuine professional advantage.

The unit's other job is to show that a test and a confidence interval are the same calculation. Once you have seen that, the interval is almost always the better thing to report.

9.1 Before you start

You should be able to:

- build a confidence interval for a mean and say what the 95% refers to — S8;
- compute a standard error and say why it is not a standard deviation — S7;
- write a conditional probability and keep it the right way round — S4.

Quick check. A 95% confidence interval for a mean comes out as (58.6, 62.4). Is the value 65 a plausible value for the population mean?

 Show the answer

Not on this evidence: 65 lies outside the interval, so it is not among the values the data are consistent with at the 95% level.

You have just performed a hypothesis test. The rest of this unit is that sentence, made precise and given a number.

9.2 The idea

Take the 57 students who came from a life-sciences background. S8 found their mean final score to be 60.51 with a 95% interval of (58.58, 62.45).

Now suppose someone claims the population these students came from has a mean of 65 — the cohort's overall average. The logic of a test is:

1. **Assume the claim is true.** Call it the **null hypothesis**, $H_0 : \mu = 65$.
2. **Work out how the data would behave if it were.** From S7, sample means from such a population would be spread around 65 with a standard error of about 0.97.
3. **See how far out the observed data actually falls.**

4. **Report how often a sample that extreme or more would occur** if the claim were true. That number is the *p*-value.

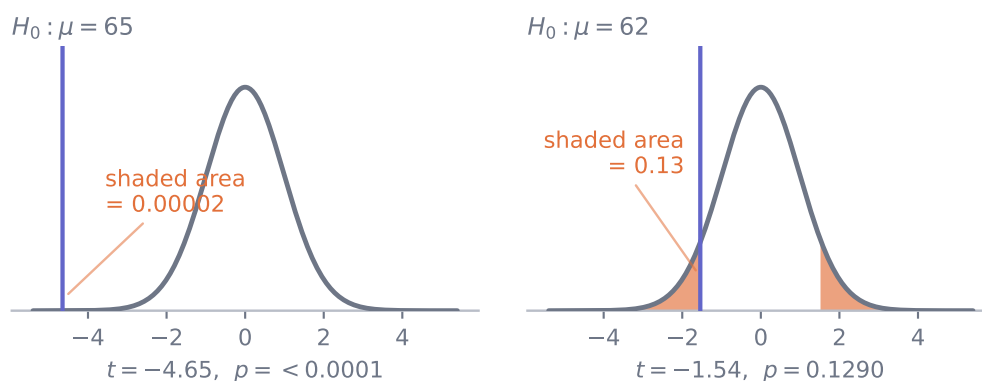


Figure 1: The same data against two null hypotheses. In each panel the curve is where the test statistic would fall if the null were true, the vertical line is where it actually fell, and the shaded area is the *p*-value — both tails, because the question was whether the mean differs, not whether it is smaller. Against 65 the observed value is far out in the tail. Against 62 it is unremarkable.

Against $H_0: \mu = 65$, data this far out would occur about twice in a hundred thousand samples. Against $H_0: \mu = 62$, it would occur about thirteen times in a hundred. The first claim looks hard to sustain; the second is perfectly consistent with what we saw.

Notice what has and has not been established. We have a statement about **how often data like this arises if the null is true**. We have said nothing about how likely the null is, and nothing at all about the second null being *correct* — only that these 57 students give no reason to doubt it.

9.3 The mechanics

The four pieces

Every test in the subject has the same four parts.

1. A **null hypothesis** H_0 : a specific claim precise enough to compute with. “The mean is 65”, “the two groups have the same mean”, “the slope is zero”.
2. An **alternative** H_1 , which fixes what counts as extreme. Two-sided ($\mu \neq 65$) is the default and the one to use unless you can justify otherwise.
3. A **test statistic**: a number summarising the distance between data and null, scaled by its own uncertainty. For a mean,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

4. A **null distribution**: where that statistic would fall if H_0 were true. For the t statistic it is the t distribution with $n - 1$ degrees of freedom, the same one S8 used to set interval widths.

The *p*-value is the probability, *computed under the null*, of a test statistic at least as extreme as the observed one.

What a p-value is, in symbols

$$p = P(\text{data at least this extreme} \mid H_0 \text{ true})$$

Everything that goes wrong with p -values goes wrong by moving the bar. It is **not** $P(H_0 \text{ true} \mid \text{data})$ — that is the reversal of S4, and getting from one to the other would need a prior probability that the test never supplies.

Three consequences worth stating explicitly.

A small p does not mean a large effect. It means the effect is distinguishable from zero given your sample size. With the cohort's spread of about 7.3 marks and 100,000 observations, a difference of a tenth of a mark gives $t = 4.3$ and $p \approx 10^{-5}$ — a decisive-looking result about a difference nobody could care about.

A large p does not mean no effect. It means your data cannot rule the null out. An underpowered study produces large p -values whether or not anything is going on, which is why “we found no significant difference” is nearly uninformative without the interval next to it.

$1 - p$ is not the probability the alternative is true. It is not a probability about any hypothesis at all.

Two ways to be wrong

Fix a threshold α — conventionally 0.05 — and reject H_0 when $p < \alpha$. Two mistakes are then possible.

Table 1: The two errors

	H_0 is true	H_0 is false
reject H_0	Type I error, rate α	correct
do not reject	correct	Type II error, rate β

Power is $1 - \beta$: the probability of detecting a real effect of a given size. It rises with the sample size and with the size of the effect, and falls as the data get noisier.

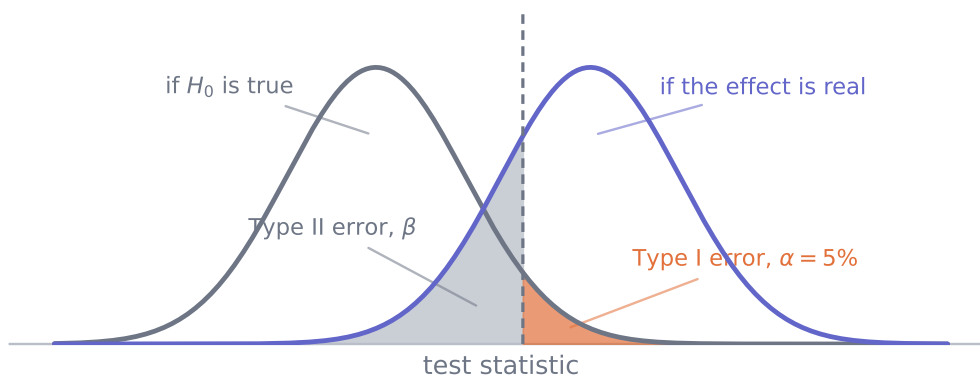


Figure 2: Two worlds. If the null is true the statistic is drawn from the left curve; if the real effect is the one on the right, from the right curve. The threshold is fixed by α alone, so the shaded slice of the left curve is 5% by construction. Everything of the right curve that falls short of the threshold is a missed real effect.

Note the asymmetry the picture makes obvious. You choose α ; you do **not** choose β , which follows from α , the sample size and the true effect. Shrinking α to be safer moves the line right and makes β worse.

A test and an interval are the same calculation

A two-sided test at level α rejects $H_0 : \mu = \mu_0$ **exactly when** μ_0 falls outside the $(1 - \alpha)$ confidence interval. They are not two methods that usually agree; they are one calculation read two ways.

$$\left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right| > t^* \iff \mu_0 \notin \left[\bar{x} - t^* \frac{s}{\sqrt{n}}, \bar{x} + t^* \frac{s}{\sqrt{n}} \right]$$

(The brackets are square because a μ_0 sitting exactly on an endpoint gives p exactly equal to α , and $p < \alpha$ does not reject. It is a boundary case of no practical consequence and of some pedagogical value: the two statements match exactly, not approximately.)

Both sides are the same inequality rearranged. This matters practically: the interval contains everything the test would tell you **and** the effect size and its precision, which the test throws away. Report the interval.

9.4 Worked example

Testing the life-sciences mean, and watching the duality hold.

```
import numpy as np
import pandas as pd
from scipy import stats

cohort = pd.read_csv("data/cohort.csv")
usable = cohort.dropna(subset=["background", "final_score"])
usable = usable[usable["final_score"].between(0, 100)]
life = usable[usable["background"] == "Life sciences"]["final_score"]
n = life.size
mean, se = life.mean(), life.std(ddof=1) / np.sqrt(n)

print(f"n = {n}    mean = {mean:.4f}    sd = {life.std(ddof=1):.4f}    se = {se:.4f}")
t_star = stats.t.ppf(0.975, n - 1)
low, high = mean - t_star * se, mean + t_star * se
print(f"95% interval for the mean: ({low:.4f}, {high:.4f})")
```

```
n = 57    mean = 60.5123    sd = 7.2893    se = 0.9655
95% interval for the mean: (58.5782, 62.4464)
```

Test the claim that this population's mean is 65, by hand and then with the library:

```
null = 65
t_stat = (mean - null) / se
p_value = 2 * stats.t.sf(abs(t_stat), n - 1)
print(f"by hand : t = {t_stat:.4f}    p = {p_value:.3e}")
print(f"scipy    : {stats.ttest_1samp(life, null)}")
```

```
by hand : t = -4.6481    p = 2.081e-05
scipy    : TtestResult(statistic=-4.648132564933887, pvalue=2.080866200706556e-05, df=56)
```

A p of 2×10^{-5} : if this population really had a mean of 65, a sample of 57 landing this far away would happen about twice in a hundred thousand studies. That is strong evidence against the claim.

The sentence to write down, though, is not the p -value. It is: *the mean final score of life-sciences students is estimated at 60.5 marks (95% CI 58.6 to 62.4), about 4.5 marks below the cohort average of 65.3*. That reports the size of the difference, its uncertainty, and — implicitly — the test.

Now the duality. Test every possible null value from 55 to 68 and see where the p -value crosses 0.05:

```
for candidate in [58.0, low, 59.0, 60.5, 62.0, high, 63.0]:
    t_c = (mean - candidate) / se
    p_c = 2 * stats.t.sf(abs(t_c), n - 1)
    verdict = "reject" if p_c < 0.05 else "do not reject"
    print(f"H0: mu = {candidate:>8.3f}    p = {p_c:.5f}    {verdict}")
```

H0: mu =	58.000	p = 0.01183	reject
H0: mu =	58.578	p = 0.05000	do not reject
H0: mu =	59.000	p = 0.12290	do not reject
H0: mu =	60.500	p = 0.98990	do not reject
H0: mu =	62.000	p = 0.12897	do not reject
H0: mu =	62.446	p = 0.05000	do not reject
H0: mu =	63.000	p = 0.01264	reject

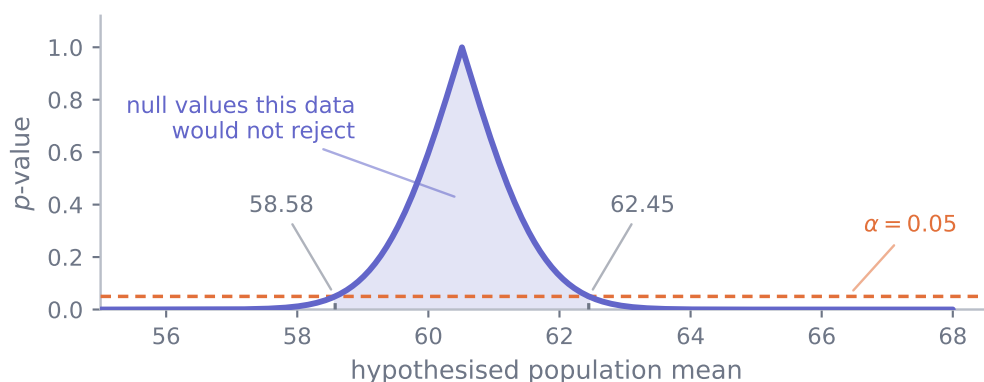


Figure 3: The p -value for every hypothesised mean between 55 and 68. It crosses 0.05 in exactly two places, and those two places are the ends of the 95% confidence interval — marked as vertical lines. The interval is the set of null values this data would not reject.

The two interval endpoints are in that list, and both come out at p exactly 0.05 — the curve crosses the threshold precisely there and nowhere else. Every null inside the interval survives; every null outside it is rejected. One calculation, two presentations.

Finally, power, because it is the thing that decides whether a non-significant result means anything at all.

```
sd = life.std(ddof=1)
effect = 2.0 # a difference of 2 marks
print(f"power to detect a {effect:.0f}-mark difference (sd {sd:.2f}, alpha = 0.05):")
for size in [10, 30, 57, 100, 200]:
    critical = stats.t.ppf(0.975, size - 1)
    ncp = (effect / sd) * np.sqrt(size)
```

```
power = stats.nct.sf(critical, size - 1, ncp) + stats.nct.cdf(-critical, size
- 1, ncp)
print(f"    n = {size:>3}    power {power:.3f}")
```

power to detect a 2-mark difference (sd 7.29, alpha 0.05):

```
n = 10    power 0.122
n = 30    power 0.306
n = 57    power 0.530
n = 100   power 0.776
n = 200   power 0.971
```

At $n = 57$ the study has a 53% chance of detecting a real 2-mark difference — barely better than a coin toss, and worse than that for anything smaller. If such a study reports $p = 0.3$, it has told you almost nothing, and saying “there is no difference” would be a serious overstatement.

9.5 Try it yourself

Exercise 1 A sample of 40 has mean 12.4 and standard deviation 3.0. Test $H_0 : \mu = 11$ against a two-sided alternative, using $t^* = 2.023$.

- Compute the standard error and the test statistic.
- Would you reject at $\alpha = 0.05$?
- Give the 95% interval, and check the two answers agree.

Show the solution

- $SE = 3.0/\sqrt{40} = 0.4743$, so $t = (12.4 - 11)/0.4743 = 2.951$.
- $|t| = 2.951 > 2.023$, so **reject** at the 5% level. (The exact p -value is about 0.005.)
- $12.4 \pm 2.023 \times 0.4743 = 12.4 \pm 0.960$, giving (11.44, 13.36). The null value 11 lies outside it, which is the same conclusion — as it must be, since the two calculations are the same inequality.
The interval is the better report: it says the data are consistent with any mean between 11.4 and 13.4, where the test says only “not 11”. (As always, the 95% belongs to the procedure — [S8](#) — not to this particular pair of endpoints.)

Exercise 2 Two studies test the same drug.

- Study A: 40 patients, mean improvement 5.0 points, 95% CI (0.2, 9.8), $p = 0.04$.
- Study B: 4,000 patients, mean improvement 0.3 points, 95% CI (0.1, 0.5), $p = 0.003$.

Which shows the stronger evidence, and which shows the larger effect?

Show the solution

Study B has the smaller p -value and by far the smaller effect. Its 0.3-point improvement is estimated precisely: the whole interval lies between 0.1 and 0.5, so the data have ruled out anything larger. Whether an improvement of that size is worth having is a clinical question, not a statistical one — but the interval is what puts the question in front of you, and the p -value of 0.003 does not. The tiny p reflects the sample size.

Study A shows the larger effect and knows almost nothing about it. The interval runs from 0.2 to 9.8, so the data are consistent with an improvement that is negligible and with one that is large. Its $p = 0.04$ crosses the conventional threshold and tells you very little.

The lesson: **p -values rank evidence against a null, not effects by importance.** Read the interval, in the units of the thing measured, and ask whether the values it contains would change what anyone does.

Exercise 3 A researcher runs a test, gets $p = 0.08$, and writes: “there is a 92% chance the treatment works”. Identify every error in that sentence and write a correct one.

💡 Show the solution

Three errors, stacked.

The conditional is reversed. p is the probability of data this extreme *if the null is true*, not the probability that the null is true given the data. Getting from one to the other requires a prior — see S4.

$1 - p$ is not the probability of the alternative. Even if p were a probability about the null — it is not — its complement would be about “not the null”, which includes effects of every size, most of them uninteresting.

“Works” is not what was tested. The null was almost certainly “no difference”. Rejecting it would be *evidence* of a difference — not proof, since 5% of tests reject a true null — and a difference is not the same thing as a benefit worth having.

A correct sentence, with the numbers the original omitted: “*The estimated effect was X units (95% CI a to b). Data this extreme would arise 8% of the time if the treatment had no effect at all, so these data do not rule out a null effect. What else they fail to rule out can be read off the interval — which is why it, and not the p -value, is the thing to report.*”

9.6 Check it in Python

The claim that a p -value is a probability computed under the null is testable: if the null really is true, p -values should be spread evenly between 0 and 1, and 5% of them should fall below 0.05.

```
rng = np.random.default_rng(2026)
trials, size = 20_000, 30
samples = rng.normal(loc=50, scale=10, size=(trials, size)) # null is TRUE: mu
               = 50
t_stats = (samples.mean(axis=1) - 50) / (samples.std(axis=1, ddof=1) /
               np.sqrt(size))
p_values = 2 * stats.t.sf(np.abs(t_stats), size - 1)

print(f"p below 0.05 : {(p_values < 0.05).mean():.4f}      (should be 0.05)")
print(f"p below 0.10 : {(p_values < 0.10).mean():.4f}      (should be 0.10)")
print(f"p below 0.50 : {(p_values < 0.50).mean():.4f}      (should be 0.50)")
```

```
p below 0.05 : 0.0494      (should be 0.05)
p below 0.10 : 0.0963      (should be 0.10)
p below 0.50 : 0.4965      (should be 0.50)
```

That is what $\alpha = 0.05$ buys: when nothing is going on, one test in twenty comes out significant anyway.

Those thousand “significant” results are not mistakes in anyone’s arithmetic — they are the designed behaviour of the procedure, and they are why S12 worries about how many tests were run.

Now the duality, checked exactly rather than by eye:

```
for candidate in np.linspace(55, 68, 27):
    rejects = 2 * stats.t.sf(abs((mean - candidate) / se), n - 1) < 0.05
    outside = not (low <= candidate <= high)
    assert rejects == outside, f"disagreement at mu0 = {candidate}"
print(f"test and interval agree at all 27 candidate values in [55, 68]")
print(f"interval ({low:.4f}, {high:.4f})")
```

```
test and interval agree at all 27 candidate values in [55, 68]
interval (58.5782, 62.4464)
```

Now generate data where the null is definitely **false** — a real effect of 2.5 marks — and run two thousand studies of the size we actually have:

```
truth, sd_pop = 62.5, 7.29 # the real mean is 2.5 marks above 60
studies = rng.normal(loc=truth, scale=sd_pop, size=(2000, 57))
p_values = stats.ttest_1samp(studies, 60.0, axis=1).pvalue
missed = np.flatnonzero(p_values >= 0.05)
print(f"real effect of 2.5 marks, n = 57: missed by {missed.size} of 2000 "
      f"studies ({100*missed.size/2000:.1f}%)")
```

```
real effect of 2.5 marks, n = 57: missed by 577 of 2000 studies (28.9%)
```

Nearly three studies in ten find nothing, and every one of them was correctly conducted on data containing a real effect. Take one of those studies and assert what a careless reader would assume — that a real effect shows up:

```
unlucky = studies[missed[0]]
result = stats.ttest_1samp(unlucky, 60.0)
assert result.pvalue < 0.05, (
    f"the null (mu = 60) is false by 2.5 marks, yet this study of 57 gives "
    f"p = {result.pvalue:.3f} and would 'find no significant difference'"
)
```

```
AssertionError: the null (mu = 60) is false by 2.5 marks, yet this study of 57
↳ gives p = 0.071 and would 'find no significant difference'
```

```
-----
AssertionError                                Traceback (most recent call last)
Cell In[11], line 3
      1 unlucky = studies[missed[0]]
      2 result = stats.ttest_1samp(unlucky, 60.0)
----> 3 assert result.pvalue < 0.05, (
      4     f"the null (mu = 60) is false by 2.5 marks, yet this study of 57
↳ gives "
      5     f"p = {result.pvalue:.3f} and would 'find no significant difference'"
      6 )
AssertionError: the null (mu = 60) is false by 2.5 marks, yet this study of 57
↳ gives p = 0.071 and would 'find no significant difference'
```

A non-significant result is consistent with no effect *and* with an effect the study was too small to see. The p -value alone cannot tell you which; only the interval and the power can.

9.7 Common wrong turns

Where this usually goes wrong

Reading p as the probability that the null is true p is computed *assuming* the null. It is $P(\text{a statistic at least this extreme} \mid H_0)$, and turning that into $P(H_0 \mid \text{data})$ needs a base rate the test does not have. This is S4's error in its most expensive form. (Note the "at least this extreme": for continuous data the probability of the exact values you observed is zero, so a p -value could not be that.)

Reading a non-significant result as "no effect" It means the data could not rule out the null. An underpowered study cannot rule out anything. Write "no significant difference was found ($p = 0.3$; the 95% interval runs from -4 to $+9$, so effects of either sign remain plausible)" — and notice how much more that says.

Confusing statistical significance with importance A large enough sample makes any non-zero difference significant. The interval, in real units, is what tells you whether the difference matters.

Choosing one-sided after seeing the data A one-sided test halves the p -value, which is exactly why it must be decided before looking. Choosing the direction afterwards is not a test at 5%; it is a test at 10% reported as 5%.

Testing many things and reporting the smallest p Twenty independent tests on pure noise produce a significant result about two-thirds of the time. If you ran many tests, say so and adjust — the machinery is in S12.

Treating $\alpha = 0.05$ as a law of nature It is a convention, and a fairly arbitrary one. There is nothing meaningfully different about $p = 0.049$ and $p = 0.051$, and reporting one as a discovery and the other as nothing is a failure of judgement, not a rule.

Reporting the test instead of the estimate The test discards the effect size and its precision, which are the two things a reader needs. Give the estimate with its interval; the test is a corollary.

9.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. A study reports $p = 0.03$. Which statement is correct?
 - a. There is a 3% chance the null hypothesis is true
 - b. There is a 97% chance the alternative is true
 - c. If the null were true, data at least this extreme would arise 3% of the time
 - d. The effect is small but real
2. A 95% confidence interval for a difference is $(-1.2, 8.6)$. Without computing anything, what do you know about the two-sided p -value for $H_0 : \text{difference} = 0$?
3. You lower α from 0.05 to 0.01. What happens to the two error rates?
4. In the simulation above, 5% of tests were significant when the null was true. Were those tests computed wrongly?
5. Study A: $p = 0.049$. Study B: $p = 0.051$. How different is the evidence?

 Show the answers

1. **(c).** That is the definition. (a) and (b) reverse the conditional; (d) confuses significance with size and cannot be read off a p -value at all — the effect could be enormous or trivial.
2. **It is greater than 0.05.** Zero lies inside the interval, and by the duality a two-sided test rejects exactly when the null value falls outside. You cannot tell *how much* greater without more information — an interval that only just contains zero corresponds to a p a little above 0.05, one that contains it centrally to a p near 1.
3. The **Type I error rate falls to 1%**, by construction — that is what α means. The **Type II error rate rises**, because the threshold has moved further out and more real effects now fall short of it. You cannot reduce both at once without more data.
4. **No.** Every one was computed correctly from data generated under the null. Five percent false positives is what a 5% threshold *is*. A procedure that never gave a false positive would also never detect anything.
5. **Essentially not at all.** The two studies provide near-identical evidence. The threshold is a convention placed at 0.05, and nothing about the underlying quantity changes as you step across it. Treating one as a finding and the other as a null result is the single most damaging convention in applied statistics, and it is a large part of why [S12](#) exists.

Where this shows up next

[S10](#) applies all of this to the comparison people actually want: two groups, and the difference between them. [S11](#) tests whether a fitted slope differs from zero, which is the same calculation again. [S12](#) asks what happens when many tests are run, and why a significant result in a low-prior field is more likely to be false than true.

10. Comparing groups, measuring association

Statistics · Unit 10

S10 · Statistics Reading: 6 pages With practice: ~ 2 hours Assumes: S9, S3

Why this unit is here

Almost every applied question is a comparison. Did the treated group do better than the untreated one? Do students from one background finish further ahead than students from another? Is the completion rate different for people who used the forum?

The quantity you care about is the **difference**, and the thing to report is that difference with an interval around it. This unit shows how to get one for two means and for two proportions, how to handle a table of counts, and — most importantly — why looking at two separate intervals and checking whether they overlap is not a substitute for computing the interval for the difference.

10.1 Before you start

You should be able to:

- read a p -value correctly and state the duality with a confidence interval — S9;
- build a confidence interval for a mean — S8;
- summarise the relationship between a categorical and a quantitative variable — S3.

Quick check. Two groups have means 40 and 46. Their standard errors are both 2. Roughly what is the standard error of the *difference* between the means?

 Show the answer

$$\sqrt{2^2 + 2^2} = \sqrt{8} = 2.83.$$

Not 2, and not 4. Standard errors, like standard deviations, combine through their squares — the variance rule from S5. Getting this wrong is the arithmetic behind the commonest mistake in the unit.

10.2 The idea

Take the two most different backgrounds in the cohort.

```
import numpy as np
import pandas as pd
from scipy import stats

cohort = pd.read_csv("data/cohort.csv")
usable = cohort.dropna(subset=["background", "final_score"])
usable = usable[usable["final_score"].between(0, 100)]

life = usable[usable["background"] == "Life sciences"]["final_score"].to_numpy()
```

```

maths = usable[usable["background"] == "Mathematics or
↳ Physics"]["final_score"].to_numpy()
for name, group in [("Life sciences", life), ("Maths or Physics", maths)]:
    se = group.std(ddof=1) / np.sqrt(group.size)
    print(f"{name:<18} n = {group.size:>3}    mean {group.mean():6.2f}    "
          f"sd {group.std(ddof=1):5.2f}    se {se:.2f}")

```

Life sciences	n = 57	mean 60.51	sd 7.29	se 0.97
Maths or Physics	n = 36	mean 75.28	sd 7.63	se 1.27

Two means, 60.51 and 75.28. The comparison people report is often “both means with their intervals, and look, they do not overlap”. That happens to work here, and in general it is the wrong tool.

The difference is a quantity in its own right. It has an estimate, a standard error and an interval, and none of the three can be read reliably off a picture of the two groups separately.

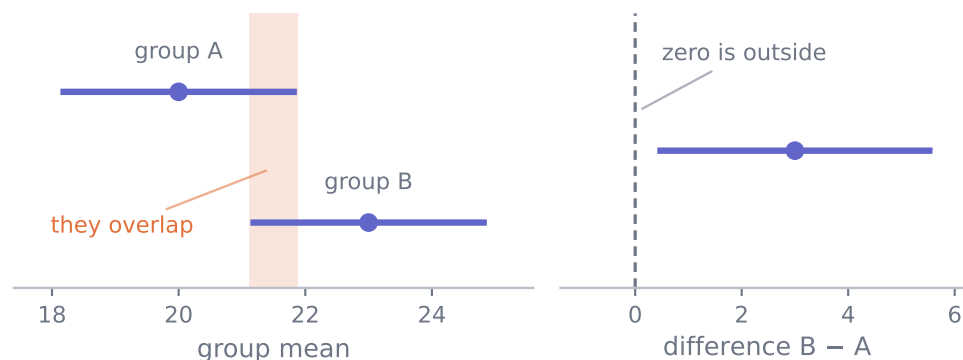


Figure 1: A constructed example, because the cohort’s own groups are too far apart to make the point. Two groups of 30 with means 20 and 23 and standard deviations of 5: their 95% intervals overlap, and yet the interval for the *difference* excludes zero and the test rejects at $p = 0.024$. Overlapping intervals do not mean no difference.

Two intervals that overlap can still correspond to a difference significantly away from zero, because the standard error of the difference is $\sqrt{SE_A^2 + SE_B^2}$ — always smaller than the sum $SE_A + SE_B$, which is roughly what the eyeball comparison uses.

The error runs one way, and it is worth knowing which. Non-overlapping intervals do imply a significant difference, so the eyeball test never invents a difference — it misses real ones. In the example above it only declares a difference when $|t| > 2.89$, which is a test at about the 0.5% level dressed up as one at 5%. Being conservative sounds safe until you remember that missing real differences is exactly what an underpowered analysis does.

10.3 The mechanics

Two means, unpaired

The estimate is $\bar{x}_A - \bar{x}_B$. Its standard error is

$$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}},$$

and the interval is the estimate plus or minus $t^* \text{ SE}$. The test statistic is the estimate divided by its standard error, exactly as in S9.

There are two versions of this test. The **pooled** (“Student”) version assumes the two populations have the same variance and combines them into one estimate. **Welch’s** version does not, and adjusts the degrees of freedom instead. Welch is the better default: it costs almost nothing when the variances really are equal and it is substantially better when they are not, and you never actually know which case you are in. `scipy.stats.ttest_ind` uses the pooled version unless you pass `equal_var=False`, which you usually should.

Two means, paired

If each observation in one group is matched to one in the other — the same person before and after, the same field split into two halves — then **do not use the two-sample test**. Take the differences within pairs and analyse those single numbers with a one-sample test.

This is not a technicality; it is usually a large gain. Pairing removes the between-subject variation, which is often the largest source of noise, so the paired analysis can be dramatically more precise. Treating paired data as unpaired throws that away — and, as S8 showed for the bootstrap, the rule is always to respect the unit that was independently sampled.

Two proportions

For a yes/no outcome, the estimate is $\hat{p}_A - \hat{p}_B$ and

$$\text{SE} = \sqrt{\frac{\hat{p}_A(1 - \hat{p}_A)}{n_A} + \frac{\hat{p}_B(1 - \hat{p}_B)}{n_B}},$$

using the Bernoulli variance $p(1 - p)$ from S6. The normal approximation behind that interval needs a reasonable number of successes *and* failures in each group — a common rule of thumb is at least ten of each — and it fails badly for rare events, where exact or score-based intervals should be used instead.

Report the difference in percentage points, and consider also reporting the ratio; they answer different questions and a change from 1% to 2% is either “one percentage point” or “double”, depending which you choose.

A table of counts: the chi-square test

With two categorical variables you have the contingency table of S3. The **chi-square test of independence** asks whether the table is compatible with the two variables being unrelated.

If they were independent, S4 says the expected count in a cell would be

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{n},$$

and the statistic accumulates the squared departures:

$$\chi^2 = \sum_{\text{cells}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Large values mean the observed table is far from what independence predicts. Two warnings. The

approximation needs the expected counts to be reasonably large — at least 5 in most cells is the usual rule — and the test tells you only *that* the table departs from independence, never how or by how much. Always look at the cell-by-cell departures as well.

When the assumptions are uncomfortable

The t -based methods lean on the sampling distribution of the mean being roughly normal, which S7 showed can fail for small, skewed samples. Two alternatives:

- the **bootstrap** of S8, applied to the difference — it assumes no shape at all and gives you an interval directly;
- a **rank-based test** such as Mann–Whitney, which asks about the probability that a randomly chosen value from one group exceeds one from the other. Note that it answers a different question: it is not a test about means, and it is not a general test of whether the two distributions differ — two distributions with the same centre and different spreads give it nothing to find.

10.4 Worked example

How far ahead do the Maths and Physics students finish?

```
difference = maths.mean() - life.mean()
se = np.sqrt(maths.var(ddof=1) / maths.size + life.var(ddof=1) / life.size)
welch = stats.ttest_ind(maths, life, equal_var=False)
t_star = stats.t.ppf(0.975, welch.df)

print(f"difference in means : {difference:.4f} marks")
print(f"standard error      : {se:.4f}")
print(f"95% interval        : ({difference - t_star*se:.4f}, "
      f"{difference + t_star*se:.4f})")
print(f"Welch t = {welch.statistic:.4f}  df = {welch.df:.2f}  p = "
      f"↪ {welch.pvalue:.3g}")
```

```
difference in means : 14.7655 marks
standard error      : 1.5969
95% interval        : (11.5821, 17.9489)
Welch t = 9.2461  df = 72.01  p = 7.28e-14
```

The sentence to report: **students arriving from Mathematics or Physics finished 14.8 marks ahead of students from life sciences (95% CI 11.6 to 17.9)**. That is an estimate, in the units of the thing measured, with its uncertainty attached. The p -value adds only that a gap this large would essentially never arise from 93 students if the two populations had the same mean — which the interval, sitting a long way from zero, had already made obvious.

Both versions of the test, for comparison, and a rank-based one:

```
print(f"Welch (unequal variances) p = {welch.pvalue:.3e}")
print(f"pooled (equal variances)  p = {stats.ttest_ind(maths,
↪ life).pvalue:.3e}")
print(f"Mann-Whitney                p = {stats.mannwhitneyu(maths,
↪ life).pvalue:.3e}")
```

Welch (unequal variances) $p = 7.277\text{e-}14$
 pooled (equal variances) $p = 6.028\text{e-}15$
 Mann-Whitney $p = 7.992\text{e-}12$

Here the group standard deviations are 7.63 and 7.29, so pooling changes almost nothing. The habit worth forming is using Welch anyway, because the case where it matters is exactly the case where you would not have noticed.

Now all five groups at once, which is what a reader actually wants to see:

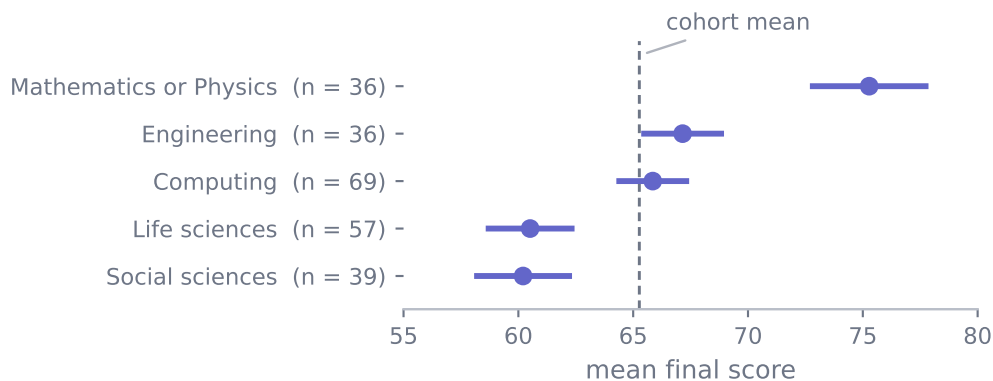


Figure 2: Every background's mean final score with a 95% interval, ordered. Reading pairwise comparisons off this picture is exactly what the first figure warned against — but as a description of where the five groups sit, and of how much less is known about the smaller ones, it is the right display.

A categorical outcome. Does the chance of scoring 70 or more depend on background at all?

```
table = pd.crosstab(usable["background"], usable["final_score"] >= 70)
table.columns = ["below 70", "70 or more"]
print(table.to_string())

result = stats.chi2_contingency(table)
print(f"\nchi-square = {result.statistic:.3f}    df = {result.dof}    "
      f"p = {result.pvalue:.3g}")
print(f"smallest expected count = {result.expected_freq.min():.2f}")
```

	below 70	70 or more
background		
Computing	49	20
Engineering	22	14
Life sciences	50	7
Mathematics or Physics	12	24
Social sciences	35	4

```
chi-square = 40.813    df = 4    p = 2.94e-08
smallest expected count = 10.48
```

The statistic is 40.8 on 4 degrees of freedom, $p \approx 3 \times 10^{-8}$: the table is a long way from independence. But the statistic alone would not have told you *how*. The departures do: Mathematics or Physics contributes 13.5 more high scorers than independence predicts, and life sciences 9.6 fewer.

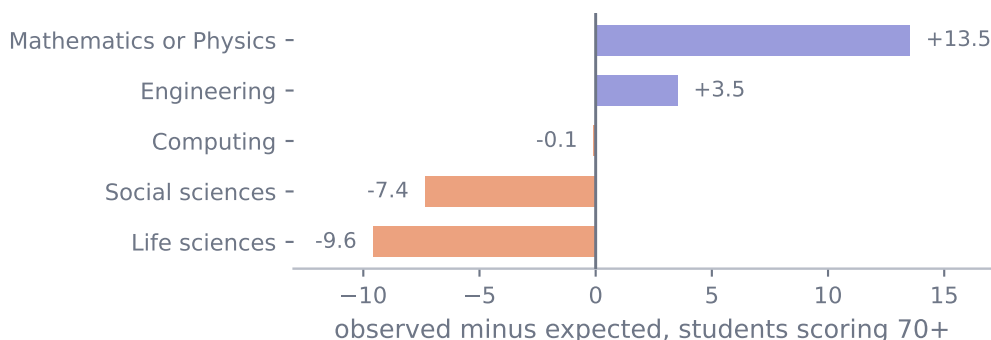


Figure 3: Observed counts minus the counts independence predicts, for the ‘70 or more’ column. Two groups have far more high scorers than independence would give them and two have far fewer — which is what the chi-square statistic adds up, and what its single number then hides.

10.5 Try it yourself

Exercise 1 Group A: $n = 50$, mean 12.0, sd 4.0. Group B: $n = 50$, mean 14.0, sd 5.0.

- Compute the standard error of the difference.
- Give an approximate 95% interval for the difference (use $t^* \approx 2$).
- Would you reject H_0 : no difference, at the 5% level?

💡 Show the solution

- $SE = \sqrt{4^2/50 + 5^2/50} = \sqrt{0.32 + 0.50} = \sqrt{0.82} = 0.906$.
- Take the difference as B minus A, so +2.0 — the sign is yours to choose and yours to state. $2.0 \pm 2 \times 0.906 = 2.0 \pm 1.81$, so (0.19, 3.81).
- Yes** — zero is outside the interval, so the test rejects. But notice how close it is: the interval reaches almost to zero, and the data are consistent with a difference as small as 0.2 units. If a difference of 0.2 would not matter in practice, “significant” is a very weak thing to have shown, and the interval is what makes that visible.

Exercise 2 A trial measures blood pressure in 40 patients before and after a treatment. Someone runs a two-sample t -test comparing the 40 “before” values with the 40 “after” values.

- What is wrong?
- What should they do?
- Which analysis is likely to give the narrower interval, and why?

💡 Show the solution

- The two samples are **not independent** — they are the same 40 people measured twice. The two-sample formula assumes independence between the groups, and here each “after” value is strongly related to its own “before” value.
- Compute the 40 within-patient differences and analyse those with a one-sample test or interval. One number per patient, 40 independent units.
- The paired analysis, usually by a wide margin.** The two-sample standard error is driven by how much patients differ from each other, which for blood pressure is a great deal. The

paired standard error is driven only by how much each patient *changed*, which is typically much smaller. Pairing removes the between-patient variation instead of measuring it. The exception is worth knowing: if before and after were *negatively* correlated, pairing would make things worse. That is rare, and the safe rule remains to analyse the design you actually ran.

Exercise 3 Two teaching methods are compared. Method A: 30 of 100 students passed. Method B: 45 of 100 passed.

- Compute the difference in proportions and its standard error.
- Give an approximate 95% interval.
- A colleague reports the result as “method B is 50% better”. Is that fair?

💡 Show the solution

- $\hat{p}_A = 0.30$, $\hat{p}_B = 0.45$. Taking the difference as B minus A — again, say which — it is +0.15.

$$SE = \sqrt{\frac{0.3 \times 0.7}{100} + \frac{0.45 \times 0.55}{100}} = \sqrt{0.0021 + 0.002475} = 0.0676.$$

- $0.15 \pm 1.96 \times 0.0676 = 0.15 \pm 0.133$, giving (0.017, 0.283) — 1.7 to 28.3 percentage points. Zero is outside, but only just, and the data are consistent with almost anything from a negligible improvement to a very large one.
- It is **arithmetically true and rhetorically misleading**. $0.45/0.30 = 1.5$, so the pass rate is indeed 50% higher in relative terms. But the same difference stated as “15 percentage points” sounds quite different, and the relative version conceals how uncertain it is: the interval on the *ratio* runs from about 1.05 to 2.1. Report the difference in percentage points, give the interval, and give the ratio too if it helps — never the ratio alone.

10.6 Check it in Python

First, that the standard error of a difference really does combine through squares. Simulate two independent groups and compare.

```
rng = np.random.default_rng(2026)
n_a, n_b, sd_a, sd_b = 40, 60, 8.0, 5.0
a = rng.normal(50, sd_a, size=(50_000, n_a)).mean(axis=1)
b = rng.normal(55, sd_b, size=(50_000, n_b)).mean(axis=1)

print(f"se of mean A      {a.std():.4f}    formula {sd_a/np.sqrt(n_a):.4f}")
print(f"se of mean B      {b.std():.4f}    formula {sd_b/np.sqrt(n_b):.4f}")
print(f"se of difference {(b - a).std():.4f}    "
      f"formula {np.sqrt(sd_a**2/n_a + sd_b**2/n_b):.4f}")
print(f"      NOT the sum of the two: {sd_a/np.sqrt(n_a) + "
      f"sd_b/np.sqrt(n_b):.4f}")
```

```
se of mean A      1.2635    formula 1.2649
se of mean B      0.6464    formula 0.6455
se of difference 1.4181    formula 1.4201
```

NOT the sum of the two: 1.9104

Now the check that fails, and it is the mistake the unit opened with. Assert that two overlapping 95% intervals mean the difference is not significant:

```
def make_group(centre, spread, size):
    """A sample of `size` values with exactly this mean and standard
    - deviation."""
    z = rng.normal(0, 1, size)
    return centre + spread * (z - z.mean()) / z.std(ddof=1)

group_a, group_b = make_group(20.0, 5.0, 30), make_group(23.0, 5.0, 30)

def own_interval(values):
    half = (stats.t.ppf(0.975, values.size - 1)
            * values.std(ddof=1) / np.sqrt(values.size))
    return values.mean() - half, values.mean() + half

lo_a, hi_a = own_interval(group_a)
lo_b, hi_b = own_interval(group_b)
overlap = lo_b <= hi_a
comparison = stats.ttest_ind(group_b, group_a, equal_var=False)

assert not (overlap and comparison.pvalue < 0.05), (
    f"intervals ({lo_a:.2f}, {hi_a:.2f}) and ({lo_b:.2f}, {hi_b:.2f}) overlap, "
    f"yet the difference has p = {comparison.pvalue:.3f}"
)
```

AssertionError: intervals (18.13, 21.87) and (21.13, 24.87) overlap, yet the
- difference has p = 0.024

```
-----
AssertionError                                Traceback (most recent call last)
Cell In[9], line 18
     15 overlap = lo_b <= hi_a
     16 comparison = stats.ttest_ind(group_b, group_a, equal_var=False)
--> 18 assert not (overlap and comparison.pvalue < 0.05), (
     19     f"intervals ({lo_a:.2f}, {hi_a:.2f}) and ({lo_b:.2f}, {hi_b:.2f})
-   overlap, "
     20     f"yet the difference has p = {comparison.pvalue:.3f}"
     21 )
```

AssertionError: intervals (18.13, 21.87) and (21.13, 24.87) overlap, yet the
- difference has p = 0.024

The two intervals share the range from 21.13 to 21.87, and the difference is significant at the 5% level anyway. The eyeball test is a *different, stricter* comparison: here it would have required the two means to differ by 3.73 rather than the 2.58 the proper test asks for. It agrees with the real answer much of the time and errs by staying silent, which is the failure mode that is easiest to miss.

Finally, the chi-square statistic, computed by hand against the library, so the formula stops being a black box:

```

observed = table.to_numpy()
row_totals = observed.sum(axis=1, keepdims=True)
col_totals = observed.sum(axis=0, keepdims=True)
expected_by_hand = row_totals * col_totals / observed.sum()
chi_by_hand = ((observed - expected_by_hand) ** 2 / expected_by_hand).sum()

print(f"by hand {chi_by_hand:.6f}    scipy {result.statistic:.6f}    "
      f"agree: {np.isclose(chi_by_hand, result.statistic)}")
print(f"df = (rows - 1)(cols - 1) = "
      f"{(observed.shape[0]-1)*(observed.shape[1]-1)}    scipy says {result.dof}")

```

```

by hand 40.812864    scipy 40.812864    agree: True
df = (rows - 1)(cols - 1) = 4    scipy says 4

```

10.7 Common wrong turns

Where this usually goes wrong

Comparing two intervals by eye instead of computing the difference Overlapping intervals can hide a significant difference and non-overlapping ones can exaggerate. The difference has its own standard error, $\sqrt{SE_A^2 + SE_B^2}$, and its own interval. Compute it.

Adding standard errors instead of adding their squares $SE_A + SE_B$ is always too big. Square, add, take the root — the variance rule of [S5](#).

Using a two-sample test on paired data Before-and-after on the same people, matched pairs, two measurements per site. Take the within-pair differences and analyse those. Ignoring the pairing throws away the design's whole advantage.

Reporting only the p -value for a comparison “The groups differed significantly ($p < 0.001$)” tells the reader nothing about *how much*. Give the difference in real units with its interval; that is the finding.

Treating a significant chi-square as an explanation It says the table departs from independence and nothing more. Look at the cell-by-cell departures, and be careful with small expected counts, where the approximation is unreliable.

Confusing a rank test with a test about means Mann–Whitney compares distributions, not means. Two distributions with the same mean can differ by it, and two with different means need not. Report what you actually tested.

Comparing many groups and reporting the biggest gap Five groups make ten pairwise comparisons, and with ten tests at 5% a “significant” result is likely from noise alone. Say how many comparisons you made — [S12](#) says what to do about it.

10.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

- Two group means have standard errors 3 and 4. What is the standard error of their difference?
 - 1
 - 5
 - 7
 - 25
- Two 95% intervals for group means overlap slightly. What can you conclude about the difference?

3. When would you prefer Welch's test to the pooled two-sample test?
4. A chi-square test on a 4×3 table gives $p = 0.001$. What have you learned?
5. A study reports that a treatment "halves the risk". The risk went from 0.02% to 0.01%. Is the reporting fair?

 Show the answers

1. **(b), 5.** $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$. Option (c) adds the standard errors, which is what you would get only if the two estimates were perfectly *negatively* correlated — the covariance term $-2\rho SE_A SE_B$ is largest when $\rho = -1$; (d) is the variance, not the standard error.
2. **Almost nothing on its own.** Overlap is compatible with a difference that is significant and with one that is not; the answer depends on the two sample sizes and standard deviations. Compute the interval for the difference — it takes one line and it is the quantity of interest anyway.
3. **Essentially always.** Welch does not assume equal variances, costs almost nothing when they happen to be equal, and is much safer when they are not — particularly when the group sizes also differ, which is when the pooled test goes wrong most badly. The one thing you should not do is test for equal variances first and choose on the result, which distorts the error rate of everything downstream.
4. That the table is **inconsistent with the two variables being independent**. You have not learned which cells drive it, in which direction, or how large the departure is in any practical sense — and with a large enough table and sample, a trivial departure produces a small p . Look at observed minus expected next.
5. **Not on its own.** The relative statement is true and the absolute change is one in ten thousand. Both should be given: "the risk fell from 0.02% to 0.01%, a halving, or one case avoided per 10,000 people treated." A relative risk with no baseline is the standard way to make a negligible effect sound dramatic.

Where this shows up next

S11 replaces the two-group comparison with a line, which lets the explanatory variable be a number rather than a label — and lets you adjust for a second variable at the same time. **S12** returns to the ten pairwise comparisons this unit resisted making, and to what happens when a difference between groups is really a difference in who is in them.

11. Simple linear regression

Statistics · Unit 11

S11 · Statistics Reading: 6 pages With practice: ☒ 2 hours Assumes: S10, M11

Why this unit is here

S10 compared groups. Regression does the same job when the explanatory variable is a *number* rather than a label — and it gives you the answer in the units people actually want: not “these groups differ”, but “**each extra hour is worth this many marks, give or take this much**”.

M11 already showed how the line is computed. This unit supplies the part that makes it a statistical statement rather than an arithmetic one: the slope is an estimate, it has a standard error, and there is a specific list of assumptions under which that standard error means what it says. It also supplies the two numbers people most often misuse — R^2 and the prediction interval.

11.1 Before you start

You should be able to:

- fit a straight line by least squares and say what it minimises — **M11**;
- build a confidence interval and read a p -value — **S9, S8**;
- interpret a correlation and say what it cannot see — **S3**.

Quick check. A fitted line is $\hat{y} = 70 - 0.24x$, where y is a final score in marks and x is hours of practice. What are the units of -0.24 , and what does the 70 mean?

💡 Show the answer

The slope is in **marks per hour**: the fitted line falls by 0.24 marks for each extra hour. The intercept is the fitted value at $x = 0$ — 70 marks for a student who practised not at all. Whether that intercept means anything depends on whether $x = 0$ is inside the range of the data. Here the smallest observed practice time is one hour, so 70 is an extrapolation, and one to be careful with.

11.2 The idea

The model behind a fitted line is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where β_0 and β_1 are fixed unknown numbers — parameters, in the sense of **S1** — and ε_i is everything else: measurement noise, omitted causes, individual variation. The ε_i are assumed to have mean zero, the same variance for every i , and to be independent of each other.

Least squares gives you $\hat{\beta}_0$ and $\hat{\beta}_1$, and the crucial point of this unit is that those hats are doing the same work they did in **S1**: **they are estimates, computed from one sample, and a different sample would have produced different ones.**

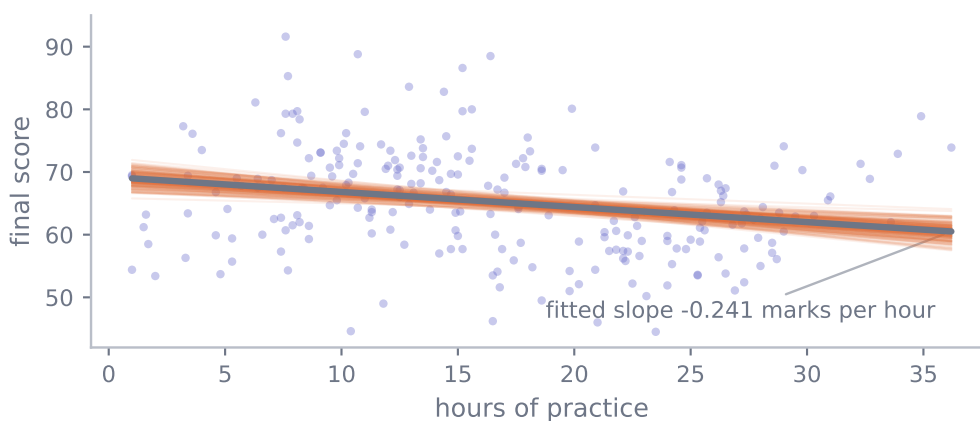


Figure 1: Two hundred lines, each fitted to a resample of the 238 students, drawn faintly behind the fit to the real data. The fan they trace out is what the standard error of the slope measures. A single line drawn on a scatterplot hides this completely, which is why the slope must be reported with an interval.

Every one of those faint lines is a slope somebody could have reported, from data generated the same way. Some are noticeably steeper than the fit and some are noticeably flatter. Quantifying that fan is what the rest of the unit does.

11.3 The mechanics

The estimates

From M11, least squares gives

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = r \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The second form for the slope is worth carrying: it is the correlation of S3, converted from unitless into marks-per-hour by the ratio of the two standard deviations. Correlation and slope carry the same sign and answer different questions.

The uncertainty

Let s_e be the **residual standard deviation**, the typical size of a vertical miss:

$$s_e = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}.$$

The divisor is $n - 2$ because two parameters have been estimated from the same data — the same argument that gave S2 its $n - 1$. Then

$$SE(\hat{\beta}_1) = \frac{s_e}{\sqrt{\sum(x_i - \bar{x})^2}}.$$

Read what makes a slope precise. **Small residuals** help, obviously. So does a **large n** . And so does a **wide spread of x** — the denominator is the total spread of the explanatory variable, so measuring people whose practice hours vary a lot pins the slope down far better than measuring the same number

of people who all practised about the same amount. That is a design lesson, and it is invisible unless you look at the formula.

The interval and test then follow the pattern of S9 exactly:

$$\hat{\beta}_1 \pm t_{n-2}^* \text{SE}(\hat{\beta}_1), \quad t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}.$$

R^2 , and what it is not

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

— the proportion of the variance in y that the fitted line accounts for. In simple regression it is exactly the square of the correlation.

Three things it does **not** tell you. It does not say the model is correct: a curved relationship can produce a respectable R^2 while the line is wrong everywhere. It does not say the slope is useful: a slope can be precisely estimated, obviously real, and account for very little of the variation, which is the normal situation with human data. And it does not say anything about cause.

A low R^2 with a tight interval on the slope means “this variable matters a little, and we know that quite precisely” — a perfectly respectable finding, and the one this unit’s worked example produces.

Two kinds of interval

Asking “what is the fitted value at x_0 ?” has two different answers, and confusing them is the commonest error in applied regression.

- A **confidence interval for the mean response** says where the *average* y for everyone at x_0 lies. It shrinks as n grows.
- A **prediction interval for a new observation** says where *one new individual* at x_0 is likely to fall. It carries the residual scatter as well, so it stays wide however much data you collect.

$$\text{SE}_{\text{mean}} = s_e \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}, \quad \text{SE}_{\text{pred}} = s_e \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

The extra 1 under the second root is the individual’s own noise, and it is usually the dominant term.

The assumptions, and how to check them

Table 1: What to check, and where to look

assumption	how it fails	how to look
the relationship is linear	curvature	residuals against fitted values
constant variance	a fan shape	residuals against fitted values
errors independent	clustering, time trends	know the design; plot in time order
errors roughly normal	skew, heavy tails	Q-Q plot of residuals

Only the last is about normality, and it is the least important of the four: the intervals rely on the sampling distribution of $\hat{\beta}_1$, which S7’s central limit theorem makes approximately normal for moderate n

even when the errors are not. Independence is the one that will hurt you most and shows up in no plot.

11.4 Worked example

How much is an hour of practice worth?

```
import numpy as np
import pandas as pd
from scipy import stats

cohort = pd.read_csv("data/cohort.csv")
scored = cohort[cohort["final_score"].between(0, 100)]
hours = scored["hours_practice"].to_numpy()
score = scored["final_score"].to_numpy()
n = hours.size
fit = stats.linregress(hours, score)

print(f"n = {n}")
print(f"slope      {fit.slope:+.4f} marks per hour      (se {fit.stderr:.4f})")
print(f"intercept {fit.intercept:+.4f} marks          (se
  ↳ {fit.intercept_stderr:.4f})")
t_star = stats.t.ppf(0.975, n - 2)
print(f"95% CI for the slope: ({fit.slope - t_star*fit.stderr:+.4f}, "
      f"{fit.slope + t_star*fit.stderr:+.4f})")
print(f"t = {fit.slope/fit.stderr:.4f}    p = {fit.pvalue:.3g}    R^2 =
  ↳ {fit.rvalue**2:.4f}")
```

```
n = 238
slope      -0.2408 marks per hour      (se 0.0652)
intercept +69.2261 marks          (se 1.1948)
95% CI for the slope: (-0.3693, -0.1123)
t = -3.6921    p = 0.000276    R^2 = 0.0546
```

The reportable sentence: **across the cohort, each additional hour of practice goes with 0.24 marks fewer on the final score (95% CI 0.11 to 0.37 marks lower), and practice hours account for 5.5% of the variation in scores.**

Three separate things are true of that sentence and worth separating.

The slope is negative and the interval excludes zero. This is not noise.

The effect is small and the fit is poor. $R^2 = 0.055$ means practice hours explain about a twentieth of the variation. Read together with the interval, the finding is “a real but modest association, estimated reasonably precisely” — which is what most findings about people look like.

It is not a statement about what practice does. S1 established that the sign reverses inside every background group. The fit above is a correct summary of the pooled cloud and a bad guide to any student’s decision.

```
fitted = fit.intercept + fit.slope * hours
residuals = score - fitted
s_e = np.sqrt((residuals**2).sum() / (n - 2))
sxx = ((hours - hours.mean())**2).sum()

print(f"residual sd s_e    {s_e:.4f}    (sd of the scores themselves
      ↳ {score.std(ddof=1):.4f})")
print(f"se by hand        {s_e/np.sqrt(sxx):.6f}")
print(f"se from scipy     {fit.stderr:.6f}")
print(f"R^2 by hand        {1 - (residuals**2).sum()/((score -
      ↳ score.mean())**2).sum():.6f}")
```

```
residual sd s_e    8.1693    (sd of the scores themselves 8.3841)
se by hand         0.065219
se from scipy      0.065219
R^2 by hand        0.054606
```

The residual standard deviation is 8.17 against the scores' own 8.38. Knowing a student's practice hours has bought you almost nothing about where they will land — which is the same fact as $R^2 = 0.055$, in marks rather than as a proportion.

The two intervals, drawn.

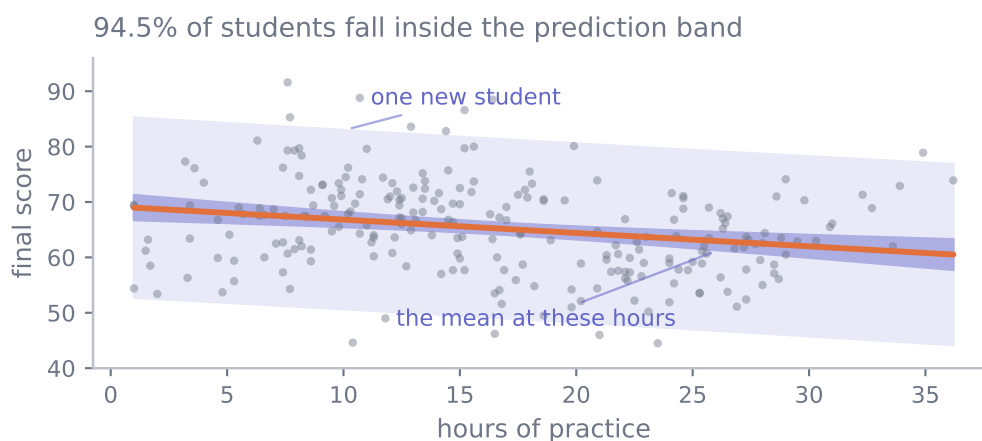


Figure 2: The fitted line with two bands. The narrow inner band is a 95% confidence interval for the *mean* score of students at each level of practice — it is not about where individual students fall, and most of them are outside it. The wide outer one is a 95% prediction interval for a single student, and 94.5% of the observed students sit inside it. Those students were used to fit the line, so that figure is a sanity check rather than an independent test.

```
x0 = 20.0
leverage0 = 1/n + (x0 - hours.mean())**2 / sxx
predicted = fit.intercept + fit.slope * x0
print(f"at {x0:.0f} hours, the fitted value is {predicted:.2f} marks")
print(f" 95% CI for the mean score there : "
      f"({predicted - t_star*s_e*np.sqrt(leverage0):.2f}, "
      f"{predicted + t_star*s_e*np.sqrt(leverage0):.2f})")
print(f" 95% prediction for one student : "
```

```
f"({predicted} - t_star*s_e*np.sqrt(1 + leverage0):.2f}, "  
f"({predicted} + t_star*s_e*np.sqrt(1 + leverage0):.2f})")
```

at 20 hours, the fitted value is 64.41 marks

95% CI for the mean score there : (63.27, 65.55)

95% prediction for one student : (48.28, 80.54)

The mean is pinned down to about a mark either way. An individual student is pinned down to about sixteen marks either way. Quoting the first when you meant the second is the standard way a regression is oversold.

Check the assumptions.

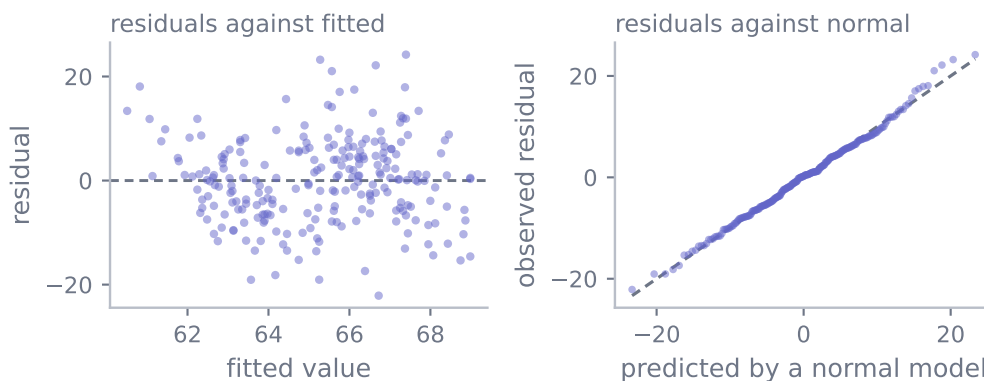


Figure 3: Diagnostics. Left: residuals against fitted values, which should show no pattern — no curve, no fan. Right: a Q-Q plot of the residuals against a normal model. Both look acceptable here, which clears the two assumptions these plots can see and says nothing about the two they cannot.

No curvature, no fan, and residuals that track a normal model closely. Linearity, constant variance and approximate normality are all in reasonable shape.

Neither plot can check the other two things that matter. **Independence** is a property of the design, not of the residuals: you establish it by knowing how the data were collected. And no residual plot at all can tell you that a variable is **missing** from the model — the fit can look immaculate and still be measuring something other than what you meant. That is precisely this dataset's situation, and the next section shows what it costs.

11.5 Try it yourself

Exercise 1 A regression of weekly sales (in £000) on advertising spend (in £000) gives $\hat{\beta}_1 = 3.2$ with a standard error of 1.5, from $n = 40$ shops.

- Give an approximate 95% interval for the slope.
- Interpret the slope in words, with units.
- The R^2 is 0.11. What does that add?

 Show the solution

- (a) With $t_{38}^* \approx 2.02$: $3.2 \pm 2.02 \times 1.5 = 3.2 \pm 3.03$, giving (0.17, 6.23). Zero is outside — just.
- (b) Each extra £1,000 of advertising goes with about **£3,200 more in weekly sales**, on average across shops. The interval says the data are consistent with anything from £170 to £6,230, which spans “not worth doing” and “extremely worth doing”.
- (c) $R^2 = 0.11$ says advertising spend accounts for 11% of the variation in sales between shops. The other 89% is location, size, season, competition and everything else. It does not weaken the slope estimate, and it does warn you that predicting one shop’s sales from its advertising alone will be poor.
- (The three numbers are not independent, incidentally: in simple regression $t^2 = (n - 2)R^2 / (1 - R^2)$, so a slope, its standard error, the sample size and R^2 determine each other. Here $t = 3.2/1.5 = 2.13$ and $R^2 = 2.13^2 / (2.13^2 + 38) = 0.11$, which is where the figure came from.)

Exercise 2 Which of these would you expect to make the standard error of a fitted slope smaller?

- (a) Collecting more observations.
 (b) Choosing x values spread over a wider range.
 (c) Reducing measurement noise in y .
 (d) Adding a constant to every x .

 Show the solution

(a), (b) and (c). Look at $SE(\hat{\beta}_1) = s_e / \sqrt{\sum (x_i - \bar{x})^2}$:

- more observations add terms to the sum in the denominator — though only if they add *spread*: an observation placed exactly at $\bar{x}$ contributes nothing to it, and one with a wild y can inflate s_e enough to make the standard error worse. In the ordinary case of more observations across the same range of x , it shrinks;
- a wider spread of x makes each term larger — this is often the cheapest of the three and the one people forget;
- less noise in y shrinks s_e .

(d) does nothing. $\sum (x_i - \bar{x})^2$ is unchanged by a shift, because $\bar{x}$ shifts with it. The slope estimate is unchanged too; only the intercept moves, since it is the fitted value at the new $x = 0$.

A caution on (b): widening the range only helps if the relationship really is straight over the wider range. Extending the x values into territory where the line stops being a good model buys precision in the estimate of the wrong thing.

Exercise 3 Using this unit’s fitted line, a student who has practised 20 hours is predicted to score 64.4 marks.

- (a) A colleague says “so a student who practises 20 hours will get about 64”. What is wrong?
 (b) The same line predicts 69.2 marks at zero hours. What is wrong with that?
 (c) Would it be reasonable to advise a student to practise less?

💡 Show the solution

- (a) It confuses the **mean response with an individual prediction**. The average score among students at 20 hours is pinned down to about (63.3, 65.6). A single student's score has a 95% prediction interval of about (48.3, 80.5) — a range of 32 marks, covering a fail and a first. The fitted value is a centre, not a forecast.
- (b) It is an **extrapolation**. No student in the data practised fewer than one hour, so nothing in the sample speaks to zero hours. The intercept is a mathematical feature of the line, not an estimate of anything observed.
- (c) **No**, and this is the important one. The fit describes an association across a cohort in which practice hours are entangled with prior background: students who arrive stronger practise less. Within every background group the slope is *positive*, at about one mark per hour. Advising an individual requires the within-group relationship, and even that is observational — students chose their own hours. S12 takes this apart properly.

11.6 Check it in Python

First, that the slope really is the correlation rescaled, and that the “by hand” standard error matches the library:

```
r = np.corrcoef(hours, score)[0, 1]
print(f"slope from r * sy/sx   {r * score.std(ddof=1) / hours.std(ddof=1):+.6f}")
print(f"slope from scipy      {fit.slope:+.6f}")
print(f"R^2 from r squared     {r**2:.6f}    from scipy {fit.rvalue**2:.6f}")
```

```
slope from r * sy/sx   -0.240793
slope from scipy      -0.240793
R^2 from r squared     0.054606    from scipy 0.054606
```

Second, the claim that the standard error measures the fan in the opening figure. Resample the students, refit, and compare the spread of the resulting slopes with the formula:

```
rng = np.random.default_rng(11)
slopes = np.empty(4000)
for i in range(slopes.size):
    pick = rng.integers(0, n, n)
    slopes[i] = stats.linregress(hours[pick], score[pick]).slope

print(f"sd of 4000 resampled slopes {slopes.std():.6f}")
print(f"formula se                    {fit.stderr:.6f}")
print(f"bootstrap 95% interval        ({np.quantile(slopes, 0.025):+.4f}, "
    f"{np.quantile(slopes, 0.975):+.4f})")
print(f"formula 95% interval          ({fit.slope - t_star*fit.stderr:+.4f}, "
    f"{fit.slope + t_star*fit.stderr:+.4f})")
```

```
sd of 4000 resampled slopes  0.065154
formula se                   0.065219
bootstrap 95% interval       (-0.3724, -0.1186)
formula 95% interval         (-0.3693, -0.1123)
```

Two completely different routes — one algebraic, one brute force — agreeing to three decimal places.

Now the check that fails, and it is the one this dataset exists to teach. A pooled slope is not the slope within any group:

```
grouped = scored.dropna(subset=["background"])
by_group = {name: stats.linregress(part["hours_practice"],
    ↳ part["final_score"]).slope
    for name, part in grouped.groupby("background")}
for name, slope in by_group.items():
    print(f" {name:<24} {slope:+.4f} marks per hour")

assert all(slope < 0 for slope in by_group.values()), (
    f"the pooled slope is {fit.slope:+.4f}, but every one of the "
    f"{len(by_group)} within-group slopes is positive"
)
```

Computing	+0.6624 marks per hour
Engineering	+0.8454 marks per hour
Life sciences	+1.1526 marks per hour
Mathematics or Physics	+1.0112 marks per hour
Social sciences	+1.3445 marks per hour

AssertionError: the pooled slope is -0.2408, but every one of the 5 within-group
↳ slopes is positive

AssertionError Traceback (most recent call last)

Cell In[9], line 7

```
4 for name, slope in by_group.items():
5     print(f" {name:<24} {slope:+.4f} marks per hour")
----> 7 assert all(slope < 0 for slope in by_group.values()), (
8     f"the pooled slope is {fit.slope:+.4f}, but every one of the "
9     f"{len(by_group)} within-group slopes is positive"
10 )
```

AssertionError: the pooled slope is -0.2408, but every one of the 5 within-group
↳ slopes is positive

Pooled, the line falls by 0.24 marks an hour. Within each background it rises by between 0.66 and 1.34 marks an hour. Both are correct least-squares fits to the data in front of them.

Adding the background information to the model — as [M11](#) does with the arrival diagnostic — flips the sign of the practice coefficient:

```
design = np.column_stack([np.ones(n), hours,
    ↳ scored["diagnostic_score"].to_numpy()])
theta, *_ = np.linalg.lstsq(design, score, rcond=None)
print(f"score ~ 1 + hours           slope {fit.slope:+.4f}")
print(f"score ~ 1 + hours + diagnostic slope {theta[1]:+.4f}")
```

score ~ 1 + hours	slope -0.2408
score ~ 1 + hours + diagnostic	slope +0.9156

Same data, same method, opposite sign. Which coefficient answers your question depends entirely on what you were asking, and nothing in the output tells you which one you wanted.

11.7 Common wrong turns

Where this usually goes wrong

Reporting a slope without an interval A slope is an estimate. “0.24 marks per hour” and “0.24 marks per hour, 95% CI 0.11 to 0.37” are different claims, and only the second can be argued with.

Confusing a confidence band with a prediction band The first is about the mean at a given x and is narrow; the second is about one new individual and is much wider, because it carries the residual scatter. In this unit’s fit they differ by a factor of about fourteen.

Reading a low R^2 as “no relationship” $R^2 = 0.055$ here, with a slope whose interval excludes zero. Small and real are compatible, and with human data they are the normal case.

Reading a high R^2 as “the model is right” A curved relationship can produce a high R^2 with a line that is wrong everywhere. Look at the residuals against the fitted values, always.

Interpreting the intercept outside the data The intercept is the fitted value at $x = 0$. If no observation is near zero, it is an extrapolation and usually meaningless — sometimes physically impossible.

Predicting outside the range of x The line is a summary of the region you observed. Nothing in the data says the relationship continues, and the further out you go the more confidently wrong the prediction becomes.

Reading the slope as an effect A regression coefficient describes an association in the data you fitted, given the other variables in the model. Making it causal needs a design that supports the claim, not a smaller p -value.

11.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. A fitted slope is 2.0 with standard error 0.5, from $n = 100$. Which is true?
 - a. The relationship explains most of the variation
 - b. A 95% interval for the slope is roughly (1.0, 3.0)
 - c. The correlation is 2.0
 - d. Each unit of x causes 2 units of y
2. You want a more precise estimate of a slope but cannot collect more data. What else could you change about the design?
3. Two regressions have identical slopes and identical standard errors, but R^2 of 0.9 and 0.2. What differs?
4. Why does a prediction interval not shrink towards zero width as n grows?
5. The cohort’s pooled slope is -0.24 and every within-background slope is positive. Which is the “real” relationship between practice and score?



Show the answers

1. **(b).** With $n = 100$ the multiplier is close to 2, so the interval is about 2.0 ± 1.0 . (a) confuses the slope with R^2 — nothing here says how much variation is explained; (c) confuses the slope with the correlation, which cannot exceed 1; (d) upgrades an association to a cause, which no regression output can license.
2. **Spread the x values further apart.** The standard error is $s_e / \sqrt{\sum (x_i - \bar{x})^2}$, so choosing

observations across a wider range of x shrinks it without a single extra data point. Reducing measurement noise in y works too. The caveat from Exercise 2 stands: the relationship must stay straight over the wider range.

3. **The residual scatter relative to the spread of y .** The regression with $R^2 = 0.2$ has much more variation in y that the line does not account for. Since the standard errors match, both slopes are estimated equally precisely — which is a clean demonstration that R^2 and the precision of a coefficient are separate things.
4. Because it contains the individual's own noise. As n grows the *line* is pinned down and the mean-response band shrinks towards nothing, but a new student still scatters about that line with standard deviation s_e . The prediction interval converges to roughly $\hat{y} \pm 2s_e$, not to a point.
5. **Neither is “the” relationship; they answer different questions.** The pooled slope describes the cohort as a whole, in which practice hours partly stand in for prior background. The within-group slopes describe students compared with others who arrived in a similar position. For advising a student the within-group figure is much closer to the question — and, since nobody assigned practice hours, still not a causal answer. [S12](#) is next.

Where this shows up next

[S12](#) takes the reversal in this unit's failing check, names it, and sets it alongside the other ways an analysis produces a confident wrong answer. [M12](#) is where regression with many correlated predictors goes when the columns start to overlap.

12. How analyses go wrong

Statistics · Unit 12

S12 · Statistics Reading: 6 pages With practice: ☒ 1.5 hours Assumes: S11

Why this unit is here

Every technique in this track can be executed perfectly and still produce a confident, reproducible, wrong answer. That is not a flaw in the techniques; it is what happens when a correct calculation is pointed at the wrong question, or run on a sample selected in a way nobody wrote down, or repeated until something interesting appears.

The failures in this unit are not exotic. They are the ordinary ones, they account for most of the retracted findings you will read about during your degree, and every one of them is visible in advance if you know the shape to look for. This is the unit that makes the previous eleven usable.

12.1 Before you start

You should be able to:

- fit and interpret a straight line with an interval on its slope — [S11](#);
- read a p -value and state what it is conditional on — [S9](#);
- name the unit, population, estimand and design of a question — [S1](#).

Quick check. A test is significant at $p < 0.05$. Twenty independent tests are run on data where nothing is going on. How many significant results would you expect, and what is the chance of at least one?

💡 Show the answer

You expect $20 \times 0.05 = 1$. The chance of at least one is $1 - 0.95^{20} = 0.64$.

So in a study with twenty comparisons and no real effects anywhere, the odds are roughly two to one that you will find something to write up. That single calculation is the engine behind a large fraction of this unit.

12.2 The idea

The cohort used throughout these notes contains a planted trap, and by now you have walked into it four times.

```
import numpy as np
import pandas as pd
from scipy import stats

cohort = pd.read_csv("data/cohort.csv")
scored = cohort[cohort["final_score"].between(0, 100)]
grouped = scored.dropna(subset=["background"])
```

```
pooled = stats.linregress(scored["hours_practice"], scored["final_score"])
print(f"pooled slope, all {len(scored)} students: {pooled.slope:+.4f} marks per
      hour")
for name, part in grouped.groupby("background"):
    within = stats.linregress(part["hours_practice"], part["final_score"])
    print(f"    {name:<24} n = {len(part):>3}    {within.slope:+.4f}")
```

pooled slope, all 238 students: -0.2408 marks per hour

Computing	n = 69	+0.6624
Engineering	n = 36	+0.8454
Life sciences	n = 57	+1.1526
Mathematics or Physics	n = 36	+1.0112
Social sciences	n = 39	+1.3445

Pooled, practice looks harmful. Within every background group, it looks helpful, by between roughly three and five times as much as the pooled slope says it hurts. This is **Simpson's paradox**: an association that reverses when a third variable is taken into account.

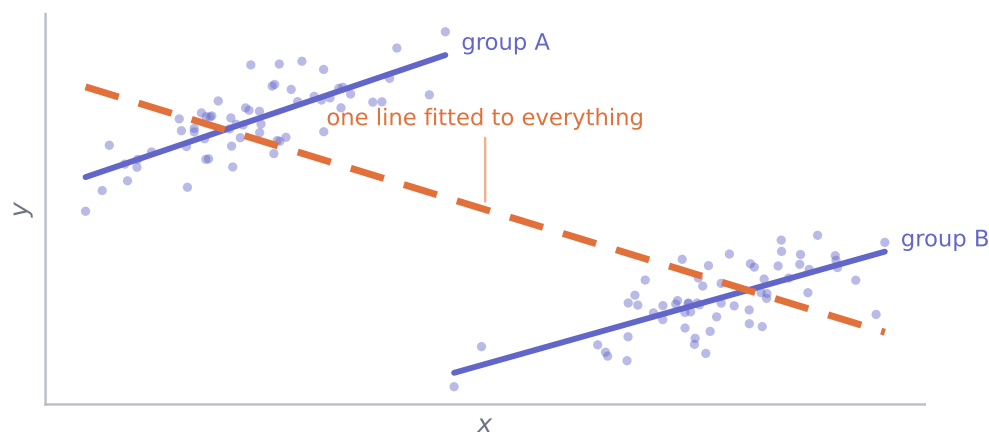


Figure 1: The mechanism, drawn cleanly. Two groups, each with a rising relationship inside it. The right-hand group starts higher on the vertical axis and lower on the horizontal one, so the two clouds are offset along a downward diagonal — and a line fitted to everything follows the offset between the groups rather than the trend inside them.

The mechanism is not mysterious. Students from quantitative backgrounds start further ahead and practise less; students from other backgrounds start further behind and practise more. Pooling the two puts a strong negative *between-group* pattern on top of a positive *within-group* one, and the between-group pattern wins.

Nothing in the pooled analysis is wrong. The slope is correctly computed, its interval is honest, its *p*-value is small. It answers a question — *among these 238 students, how does score vary with practice hours?* — and that question is almost never the one anybody wanted answered.

12.3 The mechanics

Confounding

Simpson's paradox is the dramatic case of a general problem. A **confounder** is a common cause: something that influences both the thing you varied and the thing you measured. Its influence is credited to whatever you happened to put in your model.

“Related to both” is not enough to make a variable a confounder, and the difference decides what to do about it. A variable on the causal path *from* your predictor *to* your outcome is a mediator, and adjusting for it changes which question you are answering rather than removing a bias. A variable that is a *consequence* of both is a collider, and adjusting for it manufactures a bias that was not there. Which kind you are looking at is a claim about how the world works, and no amount of data settles it.

There are three responses, in decreasing order of strength.

1. **Randomise.** If treatment is assigned by a coin, no variable — measured or not — can be systematically related to it. This is why experiments are worth their cost.
2. **Adjust.** Include the confounder in the model, or compare within its levels, as the group-by-group fits above do. This works only for confounders you thought of and measured.
3. **Say so.** If neither is available, state which confounders you could not rule out and in which direction each would bias the answer.

The third is not a failure. It is what an honest observational analysis looks like, and it is far more useful than an unqualified claim.

Selection, and the sample you did not see

S1 warned that a design decides which population an answer is about. The failures come in a family:

- **Self-selection** — people who respond differ from those who do not.
- **Survivorship** — you only see the units that made it: the funds still trading, the customers who did not leave, the aircraft that came back.
- **Collider bias** — conditioning on a *consequence* of two variables creates an association between them that does not exist in the population. Restrict attention to students who were admitted, and admission criteria will manufacture a negative relationship between the things that earn admission.

Sample size does not help with any of them; it makes the wrong answer more precise.

Multiple comparisons

Each test at $\alpha = 0.05$ has a one-in-twenty false-positive rate. Run many and false positives become the expected outcome rather than the exception.

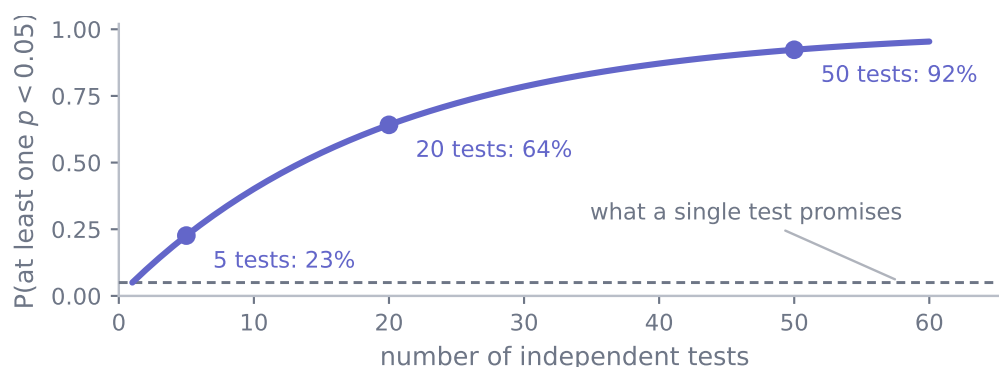


Figure 2: The probability of at least one significant result when nothing at all is going on, against the number of independent tests run. At five tests it is already better than one in five; by twenty it is nearly two in three. Nothing about any individual test has changed.

The count that matters is **every comparison you could have made**, not the ones you wrote down. Trying three outcome measures, two subgroups and two ways of excluding outliers is twelve analyses even if you report one. This is the “garden of forking paths”, and it does its damage without anyone intending to cheat: each decision felt reasonable at the time, and each was made after seeing the data.

Two repairs. **Bonferroni**: test at α/k for k comparisons, which controls the chance of any false positive and costs power. **False discovery rate** (Benjamini–Hochberg): control the *expected* proportion of your discoveries that are false — an average across many studies, not a guarantee about yours — which is usually the more sensible target when screening many hypotheses. Both are far less important than the habit of saying how many comparisons were made.

Base rates, again

S4 showed that a good test for a rare condition mostly produces false positives. Replace “condition” with “true hypothesis” and the same arithmetic applies to research.

If a fraction π of the hypotheses in a field are true, a test with power $1 - \beta$ and level α produces true positives at rate $(1 - \beta)\pi$ and false ones at rate $\alpha(1 - \pi)$, so

$$P(\text{true} \mid \text{significant}) = \frac{(1 - \beta)\pi}{(1 - \beta)\pi + \alpha(1 - \pi)}.$$

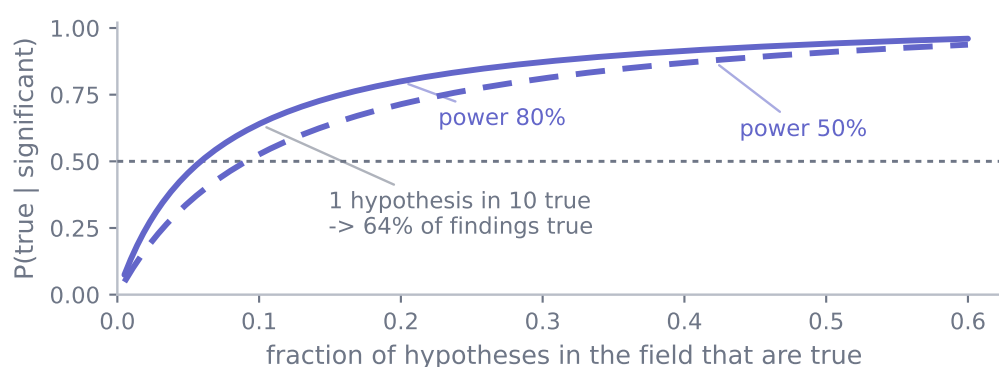


Figure 3: The chance that a significant finding is true, against the fraction of hypotheses in the field that are true, at the conventional 5% level. Even with 80% power, a field in which one hypothesis in ten holds up produces significant findings that are right only about two thirds of the time — and at 50% power, barely half.

Two things follow. **The p -value alone cannot tell you how likely a finding is to be true**, because two of the three numbers in that formula are properties of the field rather than of your study. Read them off the figure: with 80% power and one hypothesis in ten holding up, 64% of significant findings are real; drop the prior to one in twenty and it is 46%; drop the power to 50% as well and it is 35%. And **power is not just about your own study**: low power raises the share of published findings that are false, because it shrinks the numerator while leaving the false-positive rate alone.

Regression to the mean

If a measurement is part signal and part noise, the units that scored highest were partly lucky, and on a second measurement they will average closer to the middle. This happens with no cause at all, and it is routinely mistaken for one.

It is why the “worst-performing hospitals” improve after an intervention, why the sequel to a breakout

success disappoints, and why anyone who praises after a good outcome and criticises after a bad one will conclude that criticism works.

The defence is a control group. Without one you cannot separate the effect of what you did from the movement that would have happened anyway.

Overfitting

A model with enough freedom can fit any dataset exactly, including its noise. It will then perform badly on data it has not seen. The symptom is a gap between performance on the data you fitted and on data you held back; the defence is to hold data back before you start, and never to touch it until the model is final.

12.4 Worked example

Three failures, on the cohort.

One: the reversal, quantified. Fit the practice slope with and without the information about where each student started.

```
hours = scored["hours_practice"].to_numpy()
score = scored["final_score"].to_numpy()
diagnostic = scored["diagnostic_score"].to_numpy()
ones = np.ones(hours.size)

for name, design in [("score ~ 1 + hours", np.column_stack([ones, hours])),
                    ("score ~ 1 + hours + diagnostic",
                     np.column_stack([ones, hours, diagnostic]))]:
    theta, *_ = np.linalg.lstsq(design, score, rcond=None)
    print(f"{name:<32} hours coefficient {theta[1]:+.4f}")
```

```
score ~ 1 + hours           hours coefficient -0.2408
score ~ 1 + hours + diagnostic hours coefficient +0.9156
```

Adding one column moves the coefficient on practice from -0.24 to $+0.92$ marks per hour. Neither fit is a mistake. The first answers “how does score vary with practice across the cohort?”, the second “how does score vary with practice among students who arrived equally prepared?” — and only the second is close to the question a student asking whether to practise actually has.

Two: torturing the data. Split the cohort at random — a variable that cannot possibly matter — and test whether the two halves differ. Do it twenty times.

```
rng = np.random.default_rng(2026)
p_values = np.array([
    stats.ttest_ind(score[flag], score[~flag], equal_var=False).pvalue
    for flag in (rng.integers(0, 2, score.size).astype(bool) for _ in range(20))
])
print(f"20 random splits of the cohort")
print(f"  smallest p           {p_values.min():.4f}")
print(f"  number below 0.05    {(p_values < 0.05).sum()}")
print(f"  Bonferroni threshold for 20 tests: {0.05/20:.4f}")
```

```

20 random splits of the cohort
smallest p          0.0136
number below 0.05   1
Bonferroni threshold for 20 tests: 0.0025

```

One split in twenty comes out “significant” at $p = 0.014$, and a report of that split alone — “*students in group X scored significantly higher*” — would be entirely true about the arithmetic and entirely false about the world. Against the Bonferroni threshold of 0.0025 it survives nothing.

Three: regression to the mean. Take the students who scored highest on the arrival diagnostic and follow them to the final. Both variables are standardised, so the comparison is like for like.

```

z_diagnostic = (diagnostic - diagnostic.mean()) / diagnostic.std(ddof=1)
z_final = (score - score.mean()) / score.std(ddof=1)
print(f"correlation between the two: {np.corrcoef(z_diagnostic, z_final)[0,
- 1]:.4f}\n")
for label, mask in [("top 10% on arrival", z_diagnostic >=
- np.quantile(z_diagnostic, 0.9)),
                    ("bottom 10% on arrival", z_diagnostic <=
- np.quantile(z_diagnostic, 0.1))]:
    print(f"{label:<24} n = {mask.sum():>3}    "
          f"diagnostic {z_diagnostic[mask].mean():+.3f} sd    "
          f"final {z_final[mask].mean():+.3f} sd")

```

```
correlation between the two: 0.6579
```

```

top 10% on arrival      n =  24   diagnostic +1.700 sd   final +1.428 sd
bottom 10% on arrival  n =  24   diagnostic -1.491 sd   final -0.709 sd

```

The strongest starters were 1.70 standard deviations above average on arrival and 1.43 above it at the end. The weakest were 1.49 below on arrival and 0.71 below at the end. Both groups moved towards the middle, and nobody did anything to them.

Now imagine the department had run a support programme for the weakest tenth. Their improvement — most of the way back to average — would have looked like a triumph, and essentially all of it would have happened anyway.

12.5 Try it yourself

Exercise 1 A hospital reports that its overall surgical survival rate is lower than a neighbouring hospital's, but that its survival rate is higher for every individual severity category.

- Is one of the two figures wrong?
- What explains it?
- Which figure should a patient use?

Show the solution

- No.** Both can be correct simultaneously; this is Simpson's paradox with the roles of practice and background played by severity.

- (b) The hospital with the lower overall rate almost certainly takes **more severe cases**. Severity drives both which hospital you go to and whether you survive, so pooling across severity credits the case mix to the hospital.
- (c) **The category-specific figures**, matched to their own severity — that is the comparison that holds the confounder fixed. The overall figure answers a question about the two hospitals' *patients*, not about their surgeons.
Worth adding: this is exactly why publishing unadjusted mortality league tables creates an incentive to refuse difficult cases, and why the adjusted figures are the ones that get published.

Exercise 2 A team tests 4 outcome measures in 3 subgroups and reports the one result with $p = 0.03$ as their finding.

- (a) How many comparisons were made?
- (b) Roughly what is the chance of at least one $p < 0.05$ if nothing is going on?
- (c) What should they have written?

 Show the solution

- (a) $4 \times 3 = 12$ at minimum — and more if any decisions about exclusions or transformations were made after seeing the data.
- (b) $1 - 0.95^{12} = 0.46$, *if the twelve tests are independent*. Close to a coin flip, from a study in which nothing whatever is going on.
They will not be exactly independent — the same people appear in several comparisons, and the four outcomes are probably related — which pulls the number down. It does not pull it down to anything reassuring, and the honest version of the calculation is still “we made twelve comparisons”.
- (c) Something like: “*We examined 4 outcomes in 3 subgroups. One comparison reached $p = 0.03$ (uncorrected); against a Bonferroni threshold of 0.0042 it is not significant, and we report it as a hypothesis for a future study rather than as a finding.*”
Note what is **not** required: they do not have to hide the result. They have to report the denominator. A result that arrives with its number of comparisons attached is useful; the same result without it is misleading.

Exercise 3 During the Second World War, returning aircraft were examined to see where they had been hit, so that armour could be added where the damage was densest.

- (a) What is wrong with the plan?
- (b) Where should the armour go?
- (c) Name a modern analysis with the same structure.

 Show the solution

- (a) **Survivorship bias**. The sample consists only of aircraft that came back. Hits are presumably scattered fairly evenly, so the places where returning aircraft show *few* holes are the places where a hit meant not returning.
- (b) **Where the returning aircraft show no damage** — the engines, in the original study. The observed pattern is a map of survivable hits.
- (c) Many: studying only customers who did not churn, only funds that still exist, only patients

who completed a course of treatment, only startups that are still trading. In every case the units that would have been most informative are the ones missing from the file, and no amount of data on the survivors recovers them.

12.6 Check it in Python

The multiple-comparisons arithmetic is worth watching rather than believing. Here nothing is going on anywhere — two groups drawn from the same population — and we count how often at least one test in a batch comes out significant.

```
rng = np.random.default_rng(7)
for tests in [1, 5, 20]:
    hits = 0
    for _ in range(4000):
        batch = [stats.ttest_ind(rng.normal(0, 1, 30), rng.normal(0, 1,
    - 30)).pvalue
                for _ in range(tests)]
        hits += min(batch) < 0.05
    print(f"{tests}>2} tests per study: at least one significant in "
          f"{100*hits/4000:5.1f}% of studies    (formula {100*(1 -
    - 0.95**tests):5.1f}%)")
```

```
1 tests per study: at least one significant in    5.5% of studies    (formula    5.0%)
5 tests per study: at least one significant in   23.5% of studies    (formula   22.6%)
20 tests per study: at least one significant in  63.7% of studies    (formula   64.2%)
```

And now the check that fails — the one this whole unit is about. It is tempting to believe that if an association is real and strong in the data, adjusting for another variable can only change it a little:

```
simple, *_ = np.linalg.lstsq(np.column_stack([ones, hours]), score, rcond=None)
adjusted, *_ = np.linalg.lstsq(np.column_stack([ones, hours, diagnostic]),
                               score, rcond=None)
assert np.sign(simple[1]) == np.sign(adjusted[1]), (
    f"the practice coefficient is {simple[1]:+.4f} on its own and "
    f"{adjusted[1]:+.4f} once the arrival diagnostic is included – "
    f"opposite signs, same data"
)
```

```
AssertionError: the practice coefficient is -0.2408 on its own and +0.9156 once
- the arrival diagnostic is included – opposite signs, same data
```

```
-----
AssertionError                                Traceback (most recent call last)
Cell In[9], line 4
      1 simple, *_ = np.linalg.lstsq(np.column_stack([ones, hours]), score,
    - rcond=None)
      2 adjusted, *_ = np.linalg.lstsq(np.column_stack([ones, hours,
    - diagnostic]),
      3                               score, rcond=None)
----> 4 assert np.sign(simple[1]) == np.sign(adjusted[1]), (
```

```

5     f"the practice coefficient is {simple[1]:+.4f} on its own and "
6     f"{adjusted[1]:+.4f} once the arrival diagnostic is included – "
7     f"opposite signs, same data"
8 )

```

AssertionError: the practice coefficient is -0.2408 on its own and +0.9156 once
– the arrival diagnostic is included – opposite signs, same data

Not a little: the opposite direction. There is no rule that says adding a variable makes a small change, and the size of the change is exactly the size of the confounding.

Finally, regression to the mean, on data with **no relationship between the two occasions except noise around a shared underlying ability**:

```

rng = np.random.default_rng(3)
ability = rng.normal(0, 1, 5000)
first = ability + rng.normal(0, 1, 5000)      # nothing happens between the two
second = ability + rng.normal(0, 1, 5000)

# Standardise both occasions, so the two are on a common scale.
z_first = (first - first.mean()) / first.std(ddof=1)
z_second = (second - second.mean()) / second.std(ddof=1)

worst = z_first <= np.quantile(z_first, 0.1)
print(f"bottom 10% at time 1: {z_first[worst].mean():+.3f} sd"
      f" -> time 2 {z_second[worst].mean():+.3f} sd")
print(f"apparent 'improvement': "
      f"{z_second[worst].mean() - z_first[worst].mean():+.3f} sd, "
      f"from an intervention that did not exist")

```

```

bottom 10% at time 1: -1.747 sd -> time 2 -0.932 sd
apparent 'improvement': +0.815 sd, from an intervention that did not exist

```

Nothing was done to anyone. The improvement is entirely the noise that put them at the bottom in the first place, failing to repeat itself.

12.7 Common wrong turns

Where this usually goes wrong

Reporting a pooled result when the groups disagree If the direction reverses inside subgroups, the pooled number is not a summary of anything anyone asked about. Report both, and say which question each answers.

Adjusting for everything you have Adjustment is not automatically safer. Conditioning on a variable that is a *consequence* of both your predictor and your outcome creates bias rather than removing it. Which variables to adjust for is a question about how the world works, not about which columns are in the file.

Counting only the tests you reported The relevant number is every comparison you could have made, including decisions about exclusions and transformations taken after seeing the data. Report the denominator.

Explaining regression to the mean If you selected units for being extreme, expect them to move towards the middle next time. Attributing that movement to whatever you did in between requires a control group that was equally extreme and left alone.

Treating a large sample as protection Size fixes sampling noise and nothing else. Confounding, selection and survivorship all survive intact into a dataset of any size, and become harder to argue with because the intervals are so tight.

Evaluating a model on the data you fitted Any sufficiently flexible model fits its training data. Hold data back before you start, and do not look at it until the model is final — including not looking at it “just to check”.

Believing that a significant result is probably true How likely it is to be true depends on three numbers — the fraction of hypotheses in the field that hold up, the power, and α — and the p -value contains only the last. In an exploratory field with modest power the share of significant findings that are false can easily exceed half. That is base rates, and it is arithmetic rather than cynicism.

12.8 Check yourself

Answers are at the foot of the page. Give a reason as well as an answer.

1. A study finds a positive association between coffee and heart disease. Which would most change your interpretation?
 - a. A larger sample
 - b. A smaller p -value
 - c. Information about smoking
 - d. A higher R^2
2. Ten subgroups are tested and one gives $p = 0.04$. How much evidence is that?
3. A school selects its 20 weakest readers for a programme. After a term they have improved relative to the school average. Did the programme work?
4. Adding a variable to a regression changes a coefficient's sign. Which fit is right?
5. Why does low statistical power make the *published* literature less reliable, rather than just making individual studies less informative?

Show the answers

1. **(c).** Smoking is associated with both coffee drinking and heart disease, so it is a candidate confounder, and the association could be entirely its doing. (a), (b) and (d) all make the estimate more precise or the fit tighter without touching the reason it might be wrong — which is the standard shape of this unit's lesson.
2. **Very little on its own.** With ten tests, the chance of at least one $p < 0.05$ under no effect at all is $1 - 0.95^{10} = 0.40$. Reported with its denominator, and treated as a hypothesis to test on new data, it is worth something. Reported alone as a finding, it is close to worthless.
3. **You cannot tell.** They were selected for being extreme, so regression to the mean predicts improvement with no programme at all. Without a comparison group of equally weak readers who did not take part, the programme's effect and the statistical artefact are inseparable.
4. **Both are correct fits, and they answer different questions.** The unadjusted coefficient describes how the outcome varies with the predictor across the whole sample; the adjusted one describes how it varies among units alike on the added variable. Which one you want depends on what role the added variable plays: a **common cause** of predictor and

outcome (adjust, to remove confounding); something **on the path** from predictor to outcome (adjusting is legitimate but answers a narrower question — the part of the effect that does not run through it); or a **consequence of both** (adjusting creates a bias that was not there). Nothing in the regression output distinguishes the three.

5. Because power sits in the numerator of $P(\text{true} \mid \text{significant})$. Halving power halves the rate of true positives while leaving the false-positive rate at α , so the proportion of significant findings that are false goes *up*. A literature built from underpowered studies contains a higher share of false claims than the same literature built from well-powered ones, even with everyone behaving honestly.

Where this shows up next

This is the last unit of the Statistics track, and the habits in it are the ones worth carrying into everything else you do.

Write the question down before you touch the data — [S1](#). Report an estimate with an interval, in the units of the thing measured — [S8](#), [S10](#). Say how many comparisons you made. Say which confounders you could not rule out. And when a result surprises you, look for the third variable before you look for a publisher.